

Unit Functional Design Specification - Refinery 2

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1 EQUIPMENT MODULE CLASS DESIGN – EM_RF_STARTUP1 (STARTUP)

1.1 Functional Description

EM_RF_STARTUP1 is a class based equipment module that communicates with four other sequences to complete production process after checking parameters and fatty acid circuit operation. Refer FDS for the functional design requirements including faceplate and detail display. Initially Operator shall be provided only with Hold and Aux On. Production, Drain and Aux Off command shall be enabled during the execution of Aux On State. All devices used in EM shall be acquired in Aux On Command and It shall be released in Aux Off Command. Hold Request shall be enabled for all acquired devices at the end of Hold command. Commands which require operator intervention shall be configured as Operator Prompts wherever possible.

1.2 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name
EM	EM_RF_STARTUP1	RF2EQ_STARTUP1
DEV1	_MVS11_IL_130_SO	RF2_PU2734CA
DEV2	_MRSS11_130_SE	Ignored
DEV3	_MRSS11_130_SE	Ignored
DEV4	_MRSS11_130_SE	RF2_PU2741X
DEV5	_V12_IP_130_SE	Ignored
DEV6	_CMT_130_SE	RF2_FT2734CA
DEV7	EM_IC_VAC_PROD1	Ignored
DEV8	EM_IC_VAC_PROD1	Ignored
DEV9	EM_IC_VAC_PROD2	Ignored
DEV10	EM_DEOD_FA_CKT3	RF2EQ_814_23
DEV11	EM_DEOD_FA_CKT1	RF2EQ_802

Table 1: Instances

(Note: DEV7, DEV8 & DEV9 are ignored as 819IC1/2 & 878-32IC sequences are not part of Refinery 2 Area.)

1.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command

1.3.1 Command 00 – Hold

Step	Description	Action	Action Description
H0010	Command Hold Start	A0010	Store Fail Message
		A0020	Set Hold Request
T1000	Actions Complete or Time Out & Hold=True		Go to step H1000
T2000	Actions Complete or Time Out & Hold=False		Go to step H2000
H1000	Hold Command 1	A0010	Stopping and Disabling Conditional Logic for DEV1
T9900A	Actions Complete or Time Out		Go to step H9900A
H2000	General Reset	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Stopping and Disabling Conditional Logic for DEV2
		A0030	Stopping and Disabling Conditional Logic for DEV3
		A0040	Stopping and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0070	Set DEV6 Mode to Remote Output
		A0080	Set DEV6 output to Closed and Disable Conditional Logic
T9900B	Actions Complete or Time Out		Go to step H9900B
H9900A	Command Hold End	A0010	Completed
END	Command Complete		True
H9900B	Release Devices	A0010	Release Devices Message
		A0020	Release Private Devices

Step	Description	Action	Action Description
		A0030	Release Private Devices
		A0040	Release Private Devices
		A0050	Release Private Devices
		A0060	Release Private Devices
		A0070	Reset Parameters
END1	Command Complete		True

Table 2: Hold Sequence

1.3.2 Command 01 – Aux On

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Aux On Command
T0010	First Pass = False		If First Pass = False then go to step S0010
T0020A	First Pass = True		If First Pass = True then go to step S0020
S0010	Initializations	A0010	First Pass Logic
		A0020	First Pass = True
		A0030	Set Task Pointer
		A0040	Acquire EM
T0020B	Acquire Devices		If no pending actions in S0010 then go to step S0020
S0020	Acquire Devices	A0010	Acquire Devices Message
		A0020	Acquire Private Devices
		A0030	Acquire Private Devices
		A0040	Acquire Private Devices
		A0050	Acquire Private Devices
		A0060	Acquire Private Devices
		A0070	Acquire Private Devices
		A0080	Acquire Private Devices
		A0090	Acquire Private Devices
		A0100	Acquire Private Devices
		A0110	Reset Hold Request
T0030	Standby		If Task Pointer = 1 then go to step S0030
T0100B	Start Aux		If Task Pointer = 2 then go to step S0100
T0200B	Check Parameters		If Task Pointer = 3 then go to step S0200
T0300B	Start Fatty Acid Ckt		If Task Pointer = 4 then go to step S0300
T0400B	Section Ready		If Task Pointer = 5 then go to step S0400



Step	Description	Action	Action Description
S0030	Standby	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Stopping and Disabling Conditional Logic for DEV2
		A0030	Stopping and Disabling Conditional Logic for DEV3
		A0040	Stopping and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Set DEV6 Mode to Remote Output
		A0070	Set DEV4 output to Closed and Disable Conditional Logic
T0100A	Start Aux	S0030	True
S0100	Start Aux	A0010	Enabling Conditional Logic for DEV1
		A0020	Enabling Conditional Logic for DEV2
		A0030	Enabling Conditional Logic for DEV3 and DEV4
		A0040	Enabling Conditional Logic for DEV5
		A0050	Set DEV6 Mode to Remote Cascade
		A0060	Setting DEV6 setpoint (FT_734CA Setpoint) and Enabling Conditional Logic
		A0070	Start DEV7 and DEV8
		A0080	Start DEV9
		A0090	Waiting for Auxiliary Circuit to run
		A0100	1 min Delay (TP001)
T0200A	Check Parameters		If 1 min delay time (TP001) is over then go to step S0200
S0200	Check Parameters	A0010	Enabling Conditional Logic for DEV1
		A0020	Enabling Conditional Logic for DEV2
		A0030	Enabling Conditional Logic for DEV3 and DEV4
		A0040	Enabling Conditional Logic for DEV5
		A0050	Set DEV6 Mode to Cascade
		A0060	Setting DEV6 setpoint (FT_734CA Setpoint) and Enabling Conditional Logic
		A0070	Set Message: Waiting for Vacuum Setpoint
T0300A	Start Fatty Acid. Ckt		If Vacuum PT_722AB is below Vacuum set point (TP002) then go to step S0300
S0300	Start Fatty Acid. Ckt	A0010	Enabling Conditional Logic for DEV1
		A0020	Enabling Conditional Logic for DEV2

Step	Description	Action	Action Description
		A0030	Enabling Conditional Logic for DEV3 and DEV4
		A0040	Enabling Conditional Logic for DEV5
		A0050	Set DEV6 Mode to Cascade
		A0060	Setting DEV6 setpoint (FT_734CA Setpoint) and Enabling Conditional Logic
		A0070	Acquire DEV10
		A0080	Set Message: Starting DEV10
		A0090	Set Message: Checking Fatty acid tank is in step Production
T0400A	Section Ready		If Fatty acid tank DEV10 is in step <Production> then go to step S0400
S0400	Section Ready	A0010	Enabling Conditional Logic for DEV1
		A0020	Enabling Conditional Logic for DEV2
		A0030	Enabling Conditional Logic for DEV3 and DEV4
		A0040	Enabling Conditional Logic for DEV5
		A0050	Set DEV6 Mode to Cascade
		A0060	Setting DEV6 setpoint (FT_734CA Setpoint) and Enabling Conditional Logic
T9900	Complete		If no pending actions in S0400 then go to step S9900
S9900	Complete	A0010	Set Message: Aux On Command Complete
END	Command Complete		True

Table 3: Aux On Sequence

1.3.3 Command 02 – Production

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Production Command
T0100	Standby		If Task Pointer = 5 then go to step S0100
T0400B	Start Aux		If Task Pointer = 6 then go to step S0200
S0100	Startup	A0010	Enabling Conditional Logic for DEV1
		A0020	Enabling Conditional Logic for DEV2
		A0030	Enabling Conditional Logic for DEV3 and DEV4
		A0040	Enabling Conditional Logic for DEV5
		A0050	Set DEV6 Mode to Cascade
		A0060	Setting DEV6 setpoint (FT_734CA Setpoint) and Enabling Conditional Logic
		A0070	Acquire DEV11

Step	Description	Action	Action Description
		A0080	Starting DEV11
T0200	Production		If DEOD_FA_CKT2 is in step <Production> then go to step S0200
S0200	Production	A0010	Enabling Conditional Logic for DEV1
		A0020	Enabling Conditional Logic for DEV2
		A0030	Enabling Conditional Logic for DEV3 and DEV4
		A0040	Enabling Conditional Logic for DEV5
		A0050	Set DEV6 Mode to Cascade
		A0060	Setting DEV6 setpoint (FT_734CA Setpoint) and Enabling Conditional Logic
T0200	Complete		If DEOD_FA_CKT2 is in step <Production> then go to step S0200
S9900	Complete	A0010	Set Message: Production Command Complete
END	Command Complete		True

Table 4: Production Sequence

1.3.4 Command 03 – Drain

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Drain Command
T0100	Drain		If Task Pointer = 6 then go to step S0100
S0100	Drain	A0010	Enabling Conditional Logic for DEV1
		A0020	Enabling Conditional Logic for DEV2
		A0030	Enabling Conditional Logic for DEV3 and DEV4
		A0040	Enabling Conditional Logic for DEV5
		A0050	Set DEV6 Mode to Cascade
		A0060	Setting DEV6 setpoint (FT_734CA Setpoint) and Enabling Conditional Logic
T9900	Complete		If no pending actions in S0100 then go to step S9900
S9900	Complete	A0010	Set Message: Drain Command Complete
END	Command Complete		True

Table 5: Drain Sequence

1.3.5 Command 04 – Aux Off

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Aux Off Command
T0100	Aux Off		If DEV11 is in <Standby> or DEOD_FA_CKT2 is in <Standby> then go step S0100
S0100	Aux Off	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Stopping and Disabling Conditional Logic for DEV2
		A0030	Stopping and Disabling Conditional Logic for DEV3
		A0040	Stopping and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Reset AUX (OP003), RUN (OP004), DRAIN (OP005) Command
		A0070	Set DEV6 Mode to Remote Output
		A0080	Set DEV6 output to Closed and Disable Conditional Logic
T0200	Disable HM and SM		If no pending actions in S0100 then go to step S0200
S0200	Disable HM	A0010	Disable HM 01 (Module Ownership Lost)
T0300	Release Devices		If no pending actions in S0200 then go to step S0300
S0300	Release Devices	A0010	Release Devices Message
		A0020	Release Private Devices
		A0030	Release Private Devices
		A0040	Release Private Devices
		A0050	Release Private Devices
		A0060	Release Private Devices
		A0070	Release Private Devices
T9900	Complete		If no pending actions in S0300 then go to step S9900
S9900	Complete	A0010	Set Message: Aux Off Command Complete
END	Command Complete		True

Table 6: Aux Off Sequence

1.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value/Reference
Operating Parameters				
OP001	Vaccum Equipment Selection	Named Set	NA	PU1741X = 0 (EJECTOR = 1)
OP002	Vaccum Pump Selection	Named Set	NA	PU1741X1 = 0 (PU1741X2 = 1)
OP003	AUX	Named Set	NA	OFF
OP004	DRAIN	Named Set	NA	OFF
OP005	HOLD	Named Set	NA	OFF
Tuning Parameters				
TP001	Delay Timer	Floating Point	sec	60
TP002	Vacuum Setpoint	Floating Point	mbar	0.003
TP003	FT_734CA RSP	Floating Point	sec	10

Table 7: Operating and Tuning Parameters

1.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

Parameter Name	Parameter Type	Value/Refrence
MY_TAG		RF2EQ_STARTUP1
OPR_CMD_MASK		3
LT_734CA	External Refernece	RF2_LT2734CA/AI1/LO_LO_ACT
LT_734NA	External Refernece	#IGNORE
LSL_732A	External Refernece	#IGNORE
PT_722AB	External Refernece	RF2_PT2722AB/PV
819IC1	Boolean	False
819IC2	Boolean	False
DY_REF_1	Dynamic Reference	
DEOD_FA_CKT1	String	RF2EQ_802
DEOD_FA_CKT2	String	RF2EQ_822QE
DEOD_FA_CKT3	String	RF2EQ_814_23
DEOD3_STEP	Boolean	True

Table 8: Configurable Parameters

1.6 Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_RF_STARTUP1	Refinery Area: Deodorising & Ice Condensing

Table 9: Named Set Details

Details of states for NS_RF_STARTUP1 named set are as follows,

Named Set Value	Named Set Name
00	Hold
01	Aux On
02	Production
03	Drain
04	Aux Off

Table 10: Name Set State Details

2 EQUIPMENT MODULE CLASS DESIGN – EM_DEOD_FA_CKT1 (2-802)

2.1 Functional Description

EM_DEOD_FA_CKT1 is a class based equipment module that performs filling operation to produce the material and afterwards drains the material. EM_DEOD_FA_CKT1 is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. Initially Operator shall be provided only with Hold and Production command. All devices used in EM shall be acquired and released in Production Command. Hold Request shall be enabled for all acquired devices at the end of Hold command. Commands which require operator intervention shall be configured as Operator Prompts wherever possible.

2.2 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name
EM	EM_DEOD_FA_CKT1	RF2EQ_802
DEV1	EM_DEOD_FA_CKT2	RF2EQ_822QE
DEV2	_V12_IP_130_SE	RF2_V2702F
DEV3	_V12_IP_130_SE	RF2_V2721XS

Table 11: Instances

2.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

2.3.1 Command 00 – Hold

Step	Description	Action	Action Description
H0010	Command Hold Start	A0010	Store Fail Message
		A0020	Reset Hold Request
T1000	Actions Complete or Time Out & Hold=True		Go to step H1000
T2000	Actions Complete or Time Out & Hold=False		Go to step H2000

Step	Description	Action	Action Description
H1000	Hold Command 1	A0010	Set Message
T9900A	Actions Complete or Time Out		Go to step H9900A
H2000	General Reset	A0010	Closing DEV2
		A0020	Closing DEV3
T9900B	Actions Complete or Time Out		Go to step H9900B
H9900A	Command Hold End	A0010	Completed
END	Command Complete		True
H9900B	Release Devices	A0010	Release Devices Message
		A0020	Release Private Device
		A0030	Release Private Device
		A0040	Reset Parameters
END1	Command Complete		True

Table 12: Hold Sequence

2.3.2

Command 01 – Production

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Production Command
T0010	First Pass = False		If First Pass = False then go to step S0010
T0020A	First Pass = True		If First Pass = True then go to step S0020
S0010	Initializations	A0010	First Pass Logic
		A0020	First Pass = True
		A0030	Set Task pointer
T0020B	Actions Complete		If no pending actions in S0010 then go to step S0020
S0020	Acquire Devices	A0010	Acquire Devices Message
		A0020	Acquire Private Device
		A0030	Acquire Private Device
		A0040	Reset Hold Request
T0100	Start		If Task Pointer = 1 then go to step S0100

Step	Description	Action	Action Description
T0200B	Fill		If Task Pointer = 2 then go to step S0200
T0300B	Production		If Task Pointer = 3 then go to step S0300
T0400B	Drain		If Task Pointer = 4 then go to step S0400
T0500B	Drain Wait		If Task Pointer = 5 then go to step S0500
S0100	Standby	A0010	Closing and Disabling Conditional Logic for DEV2
		A0020	Closing and Disabling Conditional Logic for DEV3
T0200A	Fill		If no actions pending in S0100 then go to step S0200
S0200	Fill	A0010	Enabling Conditional Logic for DEV2
		A0020	Closing and Disabling Conditional Logic for DEV3
		A0040	Set Message: Waiting for LSL_702 is not in alarm or LSH_702 is in alarm or LT_702 is in VHL alarm
		A0050	Set Message: Scratch Time
T0300A	Production		If LSH_702 is in alarm or LT_702 is in VHL alarm or 5 min delay time (TP001) is over then go step S0300
S0300	Production	A0010	Enabling Conditional Logic for DEV2
		A0020	Enabling Conditional Logic for DEV3
		A0030	Check DEV1 is not owned by some other EM
		A0040	Acquire DEV1
		A0050	Start Heating command of DEV1
		A0060	Start Production command of DEV1
		A0070	Start DEV1 of DEV11
T0400A	Drain		If DEOD_FA_CKT2 is in <Drain 802> then go to step S0400
S0400	Drain	A0010	Closing and Disabling Conditional Logic for DEV2
		A0020	Closing and Disabling Conditional Logic for DEV3
		A0040	Set Message: LSL_702 is in alarm and LT_702 and FT_702 are in VL alarm
		A0050	10 sec Delay Time (TP002)
T0500A	Drain Wait		If 10 sec delay time (TP002) is completed then go to step S0500
S0500	Drain Wait	A0010	Closing and Disabling Conditional Logic for DEV2

Step	Description	Action	Action Description
		A0020	Closing and Disabling Conditional Logic for DEV3
		A0030	Set Message: Checking Fatty acid tank 2 is in Drain 821A2
		A0040	Set Message: 10 sec Delay
T0600	Disable HM and SM		If 10 sec delay time (TP002) is completed the go to step S0600
S0600	Disable HM	A0010	Disable HM 01 (Module Ownership Lost)
T0700	Release Devices		If no pending action in S0600 then go to step S0700
S0700	Release Devices	A0010	Release Devices Message
		A0020	Release Private Device
		A0030	Release Private Device
T9900	Complete		If no pending action in S0700 then go to step S9900
S9900	Complete	A0010	Set Message: Production Command Complete
		A0020	Self Release the EM & Reset First Pass
END1	Command Complete		True

Table 13: Production Sequence

2.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value/Reference
Operating Parameters				
OP001	Selection button to switch ON 1-821X	Named Set	NA	OFF
Tuning Parameters				
TP001	Delay Timer 1	Floating Point	sec	300
TP002	Delay Timer 2	Floating Point	sec	10

Table 14: Operating and Tuning Parameters

2.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

Parameter Name	Parameter Type	Value/Reference
MY_TAG		RF2EQ_802
OPR_CMD_MASK		3
LSH_702	External Reference	RF2_LSH2702/DI1/DISC_ACT
LT_702	String	RF2_LT2702
LSL_702	External Reference	RF2_LSL2702/DI1/DISC_ACT
FT_702	External Reference	RF2_FT2702/PID1/LO_ACT
RF_STARTUP	String	RF2EQ_STARTUP1
DEOD_FA_CKT2	String	RF2EQ_822QE
DY_REF_1	Dynamic Reference	
DY_REF_2	Dynamic Reference	
HOLD	External Reference	RF2EQ_STARTUP1/OP006_VALUE

Table 15: Configurable Parameters

2.6 Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_DEOD_FA_CKT1	Refinery Area: Deodorising & Ice Condensing

Table 16: Named Set Details

Details of states for NS_DEOD_FA_CKT1 named set are as follows,

Named Set Value	Named Set Name
00	Hold
01	Production

Table 17: Name Set State Details

3 EQUIPMENT MODULE CLASS DESIGN – EM_DEOD_FA_CKT2 (2-822QE)

3.1 Functional Description

EM_DEOD_FA_CKT2 is a class based equipment module that performs filling and heating operation and after that drains the material. EM_DEOD_FA_CKT2 is derived from the Command Driven equipment module class . Refer FDS for the functional design requirements including faceplate and detail display. Initially Operator shall be provided only with Hold and Heating. Aux Off, Production, Recirculation and Drain command shall be enabled during the execution of Heating State. Hold Request shall be enabled for all acquired devices at the end of HOLD command. Commands which require operator intervention shall be configured as Operator Prompts wherever possible.

3.2 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name
EM	EM_DEOD_FA_CKT2	RF2EQ_822QE
DEV1	_MRSS11_130_CUR	RF2_PU2702
DEV2	_V12_IP_130_SE	RF2_V2712D
DEV3	_MVS11_IL_130_SO	Ignored
DEV4	_MRSS11_130_SE	RF2_PU2722C1
DEV5	_MRSS11_130_SE	RF2_PU2722C2
DEV6	_MRSS11_130_CUR	RF2_PU27221
DEV7	_MRSS11_130_CUR	RF2_PU27222
DEV8	_V12_IP_130_SE	RF2_V2702D
DEV9	_V12_IP_130_SE	RF2_V2702B
DEV10	_V12_IP_130_SE	RF2_V2702BY
DEV11	_V12_IP_130_SE	RF2_V2722QEA
DEV12	_V12_IP_130_SE	RF2_V2722QEB
DEV13	_V12_IP_130_SE	RF2_V2722QEC
DEV14	_V12_IP_130_SE	RF2_V2722QED
DEV15	_V12_IP_130_SE	Ignored
DEV16	_V12_IP_130_SE	RF2_V2722CB
DEV17	_V12_IP_130_SE	RF2_V2780BD
DEV18	_V12_IP_130_SE	RF2_V2780BB
DEV19	_V12_IP_130_SE	RF2_V2781AB
DEV20	_CSD_130_SE	RF2_LT2780B

Device Name	Module Class	Instance Name
DEV21	_CSD_130_SE	RF2_FT2702
DEV22	_CSD_130_SE	RF2_LT2722C
DEV23	_CSD_130_SE	RF2_TT2781B
DEV24	_V12_IP_130_SE	RF2_V2721A1D
DEV25	_V12_IP_130_SE	RF2_V2721A2D
DEV26	_V12_IP_130_SE	RF2_V2722A1D
DEV27	_V12_IP_130_SE	RF2_V2722A2D
DEV28	_V12_IP_130_SE	RF2_V2722A3D
DEV29	_V12_IP_130_SE	RF2_V2722A4D
DEV30	_V12_IP_130_SE	RF2_V2722A5D
DEV31	_V12_IP_130_SE	Ignored
DEV32	_V12_IP_130_SE	RF2_V2780A1A
DEV33	_V12_IP_130_SE	RF2_V2780A1B
DEV34	_V12_IP_130_SE	RF2_V2780A2A
DEV35	_V12_IP_130_SE	RF2_V2780A2B

Table 18: Instances

(Note: DEV31 is not a part of Refinery 2 Area)

3.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

3.3.1 Command 00 – Hold

Step	Description	Action	Action Description
H0010	Command Hold Start	A0010	Store Fail Message
		A0020	Reset Hold Request
		A0030	Save Previous Step Time
T1000	Actions Complete or Time Out & Hold=True		Go to step H1000
T2000	Actions Complete or Time Out & Hold=False		Go to Step H2000
H1000	Hold Command 1	A0010	Stopping and Disabling Conditional logic for DEV1
		A0020	Stopping DEV3

Step	Description	Action	Action Description
		A0030	Stopping and Disabling Conditional Logic for DEV4
		A0040	Stopping and Disabling Conditional Logic for DEV5
		A0050	Stopping and Disabling Conditional Logic for DEV6
		A0060	Stopping and Disabling Conditional Logic for DEV7
T9900A	Actions Complete or Time Out		If no pending actions in S1000 then go step H9900B
H2000	General Reset	A0010	Stopping and Disabling Conditional logic for DEV1
		A0020	Closing DEV2
		A0030	Stopping DEV3
		A0040	Stopping and Disabling Conditional Logic for DEV4
		A0050	Stopping and Disabling Conditional Logic for DEV5
		A0060	Stopping and Disabling Conditional Logic for DEV6
		A0070	Stopping and Disabling Conditional Logic for DEV7
		A0080	Closing DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Closing and Disabling Conditional Logic for DEV14
		A0150	Closing DEV15
		A0160	Closing and Disabling Conditional Logic for DEV16
		A0170	Closing DEV17
		A0180	Closing and Disabling Conditional Logic for DEV18
		A0190	Closing and Disabling Conditional Logic for DEV19
		A0200	Set DEV20 Mode to Remote Out
		A0210	Set DEV20 output to Closed and Disable Conditional Logic
		A0220	Set DEV21 Mode to Remote Out

Step	Description	Action	Action Description
		A0230	Set DEV21 output to Closed and Disable Conditional Logic
		A0240	Set DEV22 Mode to Remote Out
		A0250	Set DEV22 output to Closed and Disable Conditional Logic
		A0260	Set DEV23 Mode to Remote Out
		A0270	Set DEV23 output to Closed and Disable Conditional Logic
		A0280	Closing and Disabling Conditional Logic for DEV24
		A0290	Closing and Disabling Conditional Logic for DEV25
		A0300	Closing and Disabling Conditional Logic for DEV26
		A0310	Closing and Disabling Conditional Logic for DEV27
		A0320	Closing and Disabling Conditional Logic for DEV28
		A0330	Closing and Disabling Conditional Logic for DEV29
		A0340	Closing and Disabling Conditional Logic for DEV30
		A0350	Closing and Disabling Conditional Logic for DEV31
		A0360	Closing and Disabling Conditional Logic for DEV32
		A0370	Closing and Disabling Conditional Logic for DEV33
		A0380	Closing and Disabling Conditional Logic for DEV34
		A0390	Closing and Disabling Conditional Logic for DEV35
		A0400	Set DEV36 Mode to Remote Out
		A0410	Set DEV36 output to Closed and Disable Conditional Logic
T9900B	Actions Complete or Time Out		If no pending actions in S2000 then go step H9900B
H9900A	Command Hold End	A0010	Completed
END	Command Complete		True
H9900B	Release Devices	A0010	Release Devices Message
		A0020	Release Private Device
		A0030	Release Private Device
		A0040	Release Private Device

Step	Description	Action	Action Description
		A0050	Release Private Device
		A0060	Release Private Device
		A0070	Release Private Device
		A0080	Release Private Device
		A0090	Release Private Device
		A0100	Release Private Device
		A0110	Release Private Device
		A0120	Release Private Device
		A0130	Release Private Device
		A0140	Release Private Device
		A0150	Release Private Device
		A0160	Release Private Device
		A0170	Release Private Device
		A0180	Release Private Device
		A0190	Release Private Device
		A0200	Release Private Device
		A0210	Release Private Device
		A0220	Release Private Device
		A0230	Release Private Device
		A0240	Release Private Device
		A0250	Release Private Device
		A0260	Release Private Device
		A0270	Release Private Device
		A0280	Release Private Device
		A0290	Release Private Device
		A0300	Release Private Device
		A0310	Release Private Device
		A0320	Release Private Device
		A0330	Release Private Device
		A0340	Release Private Device
		A0350	Release Private Device
		A0360	Release Private Device
		A0370	Reset Parameters
END1	Command Complete		True

Table 19: Hold Sequence

3.3.2 Command 01 – Heating

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Heating Command

Step	Description	Action	Action Description
T0010	First Pass = False		If First Pass = False then go to step S0010
T0020A	First Pass = True		If First Pass = True then go to step S0020
S0010	Initializations	A0010	First Pass Logic
		A0020	First Pass = True
		A0030	Set Task Pointer
T0020B	Actions Complete		If no pending actions in S0010 then go to step S0020
S0020	Acquire Devices	A0010	Acquire Devices Message
		A0020	Acquire Private Device
		A0030	Acquire Private Device
		A0040	Acquire Private Device
		A0050	Acquire Private Device
		A0060	Acquire Private Device
		A0070	Acquire Private Device
		A0080	Acquire Private Device
		A0090	Acquire Private Device
		A0100	Acquire Private Device
		A0110	Acquire Private Device
		A0120	Acquire Private Device
		A0130	Acquire Private Device
		A0140	Acquire Private Device
		A0150	Acquire Private Device
		A0160	Acquire Private Device
		A0170	Acquire Private Device
		A0180	Acquire Private Device
		A0190	Acquire Private Device
		A0200	Acquire Private Device
		A0210	Acquire Private Device
		A0220	Acquire Private Device
		A0230	Acquire Private Device
		A0240	Acquire Private Device
		A0250	Acquire Private Device
		A0260	Acquire Private Device
		A0270	Acquire Private Device
		A0280	Acquire Private Device
		A0290	Acquire Private Device
		A0300	Acquire Private Device
		A0310	Acquire Private Device

Step	Description	Action	Action Description
		A0320	Acquire Private Device
		A0330	Acquire Private Device
		A0340	Acquire Private Device
		A0350	Acquire Private Device
		A0360	Acquire Private Device
		A0370	Reset Hold Request
T0100	Standby		If Task Pointer = 1 then go to step S0100
T0200B	Fill		If Task Pointer = 2 then go to step S0200
T0300B	Heating		If Task Pointer = 3 then go to step S0300
S0100	Standby	A0010	Stopping and Disabling Conditional logic for DEV1
		A0020	Closing DEV2
		A0030	Stopping DEV3
		A0040	Stopping and Disabling Conditional Logic for DEV4
		A0050	Stopping and Disabling Conditional Logic for DEV5
		A0060	Stopping and Disabling Conditional Logic for DEV6
		A0070	Stopping and Disabling Conditional Logic for DEV7
		A0080	Closing DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Closing and Disabling Conditional Logic for DEV14
		A0150	Closing DEV15
		A0160	Closing and Disabling Conditional Logic for DEV16
		A0170	Closing DEV17
		A0180	Closing and Disabling Conditional Logic for DEV18
		A0190	Closing and Disabling Conditional Logic for DEV19
		A0200	Set DEV20 Mode to Remote Out
		A0210	Set DEV20 output to Closed and Disable Conditional Logic
		A0220	Set DEV21 Mode to Remote Out

Step	Description	Action	Action Description
		A0230	Set DEV21 output to Closed and Disable Conditional Logic
		A0240	Set DEV22 Mode to Remote Out
		A0250	Set DEV22 output to Closed and Disable Conditional Logic
		A0260	Set DEV23 Mode to Remote Out
		A0270	Set DEV23 output to Closed and Disable Conditional Logic
		A0280	Closing and Disabling Conditional Logic for DEV24
		A0290	Closing and Disabling Conditional Logic for DEV25
		A0300	Closing and Disabling Conditional Logic for DEV26
		A0310	Closing and Disabling Conditional Logic for DEV27
		A0320	Closing and Disabling Conditional Logic for DEV28
		A0330	Closing and Disabling Conditional Logic for DEV29
		A0340	Closing and Disabling Conditional Logic for DEV30
		A0350	Closing and Disabling Conditional Logic for DEV31
		A0360	Closing and Disabling Conditional Logic for DEV32
		A0370	Closing and Disabling Conditional Logic for DEV33
		A0380	Closing and Disabling Conditional Logic for DEV34
		A0390	Closing and Disabling Conditional Logic for DEV35
		A0400	Set DEV36 Mode to Remote Out
		A0410	Set DEV36 output to Closed and Disable Conditional Logic
T0200A	Fill		If EM_DEOD_FA_CKT1 is in step <Production> & No Vacuum alarm (VACUUM_ALM) then go to step S0200
S0200	Fill	A0010	Enabling Conditional logic for DEV1
		A0020	Opening DEV2
		A0030	Stopping DEV3
		A0040	Stopping and Disabling Conditional Logic for DEV4
		A0050	Stopping and Disabling Conditional Logic for DEV5
		A0060	Stopping and Disabling Conditional Logic for DEV6

Step	Description	Action	Action Description
		A0070	Stopping and Disabling Conditional Logic for DEV7
		A0080	Opening DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Opening DEV12
		A0130	Enabling Conditional Logic for DEV13
		A0140	Enabling Conditional Logic for DEV14
		A0150	Closing DEV15
		A0160	Closing and Disabling Conditional Logic for DEV16
		A0170	Closing DEV17
		A0180	Closing and Disabling Conditional Logic for DEV18
		A0190	Closing and Disabling Conditional Logic for DEV19
		A0200	Set DEV20 Mode to Remote Out
		A0210	Set DEV20 output to Closed and Disable Conditional Logic
		A0220	Set DEV21 Mode to Remote Cascade
		A0230	Setting DEV21 setpoint to (FT_702 Setpoint) and Enabling Conditional Logic
		A0240	Set DEV22 Mode to Remote Out
		A0250	Set DEV22 output to Closed and Disable Conditional Logic
		A0260	Set DEV23 Mode to Remote Out
		A0270	Set DEV23 output to Closed and Disable Conditional Logic
		A0280	Closing and Disabling Conditional Logic for DEV24
		A0290	Closing and Disabling Conditional Logic for DEV25
		A0300	Closing and Disabling Conditional Logic for DEV26
		A0310	Closing and Disabling Conditional Logic for DEV27
		A0320	Closing and Disabling Conditional Logic for DEV28
		A0330	Closing and Disabling Conditional Logic for DEV29
		A0340	Closing and Disabling Conditional Logic for DEV30
		A0350	Closing and Disabling Conditional Logic for DEV31

Step	Description	Action	Action Description
		A0360	Closing and Disabling Conditional Logic for DEV32
		A0370	Closing and Disabling Conditional Logic for DEV33
		A0380	Closing and Disabling Conditional Logic for DEV34
		A0390	Closing and Disabling Conditional Logic for DEV35
		A0400	Set DEV36 Mode to Remote Out
		A0410	Set DEV36 output to Open and Disable Conditional Logic
		A0420	Set Message: LSM_721A alarm is not active
		A0430	5 sec Delay Time (TP001)
T0300A	Heating		If 5 sec delay time (TP001) is over then go to step S0300
S0300	Heating	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Opening DEV2
		A0030	Stopping DEV3
		A0040	Stopping and Disabling Conditional Logic for DEV4
		A0050	Stopping and Disabling Conditional Logic for DEV5
		A0060	Stopping and Disabling Conditional Logic for DEV6
		A0070	Stopping and Disabling Conditional Logic for DEV7
		A0080	Opening DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Opening DEV12
		A0130	Enabling Conditional Logic for DEV13
		A0140	Enabling Conditional Logic for DEV14
		A0150	Closing DEV15
		A0160	Closing and Disabling Conditional Logic for DEV16
		A0170	Closing DEV17
		A0180	Closing and Disabling Conditional Logic for DEV18
		A0190	Closing and Disabling Conditional Logic for DEV19
		A0200	Set DEV20 Mode to Remote Out

Step	Description	Action	Action Description
		A0210	Set DEV20 output to Closed and Disable Conditional Logic
		A0220	Set DEV21 Mode to Remote Out
		A0230	Set DEV21 output to Closed and Disable Conditional Logic
		A0240	Set DEV22 Mode to Remote Out
		A0250	Set DEV22 output to Closed and Disable Conditional Logic
		A0260	Set DEV23 Mode to Remote Out
		A0270	Set DEV23 output to Closed and Disable Conditional Logic
		A0280	Closing and Disabling Conditional Logic for DEV24
		A0290	Closing and Disabling Conditional Logic for DEV25
		A0300	Closing and Disabling Conditional Logic for DEV26
		A0310	Closing and Disabling Conditional Logic for DEV27
		A0320	Closing and Disabling Conditional Logic for DEV28
		A0330	Closing and Disabling Conditional Logic for DEV29
		A0340	Closing and Disabling Conditional Logic for DEV30
		A0350	Closing and Disabling Conditional Logic for DEV31
		A0360	Closing and Disabling Conditional Logic for DEV32
		A0370	Closing and Disabling Conditional Logic for DEV33
		A0380	Closing and Disabling Conditional Logic for DEV34
		A0390	Closing and Disabling Conditional Logic for DEV35
		A0400	Set DEV36 Mode to Remote Out
		A0410	Set DEV36 output to Open and Disable Conditional Logic
		A0420	Set Message: Waiting for Deodorising Temperature Setpoint (TP011)
T9900	Complete		If Deodorising Temp. set point (TP011) reached then go to step S9900
S9900	Complete	A0010	Set Message: Heating Command Complete
END	Command Complete		True

Table 20: Heating Sequence

3.3.3 Command 02 – Production

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Production Command
T0100	Pre-Production		If Task Pointer = 3 then go to step S0100
T0200C	Confirm Production		If Task Pointer = 4 then go to step S0200
T0300B	Production		If Task Pointer = 5 then go to step S0300
T0400B	Confirm Hot Recirculation		If Task Pointer = 6 then go to step S0400
T0500B	Hot Recirculation		If Task Pointer = 7 then go to step S0500
T0600B	Confirm Cold Recirculation		If Task Pointer = 8 then go to step S0600
T0700B	Cold Recirculation		If Task Pointer = 9 then go to step S0700
S0100	Pre-Production	A0010	Enabling Conditional logic for DEV1
		A0020	Opening DEV2
		A0030	Stopping DEV3
		A0040	Enabling Conditional Logic for DEV4
		A0050	Enabling Conditional Logic for DEV5
		A0060	Enabling Conditional Logic for DEV6
		A0070	Enabling Conditional Logic for DEV7
		A0080	Opening DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Opening DEV12
		A0130	Enabling Conditional Logic for DEV13
		A0140	Enabling Conditional Logic for DEV14
		A0150	Opening DEV15
		A0160	Closing and Disabling Conditional Logic for DEV16
		A0170	Opening DEV17
		A0180	Closing and Disabling Conditional Logic for DEV18
		A0190	Closing and Disabling Conditional Logic for DEV19
		A0200	Set DEV20 Mode to Remote Cascade
		A0210	Setting DEV20 setpoint to (LT_780B Setpoint) and Enabling Conditional Logic
		A0220	Set DEV21 Mode to Remote Cascade

Step	Description	Action	Action Description
		A0230	Setting DEV21 setpoint to (FT_702 Setpoint) and Enabling Conditional Logic
		A0240	Set DEV22 Mode to Remote Cascade
		A0250	Setting DEV22 setpoint to (LT_722C Setpoint) and Enabling Conditional Logic
		A0260	Set DEV23 Mode to Remote Cascade
		A0270	Setting DEV23 setpoint to (TT_781B Setpoint) and Enabling Conditional Logic
		A0280	Closing and Disabling Conditional Logic for DEV24
		A0290	Closing and Disabling Conditional Logic for DEV25
		A0300	Closing and Disabling Conditional Logic for DEV26
		A0310	Closing and Disabling Conditional Logic for DEV27
		A0320	Closing and Disabling Conditional Logic for DEV28
		A0330	Closing and Disabling Conditional Logic for DEV29
		A0340	Closing and Disabling Conditional Logic for DEV30
		A0350	Closing and Disabling Conditional Logic for DEV31
		A0360	Closing and Disabling Conditional Logic for DEV32
		A0370	Closing and Disabling Conditional Logic for DEV33
		A0380	Closing and Disabling Conditional Logic for DEV34
		A0390	Closing and Disabling Conditional Logic for DEV35
		A0400	Set DEV36 Mode to Remote Out
		A0410	Set DEV36 output to Open and Disable Conditional Logic
T0200A	Confirm Production		If LSL_780B is not in alarm or LSH_780B is in alarm or LT_780B_VHL is in VHL alarm then go to step S0200
S0200	Confirm Production	A0010	Stopping and Disabling Conditional logic for DEV1
		A0020	Opening DEV2
		A0030	Stopping DEV3
		A0040	Stopping and Disabling Conditional Logic for DEV4
		A0050	Stopping and Disabling Conditional Logic for DEV5
		A0060	Stopping and Disabling Conditional Logic for DEV6

Step	Description	Action	Action Description
		A0070	Stopping and Disabling Conditional Logic for DEV7
		A0080	Opening DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Opening DEV12
		A0130	Enabling Conditional Logic for DEV13
		A0140	Enabling Conditional Logic for DEV14
		A0150	Opening DEV15
		A0160	Closing and Disabling Conditional Logic for DEV16
		A0170	Opening DEV17
		A0180	Closing and Disabling Conditional Logic for DEV18
		A0190	Closing and Disabling Conditional Logic for DEV19
		A0200	Set DEV20 Mode to Remote Out
		A0210	Set DEV20 output to Closed and Disable Conditional Logic
		A0220	Set DEV21 Mode to Remote Out
		A0230	Set DEV21 output to Closed and Disable Conditional Logic
		A0240	Set DEV22 Mode to Remote Out
		A0250	Set DEV22 output to Closed and Disable Conditional Logic
		A0260	Set DEV23 Mode to Remote Out
		A0270	Set DEV23 output to Closed and Disable Conditional Logic
		A0280	Closing and Disabling Conditional Logic for DEV24
		A0290	Closing and Disabling Conditional Logic for DEV25
		A0300	Closing and Disabling Conditional Logic for DEV26
		A0310	Closing and Disabling Conditional Logic for DEV27
		A0320	Closing and Disabling Conditional Logic for DEV28
		A0330	Closing and Disabling Conditional Logic for DEV29
		A0340	Closing and Disabling Conditional Logic for DEV30
		A0350	Closing and Disabling Conditional Logic for DEV31

Step	Description	Action	Action Description
		A0360	Closing and Disabling Conditional Logic for DEV32
		A0370	Closing and Disabling Conditional Logic for DEV33
		A0380	Closing and Disabling Conditional Logic for DEV34
		A0390	Closing and Disabling Conditional Logic for DEV35
		A0400	Set DEV36 Mode to Remote Cascade
		A0410	Setting DEV36 setpoint to (TT_722C Setpoint) and Enabling Conditional Logic
		A0420	Set Message: Confirm Production after closing/opening manual valves
		A0430	Reset Hot Recirculation (OP005) Parameter
T0300A	Production		If Operator Confirm <Production> (OP004) step then go to step S0300
S0300	Production	A0010	Enabling Conditional logic for DEV1
		A0020	Opening DEV2
		A0030	Stopping DEV3
		A0040	Enabling Conditional Logic for DEV4
		A0050	Enabling Conditional Logic for DEV5
		A0060	Enabling Conditional Logic for DEV6
		A0070	Enabling Conditional Logic for DEV7
		A0080	Opening DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Opening DEV12
		A0130	Enabling Conditional Logic for DEV13
		A0140	Enabling Conditional Logic for DEV14
		A0150	Opening DEV15
		A0160	Closing and Disabling Conditional Logic for DEV16
		A0170	Opening DEV17
		A0180	Closing and Disabling Conditional Logic for DEV18
		A0190	Closing and Disabling Conditional Logic for DEV19
		A0200	Set DEV20 Mode to Remote Cascade
		A0210	Setting DEV20 setpoint to (LT_780B Setpoint) and Enabling Conditional Logic
		A0220	Set DEV21 Mode to Remote Cascade

Step	Description	Action	Action Description
		A0230	Setting DEV21 setpoint to (FT_702 Setpoint) and Enabling Conditional Logic
		A0240	Set DEV22 Mode to Remote Cascade
		A0250	Setting DEV22 setpoint to (LT_722C Setpoint) and Enabling Conditional Logic
		A0260	Set DEV23 Mode to Remote Cascade
		A0270	Setting DEV23 setpoint to (TT_781B Setpoint) and Enabling Conditional Logic
		A0280	Closing and Disabling Conditional Logic for DEV24
		A0290	Closing and Disabling Conditional Logic for DEV25
		A0300	Closing and Disabling Conditional Logic for DEV26
		A0310	Closing and Disabling Conditional Logic for DEV27
		A0320	Closing and Disabling Conditional Logic for DEV28
		A0330	Closing and Disabling Conditional Logic for DEV29
		A0340	Closing and Disabling Conditional Logic for DEV30
		A0350	Closing and Disabling Conditional Logic for DEV31
		A0360	Closing and Disabling Conditional Logic for DEV32
		A0370	Closing and Disabling Conditional Logic for DEV33
		A0380	Closing and Disabling Conditional Logic for DEV34
		A0390	Closing and Disabling Conditional Logic for DEV35
		A0400	Set DEV36 Mode to Remote Cascade
		A0410	Setting DEV36 setpoint to (TT_722C Setpoint) and Enabling Conditional Logic
T0400A	Confirm Hot Recirculation		If Operator select <Confirm Hot Recirculation> (OP005) step then go to step S0400
T0600A	Confirm Cold Recirculation		If Startup is in step <Drain> then go to step S0600
S0400	Confirm Hot Recirculation	A0010	Stopping and Disabling Conditional logic for DEV1
		A0020	Opening DEV2
		A0030	Stopping DEV3
		A0040	Stopping and Disabling Conditional Logic for DEV4
		A0050	Stopping and Disabling Conditional Logic for DEV5

Step	Description	Action	Action Description
		A0060	Stopping and Disabling Conditional Logic for DEV6
		A0070	Stopping and Disabling Conditional Logic for DEV7
		A0080	Opening DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Opening DEV12
		A0130	Enabling Conditional Logic for DEV13
		A0140	Enabling Conditional Logic for DEV14
		A0150	Opening DEV15
		A0160	Closing and Disabling Conditional Logic for DEV16
		A0170	Opening DEV17
		A0180	Closing and Disabling Conditional Logic for DEV18
		A0190	Closing and Disabling Conditional Logic for DEV19
		A0200	Set DEV20 Mode to Remote Out
		A0210	Set DEV20 output to Closed and Disable Conditional Logic
		A0220	Set DEV21 Mode to Remote Out
		A0230	Set DEV21 output to Closed and Disable Conditional Logic
		A0240	Set DEV22 Mode to Remote Out
		A0250	Set DEV22 output to Closed and Disable Conditional Logic
		A0260	Set DEV23 Mode to Remote Out
		A0270	Set DEV23 output to Closed and Disable Conditional Logic
		A0280	Closing and Disabling Conditional Logic for DEV24
		A0290	Closing and Disabling Conditional Logic for DEV25
		A0300	Closing and Disabling Conditional Logic for DEV26
		A0310	Closing and Disabling Conditional Logic for DEV27
		A0320	Closing and Disabling Conditional Logic for DEV28
		A0330	Closing and Disabling Conditional Logic for DEV29
		A0340	Closing and Disabling Conditional Logic for DEV30

Step	Description	Action	Action Description
		A0350	Closing and Disabling Conditional Logic for DEV31
		A0360	Closing and Disabling Conditional Logic for DEV32
		A0370	Closing and Disabling Conditional Logic for DEV33
		A0380	Closing and Disabling Conditional Logic for DEV34
		A0390	Closing and Disabling Conditional Logic for DEV35
		A0400	Set DEV36 Mode to Remote Cascade
		A0410	Setting DEV36 setpoint to (TT_722C Setpoint) and Enabling Conditional Logic
		A0420	Set Message: Confirm Hot Recirculation after opening/closing manual valves
		A0430	Reset Production (OP004) Parameter
T0500A	Hot Recirculation		If Operator confirm <Hot Recirculation> (OP005) step then go to step S0500
S0500	Hot Recirculation (With Boiler On)	A0010	Enabling Conditional logic for DEV1
		A0020	Opening DEV2
		A0030	Stopping DEV3
		A0040	Enabling Conditional Logic for DEV4
		A0050	Enabling Conditional Logic for DEV5
		A0060	Enabling Conditional Logic for DEV6
		A0070	Enabling Conditional Logic for DEV7
		A0080	Opening DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Opening DEV12
		A0130	Enabling Conditional Logic for DEV13
		A0140	Enabling Conditional Logic for DEV14
		A0150	Opening DEV15
		A0160	Closing and Disabling Conditional Logic for DEV16
		A0170	Opening DEV17
		A0180	Closing and Disabling Conditional Logic for DEV18
		A0190	Closing and Disabling Conditional Logic for DEV19
		A0200	Set DEV20 Mode to Remote Cascade
		A0210	Setting DEV20 setpoint to (LT_780B Setpoint) and Enabling Conditional Logic

Step	Description	Action	Action Description
		A0220	Set DEV21 Mode to Remote Cascade
		A0230	Setting DEV21 setpoint to (FT_702 Setpoint) and Enabling Conditional Logic
		A0240	Set DEV22 Mode to Remote Cascade
		A0250	Setting DEV22 setpoint to (LT_722C Setpoint) and Enabling Conditional Logic
		A0260	Set DEV23 Mode to Remote Cascade
		A0270	Setting DEV23 setpoint to (TT_781B Setpoint) and Enabling Conditional Logic
		A0280	Closing and Disabling Conditional Logic for DEV24
		A0290	Closing and Disabling Conditional Logic for DEV25
		A0300	Closing and Disabling Conditional Logic for DEV26
		A0310	Closing and Disabling Conditional Logic for DEV27
		A0320	Closing and Disabling Conditional Logic for DEV28
		A0330	Closing and Disabling Conditional Logic for DEV29
		A0340	Closing and Disabling Conditional Logic for DEV30
		A0350	Closing and Disabling Conditional Logic for DEV31
		A0360	Closing and Disabling Conditional Logic for DEV32
		A0370	Closing and Disabling Conditional Logic for DEV33
		A0380	Closing and Disabling Conditional Logic for DEV34
		A0390	Closing and Disabling Conditional Logic for DEV35
		A0400	Set DEV36 Mode to Remote Cascade
		A0410	Setting DEV36 setpoint (TT_722C Setpoint) and Enabling Conditional Logic
		A0420	Set Preset Hot Recirculation time
		A0430	Start Preset Hot Recirculation Timer
		A0440	Preset Hot Recirculation time Message
T0200B	Confirm Production		If Preset Hot Recirculation time (TP013) is over then go to step S0200
S0600	Confirm Cold Recirculation	A0010	Stopping and Disabling Conditional logic for DEV1
		A0020	Opening DEV2
		A0030	Stopping DEV3
		A0040	Stopping and Disabling Conditional Logic for DEV4

Step	Description	Action	Action Description
		A0050	Stopping and Disabling Conditional Logic for DEV5
		A0060	Stopping and Disabling Conditional Logic for DEV6
		A0070	Stopping and Disabling Conditional Logic for DEV7
		A0080	Opening DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Opening DEV12
		A0130	Enabling Conditional Logic for DEV13
		A0140	Enabling Conditional Logic for DEV14
		A0150	Opening DEV15
		A0160	Closing and Disabling Conditional Logic for DEV16
		A0170	Opening DEV17
		A0180	Closing and Disabling Conditional Logic for DEV18
		A0190	Closing and Disabling Conditional Logic for DEV19
		A0200	Set DEV20 Mode to Remote Out
		A0210	Set DEV20 output to Closed and Disable Conditional Logic
		A0220	Set DEV21 Mode to Remote Out
		A0230	Set DEV21 output to Closed and Disable Conditional Logic
		A0240	Set DEV22 Mode to Remote Out
		A0250	Set DEV22 output to Closed and Disable Conditional Logic
		A0260	Set DEV23 Mode to Remote Out
		A0270	Set DEV23 output to Closed and Disable Conditional Logic
		A0280	Closing and Disabling Conditional Logic for DEV24
		A0290	Closing and Disabling Conditional Logic for DEV25
		A0300	Closing and Disabling Conditional Logic for DEV26
		A0310	Closing and Disabling Conditional Logic for DEV27
		A0320	Closing and Disabling Conditional Logic for DEV28
		A0330	Closing and Disabling Conditional Logic for DEV29

Step	Description	Action	Action Description
		A0340	Closing and Disabling Conditional Logic for DEV30
		A0350	Closing and Disabling Conditional Logic for DEV31
		A0360	Closing and Disabling Conditional Logic for DEV32
		A0370	Closing and Disabling Conditional Logic for DEV33
		A0380	Closing and Disabling Conditional Logic for DEV34
		A0390	Closing and Disabling Conditional Logic for DEV35
		A0400	Set DEV36 Mode to Remote Cascade
		A0410	Setting DEV36 setpoint to (TT_722C Setpoint) and Enabling Conditional Logic
		A0420	Set Message: Confirm Cold Recirculation after opening/closing manual valves
T0700A	Cold Recirculation		If Opeator confirm <Cold Recirculation> step (OP006) then go to step S0700
S0700	Cold Recirculation (With Boiler Off)	A0010	Enabling Conditional logic (TP02) for DEV1
		A0020	Opening DEV2
		A0030	Stopping DEV3
		A0040	Enabling Conditional Logic for DEV4
		A0050	Enabling Conditional Logic for DEV5
		A0060	Enabling Conditional Logic for DEV6
		A0070	Enabling Conditional Logic for DEV7
		A0080	Opening DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Opening DEV12
		A0130	Enabling Conditional Logic for DEV13
		A0140	Enabling Conditional Logic for DEV14
		A0150	Opening DEV15
		A0160	Closing and Disabling Conditional Logic for DEV16
		A0170	Opening DEV17
		A0180	Closing and Disabling Conditional Logic for DEV18
		A0190	Closing and Disabling Conditional Logic for DEV19
		A0200	Set DEV20 Mode to Remote Cascade

Step	Description	Action	Action Description
		A0210	Setting DEV20 setpoint to (LT_780B Setpoint) and Enabling Conditional Logic
		A0220	Set DEV21 Mode to Remote Cascade
		A0230	Setting DEV21 setpoint to (FT_702 Setpoint) and Enabling Conditional Logic
		A0240	Set DEV22 Mode to Remote Cascade
		A0250	Setting DEV22 setpoint to (LT_722C Setpoint) and Enabling Conditional Logic
		A0260	Set DEV23 Mode to Remote Cascade
		A0270	Setting DEV23 setpoint (TT_781B Setpoint) and Enabling Conditional Logic
		A0280	Closing and Disabling Conditional Logic for DEV24
		A0290	Closing and Disabling Conditional Logic for DEV25
		A0300	Closing and Disabling Conditional Logic for DEV26
		A0310	Closing and Disabling Conditional Logic for DEV27
		A0320	Closing and Disabling Conditional Logic for DEV28
		A0330	Closing and Disabling Conditional Logic for DEV29
		A0340	Closing and Disabling Conditional Logic for DEV30
		A0350	Closing and Disabling Conditional Logic for DEV31
		A0360	Closing and Disabling Conditional Logic for DEV32
		A0370	Closing and Disabling Conditional Logic for DEV33
		A0380	Closing and Disabling Conditional Logic for DEV34
		A0390	Closing and Disabling Conditional Logic for DEV35
		A0400	Set DEV36 Mode to Remote Cascade
		A0410	Setting DEV36 setpoint (TT_722C Setpoint) and Enabling Conditional Logic
		A0420	Set Message: Starting Cold Recirculation Operation
T9900	Complete		If TT_721A2 < the Set Cold Recirculation temp (TP014) and Operator confirm Proceed to <Drain> (OP007) then go to step S9900
S9900	Complete	A0010	Set Message: Production Command Complete
		A0020	Reset Cold Recirculation (OP006) Parameter
END	Command Complete		True

Table 21: Production Sequence**3.3.4 Command 03 – Recirculation**

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Recirculation Command
T0010	Drain 802		If Task Pointer = 7 then go to step S0010
T0100B	Blow 802		If Task Pointer = 8 then go to step S0100
T0200B	Drain 821A1 & Blow Coil		If Task Pointer = 9 then go to step S0200
T0300B	Drain 821A2		If Task Pointer = 10 then go to step S0300
T0400B	Drain 822C		If Task Pointer = 11 then go to step S0400
T0500B	Drain 822A1 & Blow 822C		If Task Pointer = 12 then go to step S0500
T0600B	Drain 822A2		If Task Pointer = 13 then go to step S0600
T0700B	Drain 822A3		If Task Pointer = 14 then go to step S0700
T0800B	Drain 822A4		If Task Pointer = 15 then go to step S0800
T0900B	Drain 822A5		If Task Pointer = 16 then go to step S0900
T1000B	Drain 822A6		If Task Pointer = 17 then go to step S1000
T1100B	Drain 880A1		If Task Pointer = 18 then go to step S1100
T1200B	Drain 880A2		If Task Pointer = 19 then go to step S1200
T1300B	Drain 880B		If Task Pointer = 20 then go to step S1300
T1400B	Blow 880B		If Task Pointer = 21 then go to step S1400
T1500B	Blow to Storage		If Task Pointer = 22 then go to step S1500
		A0010	Enabling Conditional logic for DEV1
		A0020	Opening DEV2
		A0030	Stopping DEV3
		A0040	Enabling Conditional Logic for DEV4
		A0050	Enabling Conditional Logic for DEV5
		A0060	Enabling Conditional Logic for DEV6
		A0070	Enabling Conditional Logic for DEV7
		A0080	Opening DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Opening DEV12
		A0130	Enabling Conditional Logic for DEV13
		A0140	Enabling Conditional Logic for DEV14
		A0150	Opening DEV15
		A0160	Closing and Disabling Conditional Logic for DEV16

Step	Description	Action	Action Description
S0010	Drain 802	A0170	Opening DEV17
		A0180	Closing and Disabling Conditional Logic for DEV18
		A0190	Closing and Disabling Conditional Logic for DEV19
		A0200	Set DEV20 Mode to Remote Cascade
		A0210	Setting DEV20 setpoint (LT_780B Setpoint) and Enabling Conditional Logic
		A0220	Set DEV21 Mode to Remote Cascade
		A0230	Setting DEV21 setpoint to (FT_702 Setpoint) and Enabling Conditional Logic
		A0240	Set DEV22 Mode to Remote Cascade
		A0250	Setting DEV22 setpoint (LT_722C Setpoint) and Enabling Conditional Logic
		A0260	Set DEV23 Mode to Remote Cascade
		A0270	Setting DEV23 setpoint (TT_781B Setpoint) and Enabling Conditional Logic
		A0280	Closing and Disabling Conditional Logic for DEV24
		A0290	Closing and Disabling Conditional Logic for DEV25
		A0300	Closing and Disabling Conditional Logic for DEV26
		A0310	Closing and Disabling Conditional Logic for DEV27
		A0320	Closing and Disabling Conditional Logic for DEV28
		A0330	Closing and Disabling Conditional Logic for DEV29
		A0340	Closing and Disabling Conditional Logic for DEV30
		A0350	Closing and Disabling Conditional Logic for DEV31
		A0360	Closing and Disabling Conditional Logic for DEV32
		A0370	Closing and Disabling Conditional Logic for DEV33
		A0380	Closing and Disabling Conditional Logic for DEV34
		A0390	Closing and Disabling Conditional Logic for DEV35
		A0400	Set DEV36 Mode to Remote Out
		A0410	Set DEV36 output to Open and Disable Conditional Logic
		A0430	Reset Cold Recirculation Button (OP006)
T0100A	Blow 802		If EM_DEOD_FA_CKT1 is in step <Drain Wait> then go to step S0100

Step	Description	Action	Action Description
S0100	Blow 802	A0010	Stopping and Disabling Conditional logic for DEV1
		A0020	Closing DEV2
		A0030	Stopping DEV3
		A0040	Enabling Conditional Logic for DEV4
		A0050	Enabling Conditional Logic for DEV5
		A0060	Enabling Conditional Logic for DEV6
		A0070	Enabling Conditional Logic for DEV7
		A0080	Opening DEV8
		A0090	Enabling Conditional Logic for DEV9
		A0100	Closing DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Opening DEV12
		A0130	Enabling Conditional Logic for DEV13
		A0140	Enabling Conditional Logic for DEV14
		A0150	Opening DEV15
		A0160	Closing and Disabling Conditional Logic for DEV16
		A0170	Opening DEV17
		A0180	Closing and Disabling Conditional Logic for DEV18
		A0190	Closing and Disabling Conditional Logic for DEV19
		A0200	Set DEV20 Mode to Remote Cascade
		A0210	Setting DEV20 setpoint (LT_780B Setpoint) and Enabling Conditional Logic
		A0220	Set DEV21 Mode to Remote Out and Disabling Conditional Logic
		A0230	Set DEV21 output to Open
		A0240	Set DEV22 Mode to Remote Cascade
		A0250	Setting DEV22 setpoint (LT_722C Setpoint) and Enabling Conditional Logic
		A0260	Set DEV23 Mode to Remote Cascade
		A0270	Setting DEV23 setpoint (TT1781B Setpoint) and Enabling Conditional Logic
		A0280	Closing and Disabling Conditional Logic for DEV24
		A0290	Closing and Disabling Conditional Logic for DEV25
		A0300	Closing and Disabling Conditional Logic for DEV26
		A0310	Closing and Disabling Conditional Logic for DEV27
		A0320	Closing and Disabling Conditional Logic for DEV28

Step	Description	Action	Action Description
		A0330	Closing and Disabling Conditional Logic for DEV29
		A0340	Closing and Disabling Conditional Logic for DEV30
		A0350	Closing and Disabling Conditional Logic for DEV31
		A0360	Closing and Disabling Conditional Logic for DEV32
		A0370	Closing and Disabling Conditional Logic for DEV33
		A0380	Closing and Disabling Conditional Logic for DEV34
		A0390	Closing and Disabling Conditional Logic for DEV35
		A0400	Set DEV36 Mode to Remote Out
		A0410	Set DEV36 output to Open and Disable Conditional Logic
		A0420	Set Message: Setting Preset blow time
		A0430	Start Preset Blow Timer
		A0440	Preset Blow time Message
T0200A	Drain 821A1 & Blow Coil		If Preset (DEV9) Blow time (TP006) is over then go to step S0200
		A0010	Stopping and Disabling Conditional logic for DEV1
		A0020	Closing DEV2
		A0030	Stopping DEV3
		A0040	Enabling Conditional Logic for DEV4
		A0050	Enabling Conditional Logic for DEV5
		A0060	Enabling Conditional Logic for DEV6
		A0070	Enabling Conditional Logic for DEV7
		A0080	Opening DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Opening DEV10
		A0110	Enabling Conditional Logic for DEV11
		A0120	Closing DEV12
		A0130	Enabling Conditional Logic for DEV13
		A0140	Enabling Conditional Logic for DEV14
		A0150	Opening DEV15
		A0160	Closing and Disabling Conditional Logic for DEV16
		A0170	Opening DEV17
		A0180	Closing and Disabling Conditional Logic for DEV18

Step	Description	Action	Action Description
S0200	Drain 821A1 & Blow Coil	A0190	Closing and Disabling Conditional Logic for DEV19
		A0200	Set DEV20 Mode to Remote Cascade
		A0210	Setting DEV20 setpoint (LT_780B Setpoint) and Enabling Conditional Logic
		A0220	Set DEV21 Mode to Remote Out
		A0230	Set DEV21 output to Closed and Disable Conditional Logic
		A0240	Set DEV22 Mode to Remote Cascade
		A0250	Setting DEV22 setpoint (LT_722C Setpoint) and Enabling Conditional Logic
		A0260	Set DEV23 Mode to Remote Cascade
		A0270	Setting DEV23 setpoint (TT_781B Setpoint) and Enabling Conditional Logic
		A0280	Enabling Conditional Logic for DEV24
		A0290	Closing and Disabling Conditional Logic for DEV25
		A0300	Closing and Disabling Conditional Logic for DEV26
		A0310	Closing and Disabling Conditional Logic for DEV27
		A0320	Closing and Disabling Conditional Logic for DEV28
		A0330	Closing and Disabling Conditional Logic for DEV29
		A0340	Closing and Disabling Conditional Logic for DEV30
		A0350	Closing and Disabling Conditional Logic for DEV31
		A0360	Closing and Disabling Conditional Logic for DEV32
		A0370	Closing and Disabling Conditional Logic for DEV33
		A0380	Closing and Disabling Conditional Logic for DEV34
		A0390	Closing and Disabling Conditional Logic for DEV35
		A0400	Set DEV36 Mode to Remote Out
		A0410	Set DEV36 output to Open and Disable Conditional Logic
		A0420	Set Preset Drain time and Blow Time
		A0430	Start Preset Drain and Blow Timer
		A0440	Preset Drain and Blow time Message
T0300A	Drain 821A2		If Preset Drain time (TP016) over and DEV11 Blow time over (TP018) then go to step S0300

Step	Description	Action	Action Description
S0300	Drain 821A2	A0010	Stopping and Disabling Conditional logic for DEV1
		A0020	Closing DEV2
		A0030	Stopping DEV3
		A0040	Enabling Conditional Logic for DEV4
		A0050	Enabling Conditional Logic for DEV5
		A0060	Enabling Conditional Logic for DEV6
		A0070	Enabling Conditional Logic for DEV7
		A0080	Closing DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Closing and Disabling Conditional Logic for DEV14
		A0150	Opening DEV15
		A0160	Closing and Disabling Conditional Logic for DEV16
		A0170	Opening DEV17
		A0180	Closing and Disabling Conditional Logic for DEV18
		A0190	Closing and Disabling Conditional Logic for DEV19
		A0200	Set DEV20 Mode to Remote Cascade
		A0210	Setting DEV20 setpoint (LT_780B Setpoint) and Enabling Conditional Logic
		A0220	Set DEV21 Mode to Remote Out
		A0230	Set DEV21 output to Closed and Disable Conditional Logic
		A0240	Set DEV22 Mode to Remote Cascade
		A0250	Setting DEV22 setpoint (LT_722C Setpoint) and Enabling Conditional Logic
		A0260	Set DEV23 Mode to Remote Cascade
		A0270	Setting DEV23 setpoint (TT_781B Setpoint) and Enabling Conditional Logic
		A0280	Opening and Disabling Conditional Logic for DEV24
		A0290	Enabling Conditional Logic for DEV25
		A0300	Closing and Disabling Conditional Logic for DEV26
		A0310	Closing and Disabling Conditional Logic for DEV27

Step	Description	Action	Action Description
		A0320	Closing and Disabling Conditional Logic for DEV28
		A0330	Closing and Disabling Conditional Logic for DEV29
		A0340	Closing and Disabling Conditional Logic for DEV30
		A0350	Closing and Disabling Conditional Logic for DEV31
		A0360	Closing and Disabling Conditional Logic for DEV32
		A0370	Closing and Disabling Conditional Logic for DEV33
		A0380	Closing and Disabling Conditional Logic for DEV34
		A0390	Closing and Disabling Conditional Logic for DEV35
		A0400	Set DEV36 Mode to Remote Out
		A0410	Set DEV36 output to Closed and Disable Conditional Logic
		A0420	Set Preset Drain time
		A0430	Start Preset Drain Timer
		A0440	Preset Drain time Message
T0400A	Drain 822C		If Preset Drain time (TP016) is over then go to step S0400
		A0010	Stopping and Disabling Conditional logic for DEV1
		A0020	Closing DEV2
		A0030	Stopping DEV3
		A0040	Enabling Conditional Logic for DEV4
		A0050	Enabling Conditional Logic for DEV5
		A0060	Enabling Conditional Logic for DEV6
		A0070	Enabling Conditional Logic for DEV7
		A0080	Closing DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Closing and Disabling Conditional Logic for DEV14
		A0150	Opening DEV15
		A0160	Closing and Disabling Conditional Logic for DEV16

Step	Description	Action	Action Description
S0400	Drain 822C	A0170	Opening DEV17
		A0180	Closing and Disabling Conditional Logic for DEV18
		A0190	Closing and Disabling Conditional Logic for DEV19
		A0200	Set DEV20 Mode to Remote Cascade
		A0210	Setting DEV20 setpoint (LT_780B Setpoint) and Enabling Conditional Logic
		A0220	Set DEV21 Mode to Remote Out
		A0230	Set DEV21 output to Closed and Disable Conditional Logic
		A0240	Set DEV22 Mode to Remote Out
		A0250	Set DEV22 output to Open and Enable Conditional logic
		A0260	Set DEV23 Mode to Remote Cascade
		A0270	Setting DEV23 setpoint (TT_781B Setpoint) and Enabling Conditional Logic
		A0280	Opening and Disabling Conditional Logic for DEV24
		A0290	Opening and Disabling Conditional Logic for DEV25
		A0300	Closing and Disabling Conditional Logic for DEV26
		A0310	Closing and Disabling Conditional Logic for DEV27
		A0320	Closing and Disabling Conditional Logic for DEV28
		A0330	Closing and Disabling Conditional Logic for DEV29
		A0340	Closing and Disabling Conditional Logic for DEV30
		A0350	Closing and Disabling Conditional Logic for DEV31
		A0360	Closing and Disabling Conditional Logic for DEV32
		A0370	Closing and Disabling Conditional Logic for DEV33
		A0380	Closing and Disabling Conditional Logic for DEV34
		A0390	Closing and Disabling Conditional Logic for DEV35
		A0400	Set DEV36 Mode to Remote Out
		A0410	Set DEV36 output to Closed and Disable Conditional Logic
T0500A	Drain 822A1 & Blow 822C		If LSL_722C is in alarm and LT_722C_VLL is in VLL alarm then go to step S0500

Step	Description	Action	Action Description
S0500	Drain 822A1 & Blow 822C	A0010	Stopping and Disabling Conditional logic for DEV1
		A0020	Closing DEV2
		A0030	Stopping DEV3
		A0040	Stopping and Disabling Conditional Logic for DEV4
		A0050	Stopping and Disabling Conditional Logic for DEV5
		A0060	Enabling Conditional Logic for DEV6
		A0070	Enabling Conditional Logic for DEV7
		A0080	Closing DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Closing and Disabling Conditional Logic for DEV14
		A0150	Closing DEV15
		A0160	Enabling Conditional Logic for DEV16
		A0170	Opening DEV17
		A0180	Closing and Disabling Conditional Logic for DEV18
		A0190	Closing and Disabling Conditional Logic for DEV19
		A0200	Set DEV20 Mode to Remote Cascade
		A0210	Setting DEV20 setpoint (LT_780B Setpoint) and Enabling Conditional Logic
		A0220	Set DEV21 Mode to Remote Out
		A0230	Set DEV21 output to Closed and Disable Conditional Logic
		A0240	Set DEV22 Mode to Remote Out
		A0250	Set DEV22 output to Open and Disable Conditional logic
		A0260	Set DEV23 Mode to Remote Cascade
		A0270	Setting DEV23 setpoint (TT_781B Setpoint) and Enabling Conditional Logic
		A0280	Opening and Disabling Conditional Logic for DEV24
		A0290	Opening and Disabling Conditional Logic for DEV25
		A0300	Enabling Conditional Logic for DEV26
		A0310	Closing and Disabling Conditional Logic for DEV27

Step	Description	Action	Action Description
		A0320	Closing and Disabling Conditional Logic for DEV28
		A0330	Closing and Disabling Conditional Logic for DEV29
		A0340	Closing and Disabling Conditional Logic for DEV30
		A0350	Closing and Disabling Conditional Logic for DEV31
		A0360	Closing and Disabling Conditional Logic for DEV32
		A0370	Closing and Disabling Conditional Logic for DEV33
		A0380	Closing and Disabling Conditional Logic for DEV34
		A0390	Closing and Disabling Conditional Logic for DEV35
		A0400	Set DEV36 Mode to Remote Out
		A0410	Set DEV36 output to Closed and Disable Conditional Logic
		A0420	Set Preset Drain time and Blow Time
		A0430	Start Preset Drain and Blow Timer
		A0440	Preset Drain and Blow time Message
T0600A	Drain 822A2		If Preset Drain time (TP016) is over and (DEV16) Blow time (TP007) is over then go to step S0600
S0600	Drain 822A2	A0010	Stopping and Disabling Conditional logic for DEV1
		A0020	Closing DEV2
		A0030	Stopping DEV3
		A0040	Stopping and Disabling Conditional Logic for DEV4
		A0050	Stopping and Disabling Conditional Logic for DEV5
		A0060	Enabling Conditional Logic for DEV6
		A0070	Enabling Conditional Logic for DEV7
		A0080	Closing DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Closing and Disabling Conditional Logic for DEV14

Step	Description	Action	Action Description
		A0150	Closing DEV15
		A0160	Closing and Disabling Conditional Logic for DEV16
		A0170	Opening DEV17
		A0180	Closing and Disabling Conditional Logic for DEV18
		A0190	Closing and Disabling Conditional Logic for DEV19
		A0200	Set DEV20 Mode to Remote Cascade
		A0210	Setting DEV20 setpoint (LT_780B Setpoint) and Enabling Conditional Logic
		A0220	Set DEV21 Mode to Remote Out
		A0230	Set DEV21 output to Closed and Disable Conditional Logic
		A0240	Set DEV22 Mode to Remote Out
		A0250	Set DEV22 output to Closed and Disable Conditional Logic
		A0260	Set DEV23 Mode to Remote Cascade
		A0270	Setting DEV23 setpoint (TT_781B Setpoint) and Enabling Conditional Logic
		A0280	Opening and Disabling Conditional Logic for DEV24
		A0290	Opening and Disabling Conditional Logic for DEV25
		A0300	Opening and Disabling Conditional Logic for DEV26
		A0310	Enabling Conditional Logic for DEV27
		A0320	Closing and Disabling Conditional Logic for DEV28
		A0330	Closing and Disabling Conditional Logic for DEV29
		A0340	Closing and Disabling Conditional Logic for DEV30
		A0350	Closing and Disabling Conditional Logic for DEV31
		A0360	Closing and Disabling Conditional Logic for DEV32
		A0370	Closing and Disabling Conditional Logic for DEV33
		A0380	Closing and Disabling Conditional Logic for DEV34
		A0390	Closing and Disabling Conditional Logic for DEV35
		A0400	Set DEV36 Mode to Remote Out
		A0410	Set DEV36 output to Closed and Disable Conditional Logic
		A0420	Set Preset Drain time
		A0430	Start Preset Drain Timer

Step	Description	Action	Action Description
		A0440	Preset Drain time Message
T0700A	Drain 822A3		If Preset Drain time (TP016) is over then go to step S0700
S0700	Drain 822A3	A0010	Stopping and Disabling Conditional logic for DEV1
		A0020	Closing DEV2
		A0030	Stopping DEV3
		A0040	Stopping and Disabling Conditional Logic for DEV4
		A0050	Stopping and Disabling Conditional Logic for DEV5
		A0060	Enabling Conditional Logic for DEV6
		A0070	Enabling Conditional Logic for DEV7
		A0080	Closing DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Closing and Disabling Conditional Logic for DEV14
		A0150	Closing DEV15
		A0160	Closing and Disabling Conditional Logic for DEV16
		A0170	Opening DEV17
		A0180	Closing and Disabling Conditional Logic for DEV18
		A0190	Closing and Disabling Conditional Logic for DEV19
		A0200	Set DEV20 Mode to Remote Cascade
		A0210	Setting DEV20 setpoint (LT_780B Setpoint) and Enabling Conditional Logic
		A0220	Set DEV21 Mode to Remote Out
		A0230	Set DEV21 output to Closed and Disable Conditional Logic
		A0240	Set DEV22 Mode to Remote Out
		A0250	Set DEV22 output to Closed and Disable Conditional Logic
		A0260	Set DEV23 Mode to Remote Cascade
		A0270	Setting DEV23 setpoint (TT_781B Setpoint) and Enabling Conditional Logic
		A0280	Opening and Disabling Conditional Logic for DEV24

Step	Description	Action	Action Description
		A0290	Opening and Disabling Conditional Logic for DEV25
		A0300	Opening and Disabling Conditional Logic for DEV26
		A0310	Opening and Disabling Conditional Logic for DEV27
		A0320	Enabling Conditional Logic for DEV28
		A0330	Closing and Disabling Conditional Logic for DEV29
		A0340	Closing and Disabling Conditional Logic for DEV30
		A0350	Closing and Disabling Conditional Logic for DEV31
		A0360	Closing and Disabling Conditional Logic for DEV32
		A0370	Closing and Disabling Conditional Logic for DEV33
		A0380	Closing and Disabling Conditional Logic for DEV34
		A0390	Closing and Disabling Conditional Logic for DEV35
		A0400	Set DEV36 Mode to Remote Out
		A0410	Set DEV36 output to Closed and Disable Conditional Logic
		A0420	Set Preset Drain time
		A0430	Start Preset Drain Timer
		A0440	Preset Drain time Message
T0800A	Drain 822A4		If Preset Drain time (TP016) is over then go to step S0800
S0800	Drain 822A4	A0010	Stopping and Disabling Conditional logic for DEV1
		A0020	Closing DEV2
		A0030	Stopping DEV3
		A0040	Stopping and Disabling Conditional Logic for DEV4
		A0050	Stopping and Disabling Conditional Logic for DEV5
		A0060	Enabling Conditional Logic for DEV6
		A0070	Enabling Conditional Logic for DEV7
		A0080	Closing DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing DEV12

Step	Description	Action	Action Description
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Closing and Disabling Conditional Logic for DEV14
		A0150	Closing DEV15
		A0160	Closing and Disabling Conditional Logic for DEV16
		A0170	Opening DEV17
		A0180	Closing and Disabling Conditional Logic for DEV18
		A0190	Closing and Disabling Conditional Logic for DEV19
		A0200	Set DEV20 Mode to Remote Cascade
		A0210	Setting DEV20 setpoint (LT_780B Setpoint) and Enabling Conditional Logic
		A0220	Set DEV21 Mode to Remote Out
		A0230	Set DEV21 output to Closed and Disable Conditional Logic
		A0240	Set DEV22 Mode to Remote Out
		A0250	Set DEV22 output to Closed and Disable Conditional Logic
		A0260	Set DEV23 Mode to Remote Cascade
		A0270	Setting DEV23 setpoint (TT_781B Setpoint) and Enabling Conditional Logic
		A0280	Opening and Disabling Conditional Logic for DEV24
		A0290	Opening and Disabling Conditional Logic for DEV25
		A0300	Opening and Disabling Conditional Logic for DEV26
		A0310	Opening and Disabling Conditional Logic for DEV27
		A0320	Opening and Disabling Conditional Logic for DEV28
		A0330	Enabling Conditional Logic for DEV29
		A0340	Closing and Disabling Conditional Logic for DEV30
		A0350	Closing and Disabling Conditional Logic for DEV31
		A0360	Closing and Disabling Conditional Logic for DEV32
		A0370	Closing and Disabling Conditional Logic for DEV33
		A0380	Closing and Disabling Conditional Logic for DEV34
		A0390	Closing and Disabling Conditional Logic for DEV35
		A0400	Set DEV36 Mode to Remote Out

Step	Description	Action	Action Description
		A0410	Set DEV36 output to Closed and Disable Conditional Logic
		A0420	Set Preset Drain time
		A0430	Start Preset Drain Timer
		A0440	Preset Drain time Message
T0900A	Drain 822A5		If Preset Drain time (TP016) is over then go to step S0900
S0900	Drain 822A5	A0010	Stopping and Disabling Conditional logic for DEV1
		A0020	Closing DEV2
		A0030	Stopping DEV3
		A0040	Stopping and Disabling Conditional Logic for DEV4
		A0050	Stopping and Disabling Conditional Logic for DEV5
		A0060	Enabling Conditional Logic for DEV6
		A0070	Enabling Conditional Logic for DEV7
		A0080	Closing DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Closing and Disabling Conditional Logic for DEV14
		A0150	Closing DEV15
		A0160	Closing and Disabling Conditional Logic for DEV16
		A0170	Opening DEV17
		A0180	Closing and Disabling Conditional Logic for DEV18
		A0190	Closing and Disabling Conditional Logic for DEV19
		A0200	Set DEV20 Mode to Remote Cascade
		A0210	Setting DEV20 setpoint (LT_780B Setpoint) and Enabling Conditional Logic
		A0220	Set DEV21 Mode to Remote Out
		A0230	Set DEV21 output to Closed and Disable Conditional Logic
		A0240	Set DEV22 Mode to Remote Out
		A0250	Set DEV22 output to Closed and Disable Conditional Logic

Step	Description	Action	Action Description
		A0260	Set DEV23 Mode to Remote Cascade
		A0270	Setting DEV23 setpoint (TT1781B Setpoint) and Enabling Conditional Logic
		A0280	Opening and Disabling Conditional Logic for DEV24
		A0290	Opening and Disabling Conditional Logic for DEV25
		A0300	Opening and Disabling Conditional Logic for DEV26
		A0310	Opening and Disabling Conditional Logic for DEV27
		A0320	Opening and Disabling Conditional Logic for DEV28
		A0330	Opening and Disabling Conditional Logic for DEV29
		A0340	Enabling Conditional Logic for DEV30
		A0350	Closing and Disabling Conditional Logic for DEV31
		A0360	Closing and Disabling Conditional Logic for DEV32
		A0370	Closing and Disabling Conditional Logic for DEV33
		A0380	Closing and Disabling Conditional Logic for DEV34
		A0390	Closing and Disabling Conditional Logic for DEV35
		A0400	Set DEV36 Mode to Remote Out
		A0410	Set DEV36 output to Closed and Disable Conditional Logic
		A0420	Set Preset Drain time
		A0430	Start Preset Drain Timer
		A0440	Preset Drain time Message
T1000A	Drain 822A6		If Preset Drain time (TP016) is over then go to step S1000
S1000	Drain 822A6	A0010	Stopping and Disabling Conditional logic for DEV1
		A0020	Closing DEV2
		A0030	Stopping DEV3
		A0040	Stopping and Disabling Conditional Logic for DEV4
		A0050	Stopping and Disabling Conditional Logic for DEV5
		A0060	Enabling Conditional Logic for DEV6
		A0070	Enabling Conditional Logic for DEV7
		A0080	Closing DEV8

Step	Description	Action	Action Description
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Closing and Disabling Conditional Logic for DEV14
		A0150	Closing DEV15
		A0160	Closing and Disabling Conditional Logic for DEV16
		A0170	Opening DEV17
		A0180	Closing and Disabling Conditional Logic for DEV18
		A0190	Closing and Disabling Conditional Logic for DEV19
		A0200	Set DEV20 Mode to Remote Cascade
		A0210	Setting DEV20 setpoint (LT_780B Setpoint) and Enabling Conditional Logic
		A0220	Set DEV21 Mode to Remote Out
		A0230	Set DEV21 output to Closed and Disable Conditional Logic
		A0240	Set DEV22 Mode to Remote Out
		A0250	Set DEV22 output to Closed and Disable Conditional Logic
		A0260	Set DEV23 Mode to Remote Cascade
		A0270	Setting DEV23 setpoint (TT_781B Setpoint) and Enabling Conditional Logic
		A0280	Opening and Disabling Conditional Logic for DEV24
		A0290	Opening and Disabling Conditional Logic for DEV25
		A0300	Opening and Disabling Conditional Logic for DEV26
		A0310	Opening and Disabling Conditional Logic for DEV27
		A0320	Opening and Disabling Conditional Logic for DEV28
		A0330	Opening and Disabling Conditional Logic for DEV29
		A0340	Opening and Disabling Conditional Logic for DEV30
		A0350	Enabling Conditional Logic for DEV31
		A0360	Closing and Disabling Conditional Logic for DEV32

Step	Description	Action	Action Description
		A0370	Closing and Disabling Conditional Logic for DEV33
		A0380	Closing and Disabling Conditional Logic for DEV34
		A0390	Closing and Disabling Conditional Logic for DEV35
		A0400	Set DEV36 Mode to Remote Out
		A0410	Set DEV36 output to Closed and Disable Conditional Logic
		A0420	Set Preset Drain time
		A0430	Start Preset Drain Timer
		A0440	Preset Drain time Message
T1100A	Drain 880A1		If Preset Drain time (TP016) is over then go to step S1100
S1100	Drain 880A1	A0010	Stopping and Disabling Conditional logic for DEV1
		A0020	Closing DEV2
		A0030	Stopping DEV3
		A0040	Stopping and Disabling Conditional Logic for DEV4
		A0050	Stopping and Disabling Conditional Logic for DEV5
		A0060	Enabling Conditional Logic for DEV6
		A0070	Enabling Conditional Logic for DEV7
		A0080	Closing DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Closing and Disabling Conditional Logic for DEV14
		A0150	Closing DEV15
		A0160	Closing and Disabling Conditional Logic for DEV16
		A0170	Opening DEV17
		A0180	Closing and Disabling Conditional Logic for DEV18
		A0190	Closing and Disabling Conditional Logic for DEV19
		A0200	Set DEV20 Mode to Remote Cascade
		A0210	Setting DEV20 setpoint (LT_780B Setpoint) and Enabling Conditional Logic

Step	Description	Action	Action Description
		A0220	Set DEV21 Mode to Remote Out
		A0230	Set DEV21 output to Closed and Disable Conditional Logic
		A0240	Set DEV22 Mode to Remote Out
		A0250	Set DEV22 output to Closed and Disable Conditional Logic
		A0260	Set DEV23 Mode to Remote Cascade
		A0270	Setting DEV23 setpoint (TT_781B Setpoint) and Enabling Conditional Logic
		A0280	Opening and Disabling Conditional Logic for DEV24
		A0290	Opening and Disabling Conditional Logic for DEV25
		A0300	Opening and Disabling Conditional Logic for DEV26
		A0310	Opening and Disabling Conditional Logic for DEV27
		A0320	Opening and Disabling Conditional Logic for DEV28
		A0330	Opening and Disabling Conditional Logic for DEV29
		A0340	Opening and Disabling Conditional Logic for DEV30
		A0350	Opening and Disabling Conditional Logic for DEV31
		A0360	Enabling Conditional Logic for DEV32
		A0370	Enabling Conditional Logic for DEV33
		A0380	Closing and Disabling Conditional Logic for DEV34
		A0390	Closing and Disabling Conditional Logic for DEV35
		A0400	Set DEV36 Mode to Remote Out
		A0410	Set DEV36 output to Closed and Disable Conditional Logic
		A0420	Set Preset Drain time
		A0430	Start Preset Drain Timer
		A0440	Preset Drain time Message
T1200A	Drain 880A2		If Preset Drain time (TP016) is over then go to step S1200
S1200	Drain 880A2	A0010	Stopping and Disabling Conditional logic for DEV1
		A0020	Closing DEV2
		A0030	Stopping DEV3
		A0040	Stopping and Disabling Conditional Logic for DEV4

Step	Description	Action	Action Description
		A0050	Stopping and Disabling Conditional Logic for DEV5
		A0060	Enabling Conditional Logic for DEV6
		A0070	Enabling Conditional Logic for DEV7
		A0080	Closing DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Closing and Disabling Conditional Logic for DEV14
		A0150	Closing DEV15
		A0160	Closing and Disabling Conditional Logic for DEV16
		A0170	Opening DEV17
		A0180	Closing and Disabling Conditional Logic for DEV18
		A0190	Closing and Disabling Conditional Logic for DEV19
		A0200	Set DEV20 Mode to Remote Cascade
		A0210	Setting DEV20 setpoint (LT_780B Setpoint) and Enabling Conditional Logic
		A0220	Set DEV21 Mode to Remote Out
		A0230	Set DEV21 output to Closed and Disable Conditional Logic
		A0240	Set DEV22 Mode to Remote Out
		A0250	Set DEV22 output to Closed and Disable Conditional Logic
		A0260	Set DEV23 Mode to Remote Cascade
		A0270	Setting DEV23 setpoint to (TT_781B Setpoint) and Enabling Conditional Logic
		A0280	Opening and Disabling Conditional Logic for DEV24
		A0290	Opening and Disabling Conditional Logic for DEV25
		A0300	Opening and Disabling Conditional Logic for DEV26
		A0310	Opening and Disabling Conditional Logic for DEV27
		A0320	Opening and Disabling Conditional Logic for DEV28
		A0330	Opening and Disabling Conditional Logic for DEV29

Step	Description	Action	Action Description
		A0340	Opening and Disabling Conditional Logic for DEV30
		A0350	Opening and Disabling Conditional Logic for DEV31
		A0360	Opening and Disabling Conditional Logic for DEV32
		A0370	Opening and Disabling Conditional Logic for DEV33
		A0380	Enabling Conditional Logic for DEV34
		A0390	Enabling Conditional Logic for DEV35
		A0400	Set DEV36 Mode to Remote Out
		A0410	Set DEV36 output to Closed and Disable Conditional Logic
		A0420	Set Preset Drain time
		A0430	Start Preset Drain Timer
		A0440	Preset Drain time Message
T1300A	Drain 880B		If Preset Drain time (TP016) is over then go to step S1300
S1300	Drain 880B	A0010	Stopping and Disabling Conditional logic for DEV1
		A0020	Closing DEV2
		A0030	Stopping DEV3
		A0040	Stopping and Disabling Conditional Logic for DEV4
		A0050	Stopping and Disabling Conditional Logic for DEV5
		A0060	Enabling Conditional Logic for DEV6
		A0070	Enabling Conditional Logic for DEV7
		A0080	Closing DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Closing and Disabling Conditional Logic for DEV14
		A0150	Closing DEV15
		A0160	Closing and Disabling Conditional Logic for DEV16
		A0170	Opening DEV17
		A0180	Closing and Disabling Conditional Logic for DEV18

Step	Description	Action	Action Description
		A0190	Closing and Disabling Conditional Logic for DEV19
		A0200	Set DEV20 Mode to Remote Out
		A0210	Set DEV20 output to Open and Enable Conditional Logic
		A0220	Set DEV21 Mode to Remote Out
		A0230	Set DEV21 output to Closed and Disable Conditional Logic
		A0240	Set DEV22 Mode to Remote Out
		A0250	Set DEV22 output to Closed and Disable Conditional Logic
		A0260	Set DEV23 Mode to Remote Cascade
		A0270	Setting DEV23 setpoint (TT_781B Setpoint) and Enabling Conditional Logic
		A0280	Opening and Disabling Conditional Logic for DEV24
		A0290	Opening and Disabling Conditional Logic for DEV25
		A0300	Opening and Disabling Conditional Logic for DEV26
		A0310	Opening and Disabling Conditional Logic for DEV27
		A0320	Opening and Disabling Conditional Logic for DEV28
		A0330	Opening and Disabling Conditional Logic for DEV29
		A0340	Opening and Disabling Conditional Logic for DEV30
		A0350	Opening and Disabling Conditional Logic for DEV31
		A0360	Opening and Disabling Conditional Logic for DEV32
		A0370	Opening and Disabling Conditional Logic for DEV33
		A0380	Opening and Disabling Conditional Logic for DEV34
		A0390	Opening and Disabling Conditional Logic for DEV35
		A0400	Set DEV36 Mode to Remote Out
		A0410	Set DEV36 output to Closed and Disable Conditional Logic
T1400A	Blow 880B		If LSL_780B is in alarm and LT_780B_VLL is in VLL alarm then go to step S1400
S1400	Blow 880B	A0010	Stopping and Disabling Conditional logic for DEV1
		A0020	Closing DEV2
		A0030	Stopping DEV3

Step	Description	Action	Action Description
		A0040	Stopping and Disabling Conditional Logic for DEV4
		A0050	Stopping and Disabling Conditional Logic for DEV5
		A0060	Stopping and Disabling Conditional Logic for DEV6
		A0070	Stopping and Disabling Conditional Logic for DEV7
		A0080	Closing DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Closing and Disabling Conditional Logic for DEV14
		A0150	Closing DEV15
		A0160	Closing and Disabling Conditional Logic for DEV16
		A0170	Closing DEV17
		A0180	Enabling Conditional Logic for DEV18
		A0190	Closing and Disabling Conditional Logic for DEV19
		A0200	Set DEV20 Mode to Remote Out
		A0210	Set DEV20 output to Open and Disable Conditional Logic
		A0220	Set DEV21 Mode to Remote Out
		A0230	Set DEV21 output to Closed and Disable Conditional Logic
		A0240	Set DEV22 Mode to Remote Out
		A0250	Set DEV22 output to Closed and Disable Conditional Logic
		A0260	Set DEV23 Mode to Remote Out
		A0270	Set DEV23 output to Closed and Disable Conditional Logic
		A0280	Opening and Disabling Conditional Logic for DEV24
		A0290	Opening and Disabling Conditional Logic for DEV25
		A0300	Opening and Disabling Conditional Logic for DEV26
		A0310	Opening and Disabling Conditional Logic for DEV27
		A0320	Opening and Disabling Conditional Logic for DEV28

Step	Description	Action	Action Description
		A0330	Opening and Disabling Conditional Logic for DEV29
		A0340	Opening and Disabling Conditional Logic for DEV30
		A0350	Opening and Disabling Conditional Logic for DEV31
		A0360	Opening and Disabling Conditional Logic for DEV32
		A0370	Opening and Disabling Conditional Logic for DEV33
		A0380	Opening and Disabling Conditional Logic for DEV34
		A0390	Opening and Disabling Conditional Logic for DEV35
		A0400	Set DEV36 Mode to Remote Out
		A0410	Set DEV36 output to Closed and Disable Conditional Logic
		A0420	Set Message: Setting Preset blow time
		A0430	Start Preset Blow Timer
		A0440	Preset Blow time Message
T1500A	Blow to Storage		If DEV18 blow time (TP008) is over then go to step S1500
S1500	Blow to Storage	A0010	Stopping and Disabling Conditional logic for DEV1
		A0020	Closing DEV2
		A0030	Stopping DEV3
		A0040	Stopping and Disabling Conditional Logic for DEV4
		A0050	Stopping and Disabling Conditional Logic for DEV5
		A0060	Stopping and Disabling Conditional Logic for DEV6
		A0070	Stopping and Disabling Conditional Logic for DEV7
		A0080	Closing DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Closing and Disabling Conditional Logic for DEV14
		A0150	Closing DEV15

Step	Description	Action	Action Description
		A0160	Closing and Disabling Conditional Logic for DEV16
		A0170	Closing DEV17
		A0180	Closing and Disabling Conditional Logic for DEV18
		A0190	Enabling Conditional Logic for DEV19
		A0200	Set DEV20 Mode to Remote Out
		A0210	Set DEV20 output to Closed and Disable Conditional Logic
		A0220	Set DEV21 Mode to Remote Out
		A0230	Set DEV21 output to Closed and Disable Conditional Logic
		A0240	Set DEV22 Mode to Remote Out
		A0250	Set DEV22 output to Closed and Disable Conditional Logic
		A0260	Set DEV23 Mode to Remote Out
		A0270	Set DEV23 output to Closed and Disable Conditional Logic
		A0280	Opening and Disabling Conditional Logic for DEV24
		A0290	Opening and Disabling Conditional Logic for DEV25
		A0300	Opening and Disabling Conditional Logic for DEV26
		A0310	Opening and Disabling Conditional Logic for DEV27
		A0320	Opening and Disabling Conditional Logic for DEV28
		A0330	Opening and Disabling Conditional Logic for DEV29
		A0340	Opening and Disabling Conditional Logic for DEV30
		A0350	Opening and Disabling Conditional Logic for DEV31
		A0360	Opening and Disabling Conditional Logic for DEV32
		A0370	Opening and Disabling Conditional Logic for DEV33
		A0380	Opening and Disabling Conditional Logic for DEV34
		A0390	Opening and Disabling Conditional Logic for DEV35
		A0400	Set DEV36 Mode to Remote Out
		A0410	Set DEV36 output to Closed and Disable Conditional Logic
		A0420	Set Message: Setting Preset blow time
		A0430	Start Preset Blow Timer
		A0440	Preset Blow time Message

Step	Description	Action	Action Description
		A0450	Reset Cold Recirculation Button
T1600	Close Devices		If DEV19 blow time (TP009) is over then go to step S1600
S1600	Close Devices	A0010	Stopping and Disabling Conditional logic for DEV1
		A0020	Closing DEV2
		A0030	Stopping DEV3
		A0040	Stopping and Disabling Conditional Logic for DEV4
		A0050	Stopping and Disabling Conditional Logic for DEV5
		A0060	Stopping and Disabling Conditional Logic for DEV6
		A0070	Stopping and Disabling Conditional Logic for DEV7
		A0080	Closing DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Closing and Disabling Conditional Logic for DEV14
		A0150	Closing DEV15
		A0160	Closing and Disabling Conditional Logic for DEV16
		A0170	Closing DEV17
		A0180	Closing and Disabling Conditional Logic for DEV18
		A0190	Closing and Disabling Conditional Logic for DEV19
		A0200	Set DEV20 Mode to Remote Out
		A0210	Set DEV20 output to Closed and Disable Conditional Logic
		A0220	Set DEV21 Mode to Remote Out
		A0230	Set DEV21 output to Closed and Disable Conditional Logic
		A0240	Set DEV22 Mode to Remote Out
		A0250	Set DEV22 output to Closed and Disable Conditional Logic
		A0260	Set DEV23 Mode to Remote Out
		A0270	Set DEV23 output to Closed and Disable Conditional Logic

Step	Description	Action	Action Description
		A0280	Closing and Disabling Conditional Logic for DEV24
		A0290	Closing and Disabling Conditional Logic for DEV25
		A0300	Closing and Disabling Conditional Logic for DEV26
		A0310	Closing and Disabling Conditional Logic for DEV27
		A0320	Closing and Disabling Conditional Logic for DEV28
		A0330	Closing and Disabling Conditional Logic for DEV29
		A0340	Closing and Disabling Conditional Logic for DEV30
		A0350	Closing and Disabling Conditional Logic for DEV31
		A0360	Closing and Disabling Conditional Logic for DEV32
		A0370	Closing and Disabling Conditional Logic for DEV33
		A0380	Closing and Disabling Conditional Logic for DEV34
		A0390	Closing and Disabling Conditional Logic for DEV35
		A0400	Set DEV36 Mode to Remote Out
		A0410	Set DEV36 output to Closed and Disable Conditional Logic
T1700	Disable HM		If no pending actions in S1600 then go to step S1700
S1700	Disable HM	A0010	Disable HM 01 (Module Ownership Lost)
T1800	Release Devices		If no pending actions in S1700 then go to step S1800
S1800	Release Devices	A0010	Release Devices Message
		A0020	Release Private Device
		A0030	Release Private Device
		A0040	Release Private Device
		A0050	Release Private Device
		A0060	Release Private Device
		A0070	Release Private Device
		A0080	Release Private Device
		A0090	Release Private Device
		A0100	Release Private Device
		A0110	Release Private Device
		A0120	Release Private Device

Step	Description	Action	Action Description
		A0130	Release Private Device
		A0140	Release Private Device
		A0150	Release Private Device
		A0160	Release Private Device
		A0170	Release Private Device
		A0180	Release Private Device
		A0190	Release Private Device
		A0200	Release Private Device
		A0210	Release Private Device
		A0220	Release Private Device
		A0230	Release Private Device
		A0240	Release Private Device
		A0250	Release Private Device
		A0260	Release Private Device
		A0270	Release Private Device
		A0280	Release Private Device
		A0290	Release Private Device
		A0300	Release Private Device
		A0310	Release Private Device
		A0320	Release Private Device
		A0330	Release Private Device
		A0340	Release Private Device
		A0350	Release Private Device
		A0360	Release Private Device
T9900	Complete		If no pending actions in S1800 then go to step S9900
S9900	Complete	A0010	Set Message: Drain Command Complete
		A0020	Self Release the EM
END	Command Complete		True

Table 22: Recirculation Sequence

3.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value/Reference
Operating Parameters				
OP001	Deodorising 1 Pump	Named Set	NA	RF1_PU1722C1

Parameter	Description	Type	Units	Default Value/Reference
OP002	Deodorising 2 Pump	Named Set	NA	RF1_PU1722
OP003	Quality Recirculation	Boolean	NA	False
OP004	Production	Boolean	NA	False
OP005	Hot Oil Recirculation (HOR)	Boolean	NA	False
OP006	Cold Recirculation (CR)	Boolean	NA	False
OP007	Start Drain Sequence - After CR	String	NA	0
OP008	Bypass 1-822QE2 Heating Coil	String	NA	0
OP009	Bypass 1-822QE2 Heating Coil	Boolean	NA	False
OP010	Manual Valve Confirmation	Boolean	NA	False
Tuning Parameters				
TP001	Delay Timer	Floating Point	sec	5
TP002	Remote Setpoint FT_702	Floating Point	kg/h	62500
TP003	Remote Setpoint LT_722C	Floating Point	%	70
TP004	Remote Setpoint LT_780B	Floating Point	%	40
TP005	Remote Setpoint TT_781B	Floating Point	°C	60
TP006	XV_802-B Blowing	Floating Point	sec	60
TP007	XV_822C-B Blowing	Floating Point	sec	60
TP008	XV_880B-B Blowing	Floating Point	sec	60
TP009	XV_881A-B Blowing	Floating Point	sec	60
TP010	Vacuum Setpoint	External Reference	mbar	RF2_PT2722AB/HI_HI_LIM
TP011	Deodorising Temperature	External Reference	°C	RF2_TT2721A2/LO_LO_LIM
TP012	Max P822 outlet Temperature	External Reference	°C	RF2_TT2781B/HI_HI_LIM
TP013	Hot recirculation Preset Time	Floating Point	min	60
TP014	Cold Recirculation Temperature	Floating Point	°C	60
TP015	Blow Coil preset time	Floating Point	sec	400

Parameter	Description	Type	Units	Default Value/Reference
TP016	Drain preset Time	Floating Point	sec	600
TP017	Bypass 1-822QE2 Heating Coil	Named Set	NA	NO
TP018	XV_822QE1-B Blowing	Floating Point	sec	0
TP019	RSP for TT_722C	Floating Point	NA	180

Table 23: Operating and Tuning Parameters

3.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

Parameter Name	Parameter Type	Value
MY_TAG		RF2EQ_822QE
OPR_CMD_MASK		31
LSM_721A	External Reference	RF2_LSM2721A/DI1/DISC_ACT
TT_721A2	External Reference	RF2_TT2721A2/AI1/PV
LSL_780B	External Reference	RF2_LSL2780B/DI1/DISC_ACT
LSH_780B	External Reference	RF2_LSH2780B/DI1/DISC_ACT
LT_780B_VLL	External Reference	RF2_LT2780B/PID1/LO_LO_ACT
LSL_722C	External Reference	RF2_LSL2722C/DI1/DISC_ACT
LT_722C_VLL	External Reference	RF2_LT2722C/PID1/LO_LO_ACT
LSH_722C	External Reference	RF2_LSH2722C/DI1/DISC_ACT
LSL_702	External Reference	RF2_LSL2702/DI1/DISC_ACT
LT_722C_VHL	External Reference	RF2_LT2722C/PID1/HI_HI_ACT
LT_780B_VHL	External Reference	RF2_LT2780B/PID1/HI_HI_ACT
BYPASS	Boolean	False
STARTUP	String	RF2EQ_STARTUP1
EM_DEOD_FA_CKT1	String	RF2EQ_802
DY_REF_1	Dynamic Reference	
VACUUM_ALM	External Reference	RF2_PT2722AB/AI1/HI_HI_ACT
HOLD	External Reference	RF2EQ_STARTUP1/OP006_VALUE
STEP_TIME	Floating point	0
COMPLETED_TIME	Floating point	0
RESET_COMP_TM	Boolean	False

Table 24: Configurable Parameters

3.6 Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_DEOD_FA_CKT2	Refinery Area: Startup EM

Table 25: Named Set Details

Details of states for NS_DEOD_FA_CKT2 named set are as follows,

Named Set Value	Named Set Name
00	Hold
01	Heating
02	Production
03	Recirculation

Table 26: Name Set State Details

4 EQUIPMENT MODULE CLASS DESIGN – EM_DEOD_FA_CKT3 (2-814/23)

4.1 Functional Description

EM_DEOD_FA_CKT3 is a class based equipment module that driven by EM_RF_STARTUP1 sequence. EM_DEOD_FA_CKT3 is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. Initially Operator shall be provided only with Hold and Production command. All devices used in EM shall be acquired in the Production command and released in Stop Command. Hold Request shall be enabled for all acquired devices at the end of Hold command.

4.2 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name
EM	EM_DEOD_FA_CKT3	RF2EQ_814_23
DEV1	_MRSS11_130_CUR	RF2_PU2714AG
DEV2	_V12_IP_130_SE	RF2_V2714AGB
DEV3	_MRSS11_130_SE	RF2_PU2750
DEV4	_CSD_130_SE	RF2_TT2781AA
DEV5	_CSD_130_SE	RF2_LT2714
DEV6	_CMT_130_SE	RF2_TT2781AB

Table 27: Instances

4.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

4.3.1 Command 00 – Hold

Step	Description	Action	Action Description
H0010	Command Hold Start	A0010	Store Fail Message
		A0020	Reset Hold Request
		A0030	Save Previous Step Time
T1000	Actions Complete or Time Out & Hold=True		Go to step T1000
T2000	Actions Complete or		Go to step T2000

Step	Description	Action	Action Description
	Time Out & Hold=False		
H1000	Hold Command 1	A0010	Stopping and Disabling Conditional Logic for DEV3
T9900A	Actions Complete or Time Out		Go to step H9900A
H2000	General Reset	A0010	Stopping DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Stopping and Disabling Conditional Logic for DEV3
		A0040	Set DEV4 Mode to Remote Output
		A0050	Set DEV4 output to Closed and Disable Conditional Logic
		A0060	Set DEV5 Mode to Remote Out
		A0070	Set DEV5 output to Closed and Disable Conditional Logic
T9900B	Actions Complete or Time Out		Go to step H9900A
H9900A	Command Hold End	A0010	Completed
END	Command Complete		True
H9900B	Release Devices	A0010	Release Devices Message
		A0020	Release Private Devices
		A0030	Release Private Devices
		A0040	Release Private Devices
		A0050	Release Private Devices
		A0060	Release Private Devices
		A0070	Release Private Devices
		A0080	Reset Parameters
END1	Command Complete		True

Table 28: Hold Sequence

4.3.2 Command 01 – Production

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Production Command
T0010	First Pass = False		If First Pass = False then go to step S0010
T0020A	First Pass = True		If First Pass = True then go to step S0020
S0010	Initializations	A0010	First Pass Logic
		A0020	First Pass = True
		A0030	Set Task Pointer
T0020B	Actions Complete		If no pending actions in S0010 then go to step S0020
S0020	Acquire Devices	A0010	Acquire Devices Message
		A0020	Acquire Private Devices
		A0030	Acquire Private Devices
		A0040	Acquire Private Devices
		A0050	Acquire Private Devices
		A0060	Acquire Private Devices
		A0070	Acquire Private Devices
		A0080	Reset Hold Request
T0100B	Standby		If Task Pointer = 1 then go to step S0100
T0200B	Fill		If Task Pointer = 2 then go to step S0200
T0300B	Blow Inlet		If Task Pointer = 3 then go to step S0300
S0100	Standby	A0010	Stopping DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Stopping and Disabling Conditional Logic for DEV3
		A0040	Set DEV4 Mode to Remote Output
		A0050	Set DEV4 output to Closed and Disable Conditional Logic
		A0060	Set DEV5 Mode to Remote Out
		A0070	Set DEV5 output to Closed and Disable Conditional Logic
T0200A	Fill		If no pending actions in S0100 then go to S0200
S0200	Fill	A0010	Stopping DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Stopping and Disabling Conditional Logic for DEV3

Step	Description	Action	Action Description
		A0040	Set DEV4 Mode to Remote Output
		A0050	Set DEV4 output to Closed and Disable Conditional Logic
		A0060	Set DEV5 Mode to Remote Out
		A0070	Set DEV5 output to Closed and Disable Conditional Logic
		A0080	Waiting for LSL_714 to not be in alarm
T0300A	Production		If LSL_714 is not in alarm then go to S0300
S0300	Production	A0010	Starting DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Enabling Conditional Logic for DEV3
		A0040	Set DEV4 Mode to Cascade and DEV6 to Remote Cascade
		A0050	Setting DEV6 setpoint (TT_781AG Setpoint) and Enabling Conditional Logic
		A0060	Set DEV5 Mode to Remote Cascade
		A0070	Setting DEV5 setpoint (LT_P714AG Setpoint) and Enabling Conditional Logic
T9900	Complete		If no pending actions in S0300 then go to S9900
S9900	Complete	A0010	Set Message: Production Command Complete
END	Command Complete		True

Table 29: Production Sequence

4.3.3

Command 02 –Drain

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Drain Command
		A0020	Reset HOLD_REQ
T0100	Drain		If Drain Command (OP001) selected by operator then go to step S0100
S0100	Drain	A0010	Starting DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Enabling Conditional Logic for DEV3
		A0040	Set DEV4 Mode to Cascade and DEV6 to Remote Cascade
		A0050	Setting DEV6 setpoint (TT_781AG Setpoint) and Enabling Conditional Logic
		A0060	Set DEV5 Mode to Remote Out

Step	Description	Action	Action Description
		A0070	Set DEV5 output to Open and Disable Conditional Logic
		A0080	Setting Auto Drain Timer (TMR1)
T9900	Complete		If no pending actions in S0100 the go to step S9900
S9900	Complete	A0010	Set Message: Drain Command Complete
END	Command Complete		True

Table 30: Drain Sequence

4.3.4 Command 03 – Stop

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Stop Drain Command
		A0020	Reset HOLD_REQ
T0100	Blow		If LSL_714 is in alarm and FT_714AG is in VLL or <Stop> command (OP002) by operator or Auto drain time finished (TMR1)
S0100	Blow	A0010	Stopping DEV1
		A0020	Enabling Conditional Logic for DEV2
		A0030	Stopping and Disabling Conditional Logic for DEV3
		A0040	Set DEV4 Mode to Remote Output
		A0050	Set DEV4 output to Closed and Disable Conditional Logic
		A0060	Set DEV5 Mode to Remote Out
		A0070	Set DEV5 output to Closed and Disable Conditional Logic
		A0080	Set Message: Setting Preset blow time
		A0090	Start Preset Blow Timer
		A0100	Preset Blow time Message
T0200	Close Devices		If Preset Blow time (TP001) is over then go to step S0200
S0200	Close Devices	A0010	Stopping DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Stopping and Disabling Conditional Logic for DEV3
		A0040	Set DEV4 Mode to Remote Output
		A0050	Set DEV4 output to Closed and Disable Conditional Logic
		A0060	Set DEV5 Mode to Remote Out

Step	Description	Action	Action Description
		A0070	Set DEV5 output to Closed and Disable Conditional Logic
T0300	Disable HM and SM		If no pending actions in S0200 the go to step S0300
S0300	Disable HM	A0010	Disable HM 01 (Module Ownership Lost)
T0400	Release Devices		If no pending actions in S0300 the go to step S0400
S0400	Release Devices	A0010	Release Devices Message
		A0020	Release Private Devices
		A0030	Release Private Devices
		A0040	Release Private Devices
		A0050	Release Private Devices
		A0060	Release Private Devices
		A0070	Release Private Devices
T9900	Complete		If no pending actions in S0400 the go to step S9900
S9900	Complete	A0010	Set Message: Stop Drain Command Complete
		A0020	Self Release the EM
END	Command Complete		True

Table 31: Stop Sequence

4.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value/Reference
Operating Parameters				
OP001	DRAIN	External Reference	NA	RF2EQ_STARTUP1/OP005_VALUE
OP002	STOP	Named Set	NA	OFF
Tuning Parameters				
TP001	Preset Blow Coil Time	Floating Point	sec	400
TP002	Remote Setpoint for TT_781AG	Floating Point	NA	55
TP003	Remote Setpoint for LT_714	Floating Point	NA	40
TP004	Auto Drain Time	Floating Point	sec	100

Table 32: Operating and Tuning Parameters

4.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

Parameter Name	Parameter Type	Value/Reference
MY_TAG		RF2EQ_814_23
OPR_CMD_MASK		15
LSL_714	External Reference	RF2_LSL2714/DI1/DISC_ACT
FT_714AG	External Reference	RF2_FT2714AG/AI1/LO_LO_ACT
RF_STARTUP	String	RF2EQ_STARTUP1
DY_REF_1	Dynamic Reference	
HOLD	External Reference	RF2EQ_STARTUP1/OP006_VALUE
EM_DEOD_CKT2		RF2EQ_822QE/A_PV
STEP_TIME	Floating point	0
COMPLETED_TIME	Floating point	0
RESET_COMP_TM	Boolean	False

Table 33: Configurable Parameters

4.6 Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_DEOD_FA_CKT3	Refinery Area: Deodorising & Ice Condensing

Table 34: Named Set Details

Details of states for NS_DEOD_FA_CKT3 named set are as follows,

Named Set Value	Named Set Name
00	Hold
01	Production
02	Drain
03	Stop

Table 35: Name Set State Details

5 EQUIPMENT MODULE CLASS DESIGN – EM_RF_STARTUP3 (STARTUP)

5.1 Functional Description

EM_RF_STARTUP3 is a class based equipment module that performs filling operation to produce the material and afterwards drains the material. EM_RF_STARTUP3 is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. Initially Operator shall be provided only with Hold and Aux On. Production, Drain and Aux off command shall be enabled during the execution of Aux On State. Hold Request shall be enabled for all acquired devices at the end of Hold command.

5.2 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name
EM	EM_RF_OF_STARTUP3	RF2EQ_STARTUP3
DEV1	_MRSS11_130_CUR	RF2_PU2641X
DEV2	_MRSS11_130_CUR	Ignored
DEV3	EM_RF_OF_BLEACH2	RF2EQ_T506

Table 36: Instances

5.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

5.3.1 Command 00 – Hold

Step	Description	Action	Action Description
H0000	Command Hold Start	A0010	Store Fail Message
		A0020	Set Hold Request
T0010	Actions Complete or Time Out		Go to step H1000
H1000	Hold Command 1	A0010	Stopping DEV1
		A0020	Stopping DEV2
T9900A	Actions Complete or Time Out & Hold=True		Go to step H9900A

Step	Description	Action	Action Description
T9900B	Actions Complete or Time Out & Hold=False		Go to step H9900B
H9900A	Command Hold End	A0010	Completed
END	Command Complete		True
H9900B	Release Device	A0010	Release Private Devices
		A0020	Release Private Devices
		A0030	Rest Parameters
END1	Command Complete		True

Table 37: Hold Sequence

5.3.2 Command 01 – Aux On

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Aux On Command
T0010	First Pass = False		If First Pass = False then go to step S0010
T0020A	First Pass = True		If First Pass = True then go to step S0020
S0010	Initializations	A0010	First Pass Logic
		A0020	First Pass = True
T0020B	Acquire Devices		If no pending actions in S0010 then go to step S0020
S0020	Acquire Devices	A0010	Acquire Devices Message
		A0020	Acquire Private Devices
		A0030	Acquire Private Devices
		A0040	Reset Hold Request
T0100	Start Aux Ckt		If Task Pointer = 1 then go to step S0100
T0200B	Ready		If Task Pointer = 2 then go to step S0200
S0100	Start Aux Ckt	A0010	Set Message: Starting pump as per operator selection
		A0020	Starting DEV1 or DEV2
		A0030	Set Message: Checking PT_622< the set Vacuum set point
T0200A	Ready		If PT_622 < Vacuum set point (TP002) then go to step S0200

Step	Description	Action	Action Description
S0200	Ready	A0010	Set Message: Starting pump as per operator selection and Set Task Pointer
		A0020	Starting DEV1 or DEV2
T9900	Complete		If no pending actions in S0200 then go to step S9900
S9900	Complete	A0010	Set Message: Aux On Command Complete.
END	Command Complete		True

Table 38: Aux On Sequence

5.3.3 Command 02 – Production

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Production Command
T0100	Startup		If Task Pointer = 2 then go to step S0100
T0200B	Production		If Task Pointer = 3 then go to step S0200
S0100	Startup	A0010	Set Message: Checking EM_RF_OF_BLEACH2 is Not Owned.
		A0020	Acquire DEV3.
		A0030	Provide Production command to EM_RF_OF_BLEACH2.
		A0040	Set Message: Starting pump as per operator selection
		A0050	Starting DEV1 or DEV2
		A0060	Set Message: Waiting for Bleach5 to be in Production step
T0200A	Production		If 622/21 (EM_622_21) is in Step <Production> then go to step S0200
S0200	Production	A0010	Set Message: Starting pump as per operator selection
		A0020	Starting DEV1 or DEV2
		A0030	Set Task Pointer
T9900	Complete		If no pending actions in S0200 then go to step S9900
S9900	Complete	A0010	Set Message: Production Command Complete
END	Actions Complete		True

Table 39: Production Sequence

5.3.4 Command 03 – Drain

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Drain Command
T0100	Drain		Go to step S0100
S0100	Drain	A0010	Set Message: Starting pump as per operator selection
		A0020	Starting DEV1 or DEV2
		A0030	Reset Production Command
T9900	Complete		If T506 (EM_T506), T503 (EM_T503), 622/21 (EM_622_21), 616A1-5 (EM_616A1/A2/A3/A4/A5), 682B (EM_682B) and 616S (EM_616S) are in step <Standby> then go to step S9900
S9900	Complete	A0010	Set Message: Drain Command Complete
		A0020	Reset Drain Command
END	Actions Complete		True

Table 40: Drain Sequence

5.3.5 Command 04 –Aux Off

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Aux Off Command
T0100	Close Device		If no pending actions in S0000 then go to step S0100
S0100	Close Device	A0010	Stopping DEV1 or DEV2
		A0020	AUX ON command
T0200	Disable HM & SM		If no pending actions in S0000 then go to step S0100
S0200	Disable HM	A0010	Disable HM 01 (Module Ownership Lost)
T0300	Release Device		If no pending actions in S0200 then go to step S0300
S0300	Release Device	A0010	Release Private Devices
		A0020	Release Private Devices
T9900	Complete		If no pending actions in S0300 then go to step S9900
S9900	Complete	A0010	Set Message: Aux Off Command Complete
END	Command Complete		True



Table 41: Aux Off Sequence

5.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value / Reference
Operating Parameters				
OP001	Pump Selection	Named Set	NA	PUMPX1
OP002	AUX	Named Set	NA	OFF
OP003	RUN	Named Set	NA	OFF
OP004	DRAIN	Named Set	NA	OFF
OP005	HOLD	Named Set	NA	OFF
Tuning Parameters				
TP001	Delay Timer	Floating Point	sec	5
TP002	Vacuum Setpoint PT_622/21	Floating Point	mbar	150

Table 42: Operating and Tuning Parameters

5.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

Parameter Name	Parameter Type	Value / Reference
MY_TAG		RF2EQ_STARTUP3
OPR_CMD_MASK		31
PT_622	External Reference	RF2_PT2622/PV
EM_622_21	String	RF2EQ_622_21
EM_T506	External Reference	RF2EQ_T506/OWNER_ID
EM_T503	External Reference	RF2EQ_T503/OWNER_ID
EM_616A1	External Reference	RF2EQ_616A1/OWNER_ID
EM_616A2	External Reference	RF2EQ_616A2/OWNER_ID
EM_616A3	External Reference	RF2EQ_616A3/OWNER_ID
EM_616A4	External Reference	RF2EQ_616A4/OWNER_ID
EM_616A5	External Reference	RF2EQ_616A5/OWNER_ID
EM_682B	External Reference	RF2EQ_682B/OWNER_ID
EM_616S	External Reference	RF2EQ_616S/OWNER_ID
DY_REF_1	Dynamic Reference	

Table 43: Configurable Parameters

5.6 Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_RF_STARTUP3	Refinery Area: Filtration and Vacuum Production

Table 44: Named Set Details

Details of states for NS_RF_STARTUP3 named set are as follows,

Named Set Value	Named Set Name
00	Hold
01	Aux On
02	Production
03	Drain
04	Aux Off

Table 45: Name Set State Details

6 EQUIPMENT MODULE CLASS DESIGN – EM_RF_OF_BLEACH1 (2-630A OR 2-630B)

6.1 Functional Description

EM_RF_OF_BLEACH1 is a class based equipment module that drives two different instances which performs filling operation using vacuum pump/fan. EM_RF_OF_BLEACH1 is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. Initially Operator shall be provided only with Hold and Fill. Hold Request shall be enabled for all acquired devices at the end of Hold command. Commands which require operator intervention shall be configured as Operator Prompts wherever possible.

6.2 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name	
EM	EM_RF_OF_BLEACH1	RF2EQ_630A	RF2EQ_630B
DEV1	_MRSS11_130_SE	RF2_BW2609A	RF2_BW2609B
DEV2	_DO_130_SE	RF2_SV2609A	RF2_SV2609B
DEV3	_MRSS11_130_SE	Ignored	RF2_RL2610B

Table 46: Instances

6.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

6.3.1 Command 00 – Hold

Step	Description	Action	Action Description
H0010	Start	A0010	Store Fail Message
		A0020	Reset HOLD_REQ
		A0030	Save Previous Step Time
T1000	Actions Complete or Time Out		Go to step H1000
H1000	Hold Command 1	A0010	Stopping DEV1
		A0020	Turning Off DEV2
		A0030	Stopping DEV3

Step	Description	Action	Action Description
T9900A	Actions Complete or Time Out and Hold=True		Go to step H9900B
T9900B	Actions Complete or Time Out and Hold=False		Go to step H9900B
H9900A	Command Hold End	A0010	Completed
END	Command Complete		True
H9900B	Release Devices	A0010	Release Devices Message
		A0020	Release Private Device
		A0030	Release Private Device
		A0040	Release Private Device
		A0050	Reset Parameters
END1	Command Complete		True

Table 47: Hold Sequence

6.3.2

Command 01 – Start Fill

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Start Fill Command
T0010	First Pass = False		If First Pass = False then go to step S0010
T0020A	First Pass = True		If First Pass = True then go to step S0020
S0010	Initializations	A0010	First Pass Logic
		A0020	First Pass = True
		A0030	Set Task Pointer
T0020B	Actions Complete		If no pending actions in S0010 then go to step S0020
S0020	Acquire Device	A0010	Acquire Devices Message
		A0020	Acquire Private Device
		A0030	Acquire Private Device
		A0040	Acquire Private Device
		A0060	Reset Hold Request
T0100	Standby		If Task Pointer = 1 then go to step S0100

Step	Description	Action	Action Description
T0200B	Start Vac. Pump/Fan		If Task Pointer = 2 then go to step S0200
TP300B	Filling		If Task Pointer = 3 then go to step S0300
S0100	Standby	A0010	Stopping DEV1
		A0020	Turning Off DEV2
		A0030	Stopping DEV3
T0200A	Standby		If LSL_630 is in alarm then go step S0200
S0200	Start Vac. Pump/Fan	A0010	Starting DEV1
		A0020	Turning On DEV2
		A0030	Stopping DEV3
		A0040	Set Start Fan time
		A0050	Start Fan Timer
		A0060	Start Fan time Message
T0300B	Filling		If DEV1 is running and Start Fan Timer (TP002) elapsed then go step S0300
S0300	Filling	A0010	Starting DEV1
		A0020	Turning On DEV2
		A0030	Starting DEV3
T9900	Complete		If Stop button (OP001) is triggered or LSH_630 is in alarm then go to step S9900
S9900	Complete	A0010	Set Message: Start Fill Command Complete
		A0020	Give A_CMD to next Command on LSH_630 is in alarm
END	Command Complete		True

Table 48: Fill Sequence

6.3.3 Command 01 – Stop Fill

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Stop Fill Command
		A0020	Check if LSH_630A is active
T0100	Stop Vac. Pump/Fan		If Task Pointer = 3 then go to step S0100
S0100	Stop Vac. Pump/Fan	A0010	Starting DEV1
		A0020	Turning On DEV2
		A0030	Stopping DEV3
		A0040	Set Stop Fan time
		A0050	Stop Fan Timer

Step	Description	Action	Action Description
		A0060	Stop Fan time Message
T0200	Step Time > TP003_VALUE (Stop Fan Time (sec))		If Stop Fan Timer elapsed (TP003) then go to step S0200
S0200	Close Devices	A0010	Stopping DEV1
		A0020	Turning Off DEV2
		A0030	Stopping DEV3
		A0040	Reset Filling Selection Button (OP001)
T0300	Disable HM and SM		If no pending actions S0200 then go to step S0300
S0300	Disable HM	A0010	Disable HM 01 (Module Ownership Lost)
T0400	Release Devices		If no pending actions S0300 then go to step S0400
S0400	Release Devices	A0010	Release Devices Message
		A0020	Release Private Device
		A0030	Release Private Device
		A0040	Release Private Device
T9900	Complete		If no pending actions S0400 then go to step S9900
S9900	Complete	A0010	Set Message: Stop Fill Command Complete
		A0020	Self release the EM
END	Command Complete		True

Table 49: Fill Sequence

6.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value/Reference
Operating Parameters				
OP001	Start Fill	Boolean	NA	False
OP002	Stop Fill	Boolean	NA	False
Tuning Parameters				
TP001	Delay Timer	Floating Point	sec	5
TP002	Start Fan Time	Floating Point	sec	600

Parameter	Description	Type	Units	Default Value/Reference
TP003	Stop Fan Time	Floating Point	sec	600

Table 50: Operating and Tuning Parameters

6.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

Parameter Name	Parameter Type	Value/Reference	
MY_TAG		RF2EQ_630A	RF2EQ_630B
OPR_CMD_MASK		7	7
LSH_630	External Reference	RF2_LSH2630A/DI1/DISC_ACT	RF2_LSH2630B/DI1/DISC_ACT
LSL_630	External Reference	RF2_LSL2630A/DI1/DISC_ACT	RF2_LSL2630B/DI1/DISC_ACT
HOLD	External Reference	RF2EQ_STARTUP3/OP005_VALUE	
STEP_TIME	Floating Point	0	
COMPLETED_TIME	Floating Point	0	
RESET_COMP_TM	Boolean	False	

Table 51: Configurable Parameters

6.6 Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_RF_OF_BLEACH1	Refinery Area: Filtration and Vacuum Production

Table 52: Named Set Details

Details of states for NS_RF_OF_BLEACH1 named set are as follows,

Named Set Value	Named Set Name
00	Hold
01	Start Fill
02	Stop Fill

Table 53: Name Set State Details

7 EQUIPMENT MODULE CLASS DESIGN – EM_RF_OF_BLEACH2 (2-T506)

7.1 Functional Description

EM_RF_OF_BLEACH2 is a class based equipment module that performs filling operation to produce the material and afterwards drains the material. EM_RF_OF_BLEACH2 is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. Initially Operator shall be provided only with Hold and Production. Hold Request shall be enabled for all acquired devices at the end of Hold command.

7.2 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name
EM	EM_RF_OF_BLEACH2	RF2EQ_T506
DEV1	_MRSS11_130_SE	RF2_PU2006
DEV2	_MRSS11_130_CUR	RF2_MX2004AC
DEV3	_MVS11_IL_130_SO	RF2_PU2034CA
DEV4	_MVS11_IL_130_SO	RF2_PU2034PA
DEV5	_V12_IP_130_SE	RF2_V2006D
DEV6	_V12_IP_130_SE	RF2_V2024F
DEV7	_V12_IP_130_SE	RF2_V2024B
DEV8	_V12_IP_130_SE	RF2_V2006B
DEV9	_CSD_130_SE	RF2_FT2011
DEV10	_CSD_130_SE	RF2_TT2021B
DEV11	_CSD_130_SE	RF2_LT2006

Table 54: Instances

7.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

7.3.1 Command 00 – Hold

Step	Description	Action	Action Description
H0010	Command Hold Start	A0010	Store Fail Message
		A0020	Reset Hold Request

Step	Description	Action	Action Description
T1000	Actions Complete or Time Out		Go to step H1000
H1000	Hold Command 1	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Stopping and Disabling Conditional Logic for DEV2
		A0030	Stopping and Disabling Conditional Logic for DEV3
		A0040	Stopping and Disabling Conditional Logic for DEV4
		A0050	Closing DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing DEV7
		A0080	Closing DEV8
		A0090	Set DEV9 Mode to Remote Out
		A0100	Set DEV9 output to Closed and Disable Conditional Logic
		A0110	Set DEV10 Mode to Remote Out
		A0120	Set DEV10 output to Closed and Disable Conditional Logic
		A0130	Set DEV11 Mode to Remote Out
		A0140	Set DEV11 output to Closed and Disable Conditional Logic
T9900A	Actions Complete or Time Out and Hold=True		Go to step H9900A
T9900B	Actions Complete or Time Out and Hold=False		Go to step H9900B
H9900A	Commom Hold End	A0010	Completed
END	Command Complete		True
H9900B	Release Devices	A0010	Release Devices Message
		A0020	Release Private Device
		A0030	Release Private Device
		A0040	Release Private Device
		A0050	Release Private Device

Step	Description	Action	Action Description
		A0060	Release Private Device
		A0070	Release Private Device
		A0080	Release Private Device
		A0090	Release Private Device
		A0100	Release Private Device
		A0110	Release Private Device
		A0120	Release Private Device
		A0130	Reset Parameters
END1	Command Complete		True

Table 55: Hold Sequence

7.3.2 Command 01 – Production

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Production Command
T0010	First Pass = False		If First Pass = False then go to step S0010
T0020A	First Pass = True		If First Pass = True then go to step S0020
S0010	Initializations	A0010	First Pass Logic
		A0020	First Pass = True
		A0030	Set Task Pointer
T0020B	Actions Complete		If no pending actions in S0010 then go to step S0020
S0020	Acquire Devices	A0010	Acquire Devices Message
		A0020	Acquire Private Device
		A0030	Acquire Private Device
		A0040	Acquire Private Device
		A0050	Acquire Private Device
		A0060	Acquire Private Device
		A0070	Acquire Private Device
		A0080	Acquire Private Device
		A0090	Acquire Private Device
		A0100	Acquire Private Device
		A0110	Acquire Private Device
		A0120	Acquire Private Device
		A0130	Reset Hold Request
T0100	Standby		If Task Pointer = 1 then go to step S0100
T0200B	Fill		If Task Pointer = 2 then go to step S0200
T0300B	Production		If Task Pointer = 3 then go to step S0300

Step	Description	Action	Action Description
T0400B	Drain		If Task Pointer = 4 then go to step S0400
T0500B	Drain Wait		If Task Pointer = 5 then go to step S0500
S0100	Standby	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Stopping and Disabling Conditional Logic for DEV2
		A0030	Stopping and Disabling Conditional Logic for DEV3
		A0040	Stopping and Disabling Conditional Logic for DEV4
		A0050	Closing DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing DEV7
		A0080	Closing DEV8
		A0090	Set DEV9 Mode to Remote Out
		A0100	Set DEV9 output to Closed and Disable Conditional Logic
		A0110	Set DEV10 Mode to Remote Out
		A0120	Set DEV10 output to Closed and Disable Conditional Logic
		A0130	Set DEV11 Mode to Remote Out
		A0140	Set DEV11 output to Closed and Disable Conditional Logic
T0200A	Fill		If Start up (EM_STARTUP3) is in step in <Start-Up> or <Production> then go to step S0200
S0200	Fill	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Stopping and Disabling Conditional Logic for DEV2
		A0030	Stopping and Disabling Conditional Logic for DEV3
		A0040	Stopping and Disabling Conditional Logic for DEV4
		A0050	Closing DEV5
		A0060	Enabling Conditional Logic for DEV6
		A0070	Closing DEV7
		A0080	Closing DEV8
		A0090	Set DEV9 Mode to Remote Cascade
		A0100	Setting DEV9 setpoint (FT_011 Setpoint) and Enabling Conditional Logic
		A0110	Set DEV10 Mode to Remote Cascade
		A0120	Set DEV10 Setpoint (TT_021B Setpoint) and Enable Conditional Logic

Step	Description	Action	Action Description
		A0130	Set DEV11 Mode to Remote Out
		A0140	Set DEV11 output to Closed and Disable Conditional Logic
		A0150	Set Message: Waiting for EM_RF_OF_BLEACH3 to be in Standby
T0300A	Production		If DEV11 >= LT_006 level reached (TP003) and EM_BLEACH3 is in step <Standby> then go to step S0300
S0300	Production	A0010	Enabling Conditional Logic for DEV1 (TP01)
		A0020	Enabling Conditional Logic for DEV2
		A0030	Enabling Conditional Logic for DEV3
		A0040	Enabling Conditional Logic for DEV4
		A0050	Opening DEV5
		A0060	Enabling Conditional Logic for DEV6
		A0070	Closing DEV7
		A0080	Closing DEV8
		A0090	Set DEV9 Mode to Remote Cascade
		A0100	Setting DEV9 setpoint (FT_011 Setpoint) and Enabling Conditional Logic
		A0110	Set DEV10 Mode to Remote Cascade
		A0120	Set DEV10 Setpoint (TT_021B Setpoint) and Enable Conditional Logic
		A0130	Set DEV11 Mode to Remote Cascade
		A0140	Setting DEV11 setpoint (LT_006 Setpoint) and Enabling Conditional Logic
		A0150	Acquire EM_RF_OF_BLEACH3
T0400A	Drain	A0170	Starting production command for EM_RF_OF_BLEACH3
		A0180	Set Message: Waiting for Startup3 to be in Step Drain
			If Start up (EM_STARTUP3) is in step <Drain> then go to step S0400
S0400	Drain	A0010	Enabling Conditional Logic for DEV1 (TP02)
		A0020	Enabling Conditional Logic for DEV2
		A0030	Enabling Conditional Logic for DEV3
		A0040	Enabling Conditional Logic for DEV4
		A0050	Opening DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Opening DEV7
		A0080	Closing DEV8
		A0090	Set DEV9 Mode to Remote Out

Step	Description	Action	Action Description
		A0100	Set DEV9 output to Open and Disable Conditional Logic
		A0110	Set DEV10 Mode to Remote Out
		A0120	Set DEV10 output to Closed and Disable Conditional Logic
		A0130	Set DEV11 Mode to Remote Out
		A0140	Set DEV11 output to Open and Enable Conditional Logic
T0500A	Drain Wait		If LAL_T506 is in alarm and LT_T506 is in VLL alarm then go to step S0500
S0500	Drain Wait	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Stopping and Disabling Conditional Logic for DEV2
		A0030	Stopping and Disabling Conditional Logic for DEV3
		A0040	Stopping and Disabling Conditional Logic for DEV4
		A0050	Closing DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing DEV7
		A0080	Opening DEV8
		A0090	Set DEV9 Mode to Remote Out
		A0100	Set DEV9 output to Closed and Disable Conditional Logic
		A0110	Set DEV10 Mode to Remote Out
		A0120	Set DEV10 output to Closed and Disable Conditional Logic
		A0130	Set DEV11 Mode to Remote Out
		A0140	Set DEV11 output to Open and Disable Conditional Logic
		A0150	Set Message: Waiting for EM_RF_OF_BLEACH3 to be in Step Drain
T0600	Close Devices		If EM_BLEACH3 is in step <Drain> then go to step S0600
S0600	Close Devices	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Stopping and Disabling Conditional Logic for DEV2
		A0030	Stopping and Disabling Conditional Logic for DEV3
		A0040	Stopping and Disabling Conditional Logic for DEV4
		A0050	Closing DEV5

Step	Description	Action	Action Description
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing DEV7
		A0080	Closing DEV8
		A0090	Set DEV9 Mode to Remote Out
		A0100	Set DEV9 output to Closed and Disable Conditional Logic
		A0110	Set DEV10 Mode to Remote Out
		A0120	Set DEV10 output to Closed and Disable Conditional Logic
		A0130	Set DEV11 Mode to Remote Out
		A0140	Set DEV11 output to Closed and Disable Conditional Logic
		A0150	Reset Acid Dosing Selection Button
T0700	Disable HM and SM		If no pending actions in S0600 then go to step S0700
S0700	Disable HM	A0010	Disable HM 01 (Module Ownership Lost)
T0800	Release Devices		If no pending actions in S0700 then go to step S0800
S0800	Release Devices	A0010	Release Devices Message
		A0020	Release Private Device
		A0030	Release Private Device
		A0040	Release Private Device
		A0050	Release Private Device
		A0060	Release Private Device
		A0070	Release Private Device
		A0080	Release Private Device
		A0090	Release Private Device
		A0100	Release Private Device
		A0110	Release Private Device
		A0120	Release Private Device
T9900	Complete		If no pending actions in S0800 then go to step S9900
S9900	Complete	A0010	Set Message: Production Command Complete
		A0020	Self release the EM
END	Command Complete		True

Table 56: Production Sequence

7.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value/Reference
Operating Parameters				
OP001	Citric Acid Selection	Boolean	NA	False
OP002	Phosphoric Acid Selection	Boolean	NA	False
Tuning Parameters				
TP001	Delay Timer	Floating Point	sec	5
TP002	Remote set point for FT_011	Floating Point	Kg/h	80
TP003	Remote set point for LT_006	Floating Point	%	80
TP004	Remote set point for TT_021B	Floating Point	°C	0

Table 57: Operating and Tuning Parameters

7.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

Parameter Name	Parameter Type	Value/Reference
MY_TAG		RF2EQ_T506
OPR_CMD_MASK		3
EM_BLEACH3	String	RF2EQ_T503
EM_STARTUP3	String	RF2EQ_STARTUP3/COMMAND_0002/ACT_STEPS
DY_REF_1	Dynamic Reference	
DY_REF_2	Dynamic Reference	
LAL_T506	External Reference	RF2_LSL2006/DI1/DISC_ACT
LAH_622_21	External Reference	RF2_LSH2622/DI1/DISC_ACT
LAH_T506	External Reference	RF2_LSH2006/DI1/DISC_ACT
LT_T534CA	External Reference	RF2_LT2044CA/AI1/LO_LO_ACT
LT_T534PA	External Reference	RF2_LT2034PA/AI1/LO_LO_ACT
LT_T506	External Reference	RF2_LT2006/DI1/DISC_ACT
BLEACH3	External Reference	RF2EQ_T503/OWNER_ID

Parameter Name	Parameter Type	Value/Reference
HOLD	External Reference	RF2EQ_STARTUP3/OP005_VALU E
STARTUP3	External Reference	RF2EQ_STARTUP3

Table 58: Configurable Parameters

7.6

Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_RF_OF_BLEACH2	Refinery Area: Filtration and Vacuum Production

Table 59: Named Set Details

Details of states for NS_RF_OF_BLEACH2 named set are as follows,

Named Set Value	Named Set Name
00	Hold
01	Production

Table 60: Name Set State Details

8 EQUIPMENT MODULE CLASS DESIGN – EM_RF_OF_BLEACH3 (2-T503)

8.1 Functional Description

EM_RF_OF_BLEACH3 is a class based equipment module that performs filling operation to produce the material and afterwards drains the material. EM_RF_OF_BLEACH3 is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. Initially Operator shall be provided only with Hold and Production. Hold Request shall be enabled for all acquired devices at the end of Hold command.

8.2 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name
EM	EM_RF_OF_BLEACH3	RF2EQ_T503
DEV1	_MRSS11_130_SE	RF2_AG2003A
DEV2	_V12_IP_130_SE	RF2_V2003F
DEV3	_V12_IP_130_SE	RF2_V2003BY
DEV4	EM_RF_OF_BLEACH6	RF2EQ_635

Table 61: Instances

8.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

8.3.1 Command 00 – Hold

Step	Description	Action	Action Description
H0010	Command Hold Start	A0010	Store Fail Message
		A0020	Reset Hold Request
		A0030	Save Previous Step Time
T1000	Actions Complete or Time Out		Go to step H1000
H1000	Hold Command 1	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Closing DEV2
		A0030	Closing DEV3
T9900A	Actions Complete or		Go to step H9900A

Step	Description	Action	Action Description
	Time Out and Hold=True		
T9900B	Actions Complete or Time Out and Hold=False		Go to step H9900B
H9900A	Command Hold End	A0010	Completed
END	Command Complete		True
H9900B	Release Devices	A0010	Release Devices Message
		A0020	Release Private Device
		A0030	Release Private Device
		A0040	Release Private Device
		A0050	Reset Parameters
END1	Command Complete		True

Table 62: Hold Sequence

8.3.2

Command 01 – Production

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Production Command
T0010	First Pass = False		If First Pass = False then go to step S0010
T0020A	First Pass = True		If First Pass = True then go to step S0020
S0010	Initializations	A0010	First Pass Logic
		A0020	First Pass = True
		A0030	Set Task Pointer
T0020B	Actions Complete		If no pending actions in S0010 then go to step S0020
S0020	Acquire Devices	A0010	Acquire Devices Message
		A0020	Acquire Private Device
		A0030	Acquire Private Device
		A0040	Acquire Private Device
		A0050	Reset Hold Request
T0100	Standby		If Task Pointer = 1 then go to step S0100
T0200B	Fill		If Task Pointer = 2 then go to step S0400
T0300B	Production		If Task Pointer = 3 then go to step S0500
T0400B	Drain		If Task Pointer = 4 then go to step S0600

Step	Description	Action	Action Description
T0500B	Drain Wait		If Task Pointer = 5 then go to step S0700
S0100	Standby	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Closing DEV2
		A0030	Closing DEV3
		A0040	Set Message: Waiting for EM_RF_OF_BLEACH2 to be in Step Production
T0200A	Fill		If EM_BLEACH2 is in step <Production> then go to step S0200
S0200	Fill	A0010	Enabling Conditional Logic for DEV1 (TP01)
		A0020	Opening DEV2
		A0030	Closing DEV3
		A0040	Set Message
T0300A	Production		If LSL_T503 is not in alarm then go to step S0300
S0300	Production	A0010	Enabling Conditional Logic for DEV1 (TP02)
		A0020	Opening DEV2
		A0030	Closing DEV3
		A0040	Acquire EM_RF_OF_BLEACH5
		A0060	Starting Production command for EM_RF_OF_BLEACH5
		A0070	Acquire DEV6
		A0075	Start DEV6
		A0080	Set Blow time
		A0090	Start Blow Timer
		A0100	Blow time Message
T0400A	Drain		If EM_T506 is in step <Drain Wait> and XV 2-T506-B blowing timer (TP001) elapsed then go to step S0400
S0400	Drain	A0010	Enabling Conditional Logic for DEV1 (TP02)
		A0020	Opening DEV2
		A0030	Opening DEV3
		A0040	Set Message: Waiting for LSL_003 to be in alarm
		A0050	Set Scratch Time
T0500A	Drain Wait		If 30 secs delay time completed then go to step S0500
S0500	Drain Wait	A0010	Stopping and Disabling Conditional Logic for DEV1

Step	Description	Action	Action Description
		A0020	Closing DEV2
		A0030	Closing DEV3
		A0040	Set Drain 635A & 635B preset time
		A0050	Start Drain 635A & 635B preset Timer
		A0060	Blow time Message
T0600	Disable HM		If Drain 635 A&B timer (TP002) is elapsed then go to step S0600
S0600	Disable HM	A0010	Disable HM 01 (Module Ownership Lost)
T0700	Release Devices		If no pending actions in S0600 then go to step S0700
S0700	Release Devices	A0010	Release Devices Message
		A0020	Release Private Device
		A0030	Release Private Device
		A0040	Release Private Device
T9900	Complete		If no pending actions in S0700 then go to step S9900
S9900	Complete	A0010	Set Message: Production Command Complete
		A0020	Self release the EM
END	Command Complete		True

Table 63: Production Sequence

8.4

Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value/Reference
Operating Parameters				
NA				
Tuning Parameters				
TP001	Blowing Preset Time	Floating Point	sec	5
TP002	Drain 635A & 635B preset Time (sec)	Floating Point	sec	20

Table 64: Operating and Tuning Parameters

8.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

Parameter Name	Parameter Type	Value/Reference
MY_TAG		RF2EQ_T503
OPR_CMD_MASK		3
LSL_T503	String	RF2_LSL2003/DI1/DISC_ACT
MX_004AC	String	RF2_MX2004AC/PV_D
LSL_T503_CM_NAME	String	RF2_LSL2003
DY_REF_1	Dynamic Reference	
HOLD	External Reference	RF2EQ_STARTUP3/OP005_VALUE
EM_BLEACH2	String	RF2EQ_T506
EM_BLEACH5	String	RF2EQ_622_21
BLEACH5	External Reference	RF2EQ_622_21/OWNER_ID
STEP_TIME	Floating Point	0
COMPLETED_TIME	Floating Point	0
RESET_COMP_TM	Boolean	False
EM_T506	External Reference	RF2EQ_T506/COMMAND_00001/S0500/ACTIVE

Table 65: Configurable Parameters

8.6 Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_RF_OF_BLEACH3	Refinery Area: Filtration and Vacuum Production

Table 66: Named Set Details

Details of states for NS_RF_OF_BLEACH3 named set are as follows,

Named Set Value	Named Set Name
00	Hold
01	Production

Table 67: Name Set State Details

9 EQUIPMENT MODULE CLASS DESIGN – EM_RF_OF_BLEACH4 (2-606A1~A6)

9.1 Functional Description

EM_RF_OF_BLEACH4 is a class based equipment module that performs filling operation after evacuation and then discharge the material. EM_RF_OF_BLEACH4 is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. Initially Operator shall be provided only with Hold and Fill commands. Discharge command will enable while execution of filling operation. Hold Request shall be enabled for all acquired devices at the end of Hold command.

9.2 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name					
EM	EM_RF_OF_BLEACH4	RF2E Q_606 A1	RF2E Q_606 A2	RF2E Q_606 A3	RF2E Q_606 A4	RF2E Q_606 A5	RF2E Q_606 A6
DEV1	_V12_IP_130_SE	RF2_V 2606A 1F	RF2_V 2606A 2F	RF2_V 2606A 3F	RF2_V 2606A 4F	RF2_V 2606A 5F	RF2_V 2606A 6F
DEV2	_V12_IP_130_SE	RF2_V 2606A 1D	RF2_V 2606A 2D	RF2_V 2606A 3D	RF2_V 2606A 4D	RF2_V 2606A 5D	RF2_V 2606A 6D
DEV3	_V12_IP_130_SE	RF2_V2630A12					
DEV4	_V12_IP_130_SE	RF2_V2630A34					

Table 68: Instances

9.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

9.3.1 Command 00 – Hold

Step	Description	Action	Action Description
H0010	Start	A0010	Store Fail Message
		A0020	Reset Hold Request
		A0030	Save Previous Step Time

Step	Description	Action	Action Description
T1000	Actions Complete or Time Out		Go to step H1000
H1000	Hold Command 1	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
T9900A	Actions Complete or Time Out & Hold=True		Go to step H9900A
T9900B	Actions Complete or Time Out & Hold=False		Go to step H9900B
H9900A	Command Hold End	A0010	Completed
END	Command Complete		True
H9900B	Release Devices	A0010	Release Devices Message
		A0020	Release Private Device
		A0030	Release Private Device
		A0040	Release Shared Device
		A0050	Release Shared Device
END1	Command Complete		True

Table 69: Hold Sequence

9.3.2

Command 01 – Fill

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Fill Command
		A0020	Set OWNER_ID
T0010	First Pass = False		If First Pass = False then go to step S0010
T0020A	First Pass = True		If First Pass = True then go to step S0020
S0010	Initializations	A0010	First Pass Logic
		A0020	First Pass = True

Step	Description	Action	Action Description
		A0030	Set Task Pointer
T0020B	Actions Complete		If no pending actions in S0010 then go to step S0020
S0020	Acquire Devices	A0010	Acquire Devices Message
		A0020	Acquire Private Device
		A0030	Acquire Private Device
		A0040	Acquire Private Device
		A0050	Acquire Private Device
		A0060	Reset Hold Request
T0100	Standby		If Task Pointer = 1 then go to step S0100
T0200B	Evacuate		If Task Pointer = 2 then go to step S0200
T0300B	Fill		If Task Pointer = 3 then go to step S0300
S0100	Standby	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
T0200A	Evacuate		If no pending actions in S0100 then go to step S0200
S0200	Evacuate	A0010	Enabling Conditional Logic for DEV2
		A0020	Integrate Quantity Dosed
		A0030	Set Preset Evacuate time
		A0040	Start Preset Evacuate Timer
		A0050	Preset Evacuate time Message
		A0060	Increment Arms in Discharge Counter by 1
T0300A	Fill		If Preset Evacuate time (TP002) is over then go to step S0300
S0300	Fill	A0010	Closing and Disabling Conditional Logic for DEV2
		A0020	Enabling Conditional Logic for DEV1
		A0030	Enabling Conditional Logic for DEV4
		A0040	Enabling Conditional Logic for DEV3
		A0050	Set Preset Filling time
		A0060	Start Preset Filling Timer
		A0070	Preset Waiting Fast time Message
T09900	Complete		If Preset Fill time (TP003) is over then go to step S9900

Step	Description	Action	Action Description
S9900	Complete	A0010	Set Message: Fill Command Complete
END	Command Complete		True

Table 70: Fill Sequence

9.3.3 Command 02 – Discharge

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Discharge Command
		A0020	Reset Hold Request
T0100	Wait		If Task Pointer = 3 then go to step S0100
T0200B	Discharge		If Task Pointer = 4 then go to step S0200
S0100	Wait	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Set Preset Waiting time
		A0060	Start Preset Waiting Timer
		A0070	Preset Waiting Fast time Message
T0200A	Discharge		If Preset Wait time (TP005) is over and DEMAND_DOSING > TP008 then go to step S0200
S0200	Discharge	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Enabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Integrate Quantity Dosed
		A0060	Set Preset Discharge time
		A0070	Start Preset Discharge Timer
		A0080	Preset Discharge time Message
		A0090	Decrement Arms in Discharge Counter by 1
T0300	Close Devices		If Preset Discharge time (TP006) is over then go to step S0300

Step	Description	Action	Action Description
S0300	Close Devices	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
S0400	Disable HM	A0010	Disable HM 01 (Module Ownership Lost)
T0500	Close Devices		If no pending actions in S0400 then go to step S0500
S0500	Release Devices	A0010	Release Devices Message
		A0020	Release Private Device
		A0030	Release Private Device
		A0040	Release Shared Device
		A0050	Release Shared Device
T9900	Close Devices		If no pending actions in S0500 then go to step S9900
S9900	Complete	A0010	Set Message: Discharge Command Complete
		A0020	Self Release the EM
END	Command Complete		True

Table 71: Discharge Sequence

9.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value/Reference
Operating Parameters				
OP001	Flow	Floating Point	kg/h	62500
OP002	Dosing aid	Floating Point	%	False
OP003	Dosing Arm	String	NA	606A
OP004	Mode	Boolean	NA	MAN = 0 (AUTO = 1)
OP005	Manual Set Point	Floating Point	kg/h	0
OP006	Actual Step	Floating Point		0
OP007	Remain Time	Floating Point	sec	0
OP008	Dosed Quantity	Floating Point	kg	100

Parameter	Description	Type	Units	Default Value/Reference
OP009	Reset dosed quantity	Named Set	NA	NO
Tuning Parameters				
TP001	Delay Timer	Floating Point	sec	5
TP002	Preset Evacuate Time	Floating Point	sec	5
TP003	Preset Filling Time	Floating Point	sec	10
TP004	Preset Waiting Time Slow	Floating Point	sec	5
TP005	Preset Waiting Time Fast	Floating Point	sec	2
TP006	Discharge Time	Floating Point	sec	10
TP007	Dosing (Ax)	Floating Point	%	0.8
TP008	Delta Demand	Floating Point	kg	5
TP009	Volume 606A	Floating Point	kg	4
TP010	Call Demand/hour	Floating Point	kg/h	500
TP011	Call Demand	Floating Point	kg	0.2
TP012	Total Dosed	Floating Point	kg	234
TP013	Volume 606Bx	Floating Point	kg	0
TP014	Demand Limit	Floating Point	NA	0

Table 72: Operating and Tuning Parameters

9.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

Parameter Name	Parameter Type	Value/Reference					
MY_TAG		RF2 EQ_ 606A 1	RF2 EQ_ 606A 2	RF2 EQ_ 606A 3	RF2 EQ_ 606A 4	RF2 EQ_ 606A 5	RF2 EQ_ 606A 6
OPR_CMD_MASK		7					
DEMAND_DOSIN G	Floating Point						
DEMAND_IN_KG_ SEC	Floating Point						
HOLD		RF2EQ_STARTUP3/OP005_VALUE					
STEP_TIME	Floating Point	0					

Parameter Name	Parameter Type	Value/Reference
COMPLETED_TIME	Floating Point	0
RESET_COMP_TM	Boolean	False
EM_BLEACH3	External Reference	RF2EQ_T503/COMMAND_00001/ACT_STEPS
OWNER	String	RF2EQ_T503

Table 73: Configurable Parameters

9.6 Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_RF_OF_BLEACH4	Refinery Area: Filtration and Vacuum Production

Table 74: Named Set Details

Details of states for NS_RF_OF_BLEACH4 named set are as follows,

Named Set Value	Named Set Name
00	Hold
01	Fill
02	Discharge

Table 75: Name Set State Details

10 EQUIPMENT MODULE CLASS DESIGN – EM_RF_OF_BLEACH5 (2-622/21)

10.1 Functional Description

EM_RF_OF_BLEACH5 is a class based equipment module that performs filling operation to produce the material and afterwards drains the material. EM_RF_OF_BLEACH5 is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. Initially Operator shall be provided only with Hold and Production. Drain command shall be enabled during the execution of Production State. Hold Request shall be enabled for all acquired devices at the end of Hold command.

10.2 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name
EM	EM_RF_OF_BLEACH5	RF2EQ_622_21
DEV1	EM_RF_FILTRATION	RF2EQ_616A1
DEV2	EM_RF_FILTRATION	RF2EQ_616A2
DEV3	EM_RF_FILTRATION	RF2EQ_616A3
DEV4	EM_RF_FILTRATION	RF2EQ_616A4
DEV5	EM_RF_FILTRATION	RF2EQ_616A5
DEV6	_V12_IP_130_SE	RF2_V2616A15
DEV7	_V12_IP_130_SE	RF2_V2622C
DEV8	_V12_IP_130_SE	RF2_V2622B
DEV9	_V12_IP_130_SE	RF2_V2622A
DEV10	_CSD_130_SE	RF2_TT2622A

Table 76: Instances

10.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

10.3.1 Command 00 – Hold

Step	Description	Action	Action Description
H0010	Command Hold Start	A0010	Store Fail Message
		A0020	Reset Hold Request
		A0030	Save Previous Step Time

Step	Description	Action	Action Description
T1000	Actions Complete or Time Out		Go to step H1000
H1000	Hold Command 1	A0010	Closing DEV6
		A0020	Closing DEV7
		A0030	Closing and Disabling Conditional Logic for DEV8
		A0040	Closing DEV9
		A0050	Set DEV10 Mode to Remote Out
		A0060	Set DEV10 output to Closed
		A0070	Step Name
T9900A	Actions Complete or Time Out and Hold=True		Go to step H9900A
T9900B	Actions Complete or Time Out and Hold=False		Go to step H9900B
H9900A	Command Hold End	A0010	Completed
END	Command Complete		True
H9900B	Release Devices	A0010	Release Devices Message
		A0020	Release Private Device
		A0030	Release Private Device
		A0040	Release Private Device
		A0050	Release Private Device
		A0060	Release Private Device
		A0070	Release Private Device
		A0080	Release Private Device
		A0090	Release Private Device
		A0100	Release Private Device
		A0110	Release Private Device
		A0120	Reset Parameters
END1	Command Complete		True

Table 77: Hold Sequence

10.3.2 Command 01 – Production

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Production Command
T0010	First Pass = False		If First Pass = False then go to step S0010
T0020A	First Pass = True		If First Pass = True then go to step S0020
S0010	Initializations	A0010	First Pass Logic
		A0020	First Pass = True
		A0030	Set Task Pointer
		A0040	Step Name
T0020B	Actions Complete		If no pending actions in S0010 then go to step S0020
S0020	Acquire Devices	A0010	Acquire Devices Message
		A0020	Acquire Private Device
		A0030	Acquire Private Device
		A0040	Acquire Private Device
		A0050	Acquire Private Device
		A0060	Acquire Private Device
		A0070	Reset Hold Request
T0100	Standby		If Task Pointer = 1 then go to step S0100
T0200B	Fill		If Task Pointer = 2 then go to step S0200
T0300B	Production		If Task Pointer = 3 then go to step S0300
T0400B	Pre Drain		If Task Pointer = 4 then go to step S0400
T0500B	Final Recirculation		If Task Pointer = 5 then go to step S0500
S0100	Standby	A0010	Closing DEV6
		A0020	Closing DEV7
		A0030	Closing and Disabling Conditional Logic for DEV8
		A0040	Closing DEV9
		A0050	Set DEV10 Mode to Remote Out
		A0060	Set DEV10 output to Closed
T0200A	Fill		If no pending actions in S0100 then go to step S0200
S0200	Fill	A0010	Closing DEV6
		A0020	Closing DEV7
		A0030	Closing and Disabling Conditional Logic for DEV8
		A0040	Closing DEV9

Step	Description	Action	Action Description
		A0050	Set DEV10 Mode to Remote Cascade
		A0060	Set DEV10 setpoint to Remote Setpoint for TT_622A
		A0070	Step Name
T0300A	Production		If LSH_622_21 is in alarm and any one 616A Filter in step Standby ON (STANDBY_C > 0) then go to step S0300
S0300	Production		
		A0010	Closing DEV6
		A0020	Opening DEV7
		A0030	Enabling Conditional Logic for DEV8
		A0040	Closing DEV9
		A0050	Set DEV10 Mode to Remote Cascade
		A0060	Set DEV10 setpoint to Remote Setpoint for TT_622A
		A0070	Acquire Filters 1
		A0080	Starting Fill command for DEV1
		A0090	Acquire Filters 2
		A0100	Starting Fill command for DEV2
		A0110	Acquire Filters 3
		A0120	Starting Fill command for DEV3
		A0130	Acquire Filters 4
		A0140	Starting Fill command for DEV4
		A0150	Acquire Filters 5
		A0160	Starting Fill command for DEV5
		A0170	Set Message
		A0180	Step Name
T0400A	Pre Drain		If EM_BLEACH3 is in step <Drain Wait> and any one filter is in <Filtration> (FILTRATION_C > 0) then go to step S0400
S0400	Pre Drain		
		A0010	Closing DEV6
		A0020	Opening DEV7
		A0030	Enabling Conditional Logic for DEV8
		A0040	Closing DEV9
		A0050	Set DEV10 Mode to Remote Out
		A0060	Set DEV10 output to Closed
		A0070	Step Name
T0500A	Final Recirculation		If Final Recirculation Level Reached (LT_622_PV >= TP003) and no filter is in step <DRAIN TO 622> (DRAIN_TO_622C) & number of filters is in <Filtration> is not > 1 (FILTRATION_C) and number of filters is in

Step	Description	Action	Action Description
			step <Fill> = 0 (FILL_C) then go to step S0500
S0500	Final Recirculation	A0010	Closing DEV6
		A0020	Opening DEV7
		A0030	Opening and Disabling Conditional Logic for DEV8
		A0040	Closing DEV9
		A0050	Set DEV10 Mode to Remote Out
		A0060	Set DEV10 output to Closed
		A0070	Step Name
T9900	Complete		If no filter is in step <Final Recirculation> and any one filter is in step <Final Filtration> then go to step S9900
S9900	Complete	A0010	Set Message: Production Command Complete
END	Command Complete		True

Table 78: Production Sequence

10.3.3

Command 02 – Drain

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Drain Command
T0100	Drain		If Task Pointer = 5 then go to step S0100
T0200B	Fill from Filter 616Ax		If Task Pointer = 6 then go to step S000
T0300B	Empty		If Task Pointer = 7 then go to step S0500
T0400B	Blow		If Task Pointer = 8 then go to step S0600
T0500B	Drain Wait		If Task Pointer = 9 then go to step S0700
S0100	Drain	A0010	Closing DEV6
		A0020	Opening DEV7
		A0030	Opening and Disabling Conditional Logic for DEV8
		A0040	Closing DEV9
		A0050	Set DEV10 Mode to Remote Out
		A0060	Set DEV10 output to Closed
		A0070	Check if LT_622 is active
		A0080	Set Message: Adding 10 secs delay
T0200A	Fill from Filter 616Ax		If 10 seconds completed then go to step S0200

Step	Description	Action	Action Description
S0200	Fill from Filter 616Ax	A0010	Closing DEV6
		A0020	Opening DEV7
		A0030	Opening and Disabling Conditional Logic for DEV8
		A0040	Closing DEV9
		A0050	Set DEV10 Mode to Remote Out
		A0060	Set DEV10 output to Closed
T0300A	Empty		If no filter is in step <DRAIN TO 622> (DRAIN_TO_622C) or <Drying> (DRYING_C) <Sec. Drain> (SEC_DRAIN_C) or <Depress.> (DEPRESS_C) and 682B in <Standby> and Empty Confirmation Button (OP001) is Pressed then go to step S0300
S0300	Empty	A0010	Opening DEV6
		A0020	Opening DEV7
		A0030	Opening and Disabling Conditional Logic for DEV8
		A0040	Closing DEV9
		A0050	Set DEV10 Mode to Remote Out
		A0060	Set DEV10 output to Closed
		A0070	Check if LSL_622 is active
		A0080	Set Message: Adding 10 secs delay
		A0090	Step Name
		A0100	Enabling Conditional Logic for PU_6225
T0400A	Blow		If no action pending in S0300 then go to step S0400
S0400	Blow	A0010	Opening DEV6
		A0020	Closing DEV7
		A0030	Opening and Disabling Conditional Logic for DEV8
		A0040	Opening DEV9
		A0050	Set DEV10 Mode to Remote Out
		A0060	Set DEV10 output to Closed
		A0070	Step Name
		A0080	Set Preset Blow time
		A0090	Start Preset Blow Timer
		A0100	Preset Blow time Message
T0500A	Drain Wait		If XV 2-622/21-BS Blowing timer elapsed (TP005) and 5 secs completed then go to step S0500
S0500	Drain Wait	A0010	Opening DEV6

Step	Description	Action	Action Description
		A0020	Closing DEV7
		A0030	Opening and Disabling Conditional Logic for DEV8
		A0040	Closing DEV9
		A0050	Set DEV10 Mode to Remote Out
		A0060	Set DEV10 output to Closed
		A0070	Disable VPRD1
T0600	Close Devices		If 5 secs (TP001) completed then go to step S0600
S0600	Close Devices	A0010	Closing DEV6
		A0020	Closing DEV7
		A0030	Closing and Disabling Conditional Logic for DEV8
		A0040	Closing DEV9
		A0050	Set DEV10 Mode to Remote Out
		A0060	Set DEV10 output to Closed
T0700	Disable HM		If no pending actions in S0600 then go to step S0700
S0700	Disable HM	A0010	Disable HM 01 (Module Ownership Lost)
T0800	Release Devices		If no pending actions in S0700 then go to step S0800
S0800	Release Devices	A0010	Release Devices Message
		A0020	Release Private Device
		A0030	Release Private Device
		A0040	Release Private Device
		A0050	Release Private Device
		A0060	Release Private Device
		A0070	Release Private Device
		A0080	Release Private Device
		A0090	Release Private Device
		A0100	Release Private Device
		A0110	Release Private Device
T9900	Complete		If no pending actions in S0800 then go to step S9900
S9900	Complete	A0010	Set Message: Drain Command Complete
		A0020	Self release the EM
END	Command Complete		True

Table 79: Drain Sequence

10.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value/Reference
Operating Parameters				
OP001	Empty Confirmation Button	Named Set	NA	NO
Tuning Parameters				
TP001	Delay Timer	Floating Point	sec	5
TP002	Safe Drain level	Floating Point	sec	70
TP003	Final Recirculation level	Floating Point	%	5
TP004	Vacuum	Floating Point	mbar	100
TP005	Blow Time for XV_622/21-BS	Floating Point	sec	0
TP006	Remote Setpoint for TCV_622/21	Floating Point	°C	105

Table 80: Operating and Tuning Parameters

10.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

Parameter Name	Type	Value/Reference
MY_TAG	String	RF2EQ_622_21
OPR_CMD_MASK		7
LSH_622_21	External Reference	RF2_LSH2622/DI1/DISC_ACT
LT_622_PV	External Reference	RF2_LT2622/LO_LO_ALM
LT_622_ALM	External Reference	RF2_LT2622/PV
LSLL_622	External Reference	RF2_LSLL2622/DI1/DISC_ACT
EM_BLEACH3	String	RF2EQ_T503
EM_VPRD2	External Reference	RF2EQ_682B/OWNER_ID
HOLD	External Reference	RF2EQ_STARTUP3/OP005_VALUE
STEP_NAME	String	
STANDBY_C	External Reference	RF2_MISC/STANDBY_C
FILTRATION_C	External Reference	RF2_MISC/FILTRATION_C
DRAIN_TO_622C	External Reference	RF2_MISC/DRAIN_TO_622_C
FILL_C	External Reference	RF2_MISC/FILL_C

Parameter Name	Type	Value/Reference
F_FILTRATION_C	External Reference	RF2_MISC/F_FILTRATION_C
F_RECIRCULATN_C	External Reference	RF2_MISC/F_RECIRCULATN_C
DRYING_C	External Reference	RF2_MISC/DRYING_C
SEC_DRAIN_C	External Reference	RF2_MISC/SEC_DRAIN_C
DEPRESS_C	External Reference	RF2_MISC/DEPRESS_C
STEP_TIME	Floating Point	0
COMPLETED_TIME	Floating Point	0
RESET_COMP_TM	Boolean	False
PU_6225	External Reference	

Table 81: Configurable Parameters

10.6 Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_RF_OF_BLEACH5	Refinery Area: Filtration and Vacuum Production

Table 82: Named Set Details

Details of states for NS_RF_OF_BLEACH5 named set are as follows,

Named Set Value	Named Set Name
00	Hold
01	Production
02	Drain

Table 83: Name Set State Details

11 EQUIPMENT MODULE CLASS DESIGN – EM_RF_FILTRATION (2-616A1-5)

11.1 Functional Description

EM_RF_FILTRATION is a class based equipment module that performs filtration and final recirculation and afterwards discharge the material. EM_RF_FILTRATION is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. Initially Operator shall be provided only with Hold and Fill. Filtration and Discharge command shall be enabled during the execution of Fill State. Hold Request shall be enabled for all acquired devices at the end of Hold command. Commands which require operator intervention shall be configured as Operator Prompts wherever possible.

11.2 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name				
EM	EM_RF_FILTRATION	RF2EQ_6 16A1	RF2EQ_6 16A2	RF2EQ_6 16A3	RF2EQ_6 16A4	RF2EQ_6 16A5
DEV1	_MRSS1 1_130_SE	RF2_PU2 6221	RF2_PU2 6222	RF2_PU2 6223	RF2_PU2 6224	RF2_PU2 6225
DEV2	_V12_IP_ 130_SE	RF2_V26 16A1A	RF2_V26 16A2A	RF2_V26 16A3A	RF2_V26 16A4A	RF2_V26 16A5A
DEV3	_V12_IP_ 130_SE	RF2_V26 16A1C	RF2_V26 16A2C	RF2_V26 16A3C	RF2_V26 16A4C	RF2_V26 16A5C
DEV4	_V12_IP_ 130_SE	RF2_V26 16A1D	RF2_V26 16A2D	RF2_V26 16A3D	RF2_V26 16A4D	RF2_V26 16A5D
DEV5	_V12_IP_ 130_SE	RF2_V26 16A1F	RF2_V26 16A2F	RF2_V26 16A3F	RF2_V26 16A4F	RF2_V26 16A5F
DEV6	_V12_IP_ 130_SE	Ignored	Ignored	Ignored	Ignored	Ignored
DEV7	_V12_IP_ 130_SE	RF2_V26 16A1G	RF2_V26 16A2G	RF2_V26 16A3G	RF2_V26 16A4G	RF2_V26 16A5G
DEV8	_V12_IP_ 130_SE	RF2_V26 16A1H	RF2_V26 16A2H	RF2_V26 16A3H	RF2_V26 16A4H	RF2_V26 16A5H
DEV9	_V12_IP_ 130_SE	RF2_V26 16A1I	RF2_V26 16A2I	RF2_V26 16A3I	RF2_V26 16A4I	RF2_V26 16A5I
DEV10	_V12_IP_ 130_SE	RF2_V26 16A1J	RF2_V26 16A2J	RF2_V26 16A3J	RF2_V26 16A4J	RF2_V26 16A5J
DEV11	_V12_IP_ 130_SE	RF2_V26 16A1K	RF2_V26 16A2L	RF2_V26 16A3L	RF2_V26 16A4K	RF2_V26 16A5L
DEV12	_V12_IP_ 130_SE	RF2_V26 16A1L	RF2_V26 16A2M	RF2_V26 16A3M	RF2_V26 16A4L	RF2_V26 16A5M
DEV13	_V12_IP_ 130_SE	RF2_V26 16A1M	RF2_V26 16A2K	RF2_V26 16A3K	RF2_V26 16A4M	RF2_V26 16A5K



Device Name	Module Class	Instance Name				
DEV14	_V12_IP_130_SE	RF2_V2616A1B	RF2_V2616A2B	RF2_V2616A3B	RF2_V2616A4B	RF2_V2616A5B
DEV15	_V12_IP_130_SE	RF2_V2616A1E	RF2_V2616A2E	RF2_V2616A3E	RF2_V2616A4E	RF2_V2616A5E
DEV16	_MRSS11_130_SE	Ignored				
DEV17	_V12_IP_130_SE	RF2_V2623				

Table 84: Instances

11.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

11.3.1 Command 00 – Hold

Step	Description	Action	Action Description
H0010	Start	A0010	Store Fail Message
		A0020	Reset Hold Request
T1000	Actions Complete or Time Out		Go to step H1000
H1000	Hold Command 1	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11

Step	Description	Action	Action Description
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Closing and Disabling Conditional Logic for DEV14
		A0150	Closing and Disabling Conditional Logic for DEV15
		A0160	Stopping and Disabling Conditional Logic for DEV16
		A0170	Closing DEV17
T9900A	Actions Complete or Time Out & Hold=True		Go to step H9900A
T9900B	Actions Complete or Time Out & Hold=False		Go to step H9900B
H9900A	Command Hold End	A0010	Completed
END	Command Complete		True
H9900B	Release Devices	A0010	Release Devices Message
		A0020	Release Private Devices
		A0030	Release Private Devices
		A0040	Release Private Devices
		A0050	Release Private Devices
		A0060	Release Private Devices
		A0070	Release Private Devices
		A0080	Release Private Devices
		A0090	Release Private Devices
		A0100	Release Private Devices
		A0110	Release Private Devices
		A0120	Release Private Devices
		A0130	Release Private Devices
		A0140	Release Private Devices
		A0150	Release Private Devices
		A0160	Release Private Devices
		A0170	Reset Parameters
END1	Command Complete		True

Table 85: Hold Sequence

11.3.2 Command 01 – Fill

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Fill Command
T0010	First Pass = False		If First Pass = False then go to step S0010
T0020A	First Pass = True		If First Pass = True then go to step S0020
S0010	Initializations	A0010	First Pass Logic
		A0020	First Pass = True
		A0030	Set Task Pointer
T0020B	Actions Complete		If no pending actions in S0010 then go to step S0020
S0020	Acquire Devices	A0010	Acquire Devices Message
		A0020	Acquire Private Devices
		A0030	Acquire Private Devices
		A0040	Acquire Private Devices
		A0050	Acquire Private Devices
		A0060	Acquire Private Devices
		A0070	Acquire Private Devices
		A0080	Acquire Private Devices
		A0090	Acquire Private Devices
		A0100	Acquire Private Devices
		A0110	Acquire Private Devices
		A0120	Acquire Private Devices
		A0130	Acquire Private Devices
		A0140	Acquire Private Devices
		A0150	Acquire Private Devices
		A0160	Acquire Private Devices
		A0170	Reset Hold Request
T0100B	Standby Off		If Task Pointer = 1 then go to step S0100
T0200B	Standby On		If Task Pointer = 2 then go to step S0200
T0300B	Close V9		If Task Pointer = 3 then go to step S0300
T0400B	Close V9		If Task Pointer = 4 then go to step S0400
T0500B	Filling		If Task Pointer = 5 then go to step S0500
T0600B	Fillinf from 604C		If Task Pointer = 6 then go to step S0600
S0100	Standby Off	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing DEV3

Step	Description	Action	Action Description
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Closing and Disabling Conditional Logic for DEV14
		A0150	Closing and Disabling Conditional Logic for DEV15
		A0160	Stopping and Disabling Conditional Logic for DEV16
		A0170	Closing DEV17
		A0180	Operator Prompt
T0200A	Standby On		If (Auto Mode selected) or (Manual Mode and <Standby On> selected) then go to step S0200
S0200	Standby On	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Opening DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10

Step	Description	Action	Action Description
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Closing and Disabling Conditional Logic for DEV14
		A0150	Closing and Disabling Conditional Logic for DEV15
		A0160	Stopping and Disabling Conditional Logic for DEV16
		A0170	Closing DEV17
		A0180	Operator Prompt
T0100B	Standby Off		If Manual mode (OP001) & step <Standby Off> selected then go to step S0100
T0300A	Close V9		If Auto mode (OP001) selected and no. of filters in step <Fill> = 0 (FILL_C) and no. of filters in step <Filtration> < 4 (FILTRATION_C) and EM_BLEACH5 is in step <Production> and Fill From 2-604C button (OP002) is not pressed then go to step S0300
T0400A	Close V9		If Auto mode (OP001) selected and no. of filters in step <Fill> = 0 (FILL_C) and no. of filters in step <Filtration> < 4 (FILTRATION_C) and EM_BLEACH5 is in step <Production> and Fill From 2-604C button (OP002) is pressed then go to step S0400
S0300	Close V9	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Opening DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10

Step	Description	Action	Action Description
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Closing and Disabling Conditional Logic for DEV14
		A0150	Closing and Disabling Conditional Logic for DEV15
		A0160	Stopping and Disabling Conditional Logic for DEV16
		A0170	Closing DEV17
		A0180	Setting Delay Time
T0500A	Filling		If 10 secs completed (TP002) then go to step S0500
S0400	Close V9	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Opening DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Closing and Disabling Conditional Logic for DEV14
		A0150	Closing and Disabling Conditional Logic for DEV15
		A0160	Stopping and Disabling Conditional Logic for DEV16
		A0170	Closing DEV17

Step	Description	Action	Action Description
		A0180	Setting Delay Time
		A0190	Acquire Shared Device
T0600A	Filling		If 10 secs completed (TP002) then go to step S0600
S0500	Filling	A0010	Enabling Conditional Logic for DEV1
		A0020	Opening and Disabling Conditional Logic for DEV2
		A0030	Opening DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Closing and Disabling Conditional Logic for DEV14
		A0150	Closing and Disabling Conditional Logic for DEV15
		A0160	Stopping and Disabling Conditional Logic for DEV16
		A0170	Closing DEV17
		A0180	Operator Prompt
		A0190	Setting Delay
T9900A	Actions Complete		If (Auto mode (OP001) selected & LSH_616AX is in alarm and 5 secs (TP001)) Or (Manual mode (OP001) selected & <Precoat> selected) then go to step S9900
S0600	Filling from 604C	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Opening DEV3

Step	Description	Action	Action Description
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Opening and Disabling Conditional Logic for DEV14
		A0150	Closing and Disabling Conditional Logic for DEV15
		A0160	Enabling Conditional Logic for DEV16
		A0170	Closing DEV17
		A0180	Operator Prompt
		A0190	Setting Delay
T9900B	Actions Complete		If (Auto mode (OP001) selected & LSH_616AX is in alarm and 5 secs (TP001)) Or (Manual mode (OP001) selected & <Precoat From 1-604C> selected) then go to step S9900
S9900	Complete	A0010	Set Message: Fill Command Complete
END	Command Complete		True

Table 86: Fill Sequence

11.3.3 Command 02 – Filtration

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Filtration Command
T0100	Initializations		If no pending actions in S0000 then go to step S0100
T0200C	Precoat		If Task Pointer = 7 then go to step S0200
T0300B	Precoat from 604C		If Task Pointer = 8 then go to step S0300

Step	Description	Action	Action Description
T0400C	Xfer1		If Task Pointer = 9 then go to step S0400
T0500B	Xfer1		If Task Pointer = 10 then go to step S0500
T0600B	Xfer 2		If Task Pointer = 11 then go to step S0600
T0700B	Xfer 2		If Task Pointer = 12 then go to step S0700
T0800C	Filtration		If Task Pointer = 13 then go to step S0800
TP900B	Final Filtration		If Task Pointer = 14 then go to step S0900
T1000B	Empty 604C		If Task Pointer = 15 then go to step S1000
T1100B	Final Recirculation		If Task Pointer = 16 then go to step S1100
T1200E	Drain to 622		If Task Pointer = 17 then go to step S1200
T1300D	Drying		If Task Pointer = 18 then go to step S1300
T1400C	Sec. Drain		If Task Pointer = 19 then go to step S1400
T1500B	Depressurization		If Task Pointer = 20 then go to step S1500
S0100	Initializations		Empty
T0200A	Precoat		If Task Pointer = 5 then go to step S0200
T0300A	Precoat from 604C		If Task Pointer = 6 then go to step S0300
S0200	Precoat	A0010	Enabling Conditional Logic for DEV1
		A0020	Enabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Enabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Enabling Conditional Logic for DEV13
		A0140	Closing and Disabling Conditional Logic for DEV14
		A0150	Closing and Disabling Conditional Logic for DEV15
		A0160	Stopping and Disabling Conditional Logic for DEV16

Step	Description	Action	Action Description
		A0170	Closing DEV17
		A0180	Operator Prompt
		A0190	Set Precoat time
		A0200	Start Precoat Timer
		A0210	Precoat time Message
		A0220	Set bit "From Precoat"
T0400A	Xfer1		If (Auto mode (OP001) and Precoat time elapsed (TP003)) Or (Manual mode (OP001) and <Filtration> step selected by operator) then go to step S0400
S0300	Precoat from 604C	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Enabling Conditional Logic for DEV13
		A0140	Enabling Conditional Logic for DEV14
		A0150	Enabling Conditional Logic for DEV15
		A0160	Enabling Conditional Logic for DEV16
		A0170	Closing DEV17
		A0180	Operator Prompt
		A0190	Set Precoat time
		A0200	Start Precoat Timer
		A0210	Precoat time Message
		A0220	Set bit "From Precoat From 1-604C"
T0500A	Xfer1		If (Auto Mode (OP001) and Precoat time elapsed (TP013)) Or (Manual mode (OP001)

Step	Description	Action	Action Description
			and <Filtration> step selected by operator) then go to step S0500
S0400	Xfer 1	A0010	Enabling Conditional Logic for DEV1
		A0020	Opening and Disabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Opening and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Opening and Disabling Conditional Logic for DEV13
		A0140	Closing and Disabling Conditional Logic for DEV14
		A0150	Closing and Disabling Conditional Logic for DEV15
		A0160	Stopping and Disabling Conditional Logic for DEV16
		A0170	Closing DEV17
		A0180	Setting Delay Time
T0600A	Xfer 2		If 10 secs (TP002) completed then go to step S0600
S0500	Xfer 1	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6

Step	Description	Action	Action Description
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Opening and Disabling Conditional Logic for DEV13
		A0140	Opening and Disabling Conditional Logic for DEV14
		A0150	Opening and Disabling Conditional Logic for DEV15
		A0160	Enabling Conditional Logic for DEV16
		A0170	Closing DEV17
		A0180	Setting Delay Time
T0700A	Xfer 2		If 10 secs (TP002) completed then go to step S0700
S0600	Xfer 2	A0010	Enabling Conditional Logic for DEV1
		A0020	Opening and Disabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Opening and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Opening and Disabling Conditional Logic for DEV13
		A0140	Closing and Disabling Conditional Logic for DEV14

Step	Description	Action	Action Description
		A0150	Closing and Disabling Conditional Logic for DEV15
		A0160	Stopping and Disabling Conditional Logic for DEV16
		A0170	Closing DEV17
		A0180	Setting Delay Time
T0800A	Filtration		If bit "From Precoat" (FROM_PRECOAT) is set and 10 secs (TP002) completed then go to step S0800
T0900A	Final Filtration		If bit "From Final Recirculation" (FROM_FNL_RECIRC) is set and 10 secs (TP002) completed then go to step S0900
S0700	Xfer 2	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Opening and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Opening and Disabling Conditional Logic for DEV13
		A0140	Opening and Disabling Conditional Logic for DEV14
		A0150	Closing and Disabling Conditional Logic for DEV15
		A0160	Enabling Conditional Logic for DEV16
		A0170	Closing DEV17
		A0180	Setting Delay Time
T0900A	Empty 604C		If bit " From Precoat from 1-604C" (FROM_PRECOAT604C) is set and 10 secs (TP002) completed then go to step S0900

Step	Description	Action	Action Description
S0800	Filtration	A0010	Enabling Conditional Logic for DEV1
		A0020	Enabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Enabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Enabling Conditional Logic for DEV13
		A0140	Closing and Disabling Conditional Logic for DEV14
		A0150	Closing and Disabling Conditional Logic for DEV15
		A0160	Stopping and Disabling Conditional Logic for DEV16
		A0170	Closing DEV17
		A0180	Operator Prompt
		A0190	Reset bit "From Precoat" (FROM_PRECOAT)
		A0200	Release Shared Device
T0200B	Precoat		If Manual mode (OP001) selected and operator select <Precoat> and EM_BLEACH5 is in step <Production> or <Pre Drain> then go to step S0200
T1100A	Final Recirculation		If Auto mode (OP001) selected and EM_BLEACH5 is in step <Final Recirculation> then go to step S1100
T1200A	Drain to 622		If (Auto mode (OP001) selected and Max. BE Qty (MAX_QTY_FILTER) in filter or Max Pressure (PT_616A >= TP012) in filter and EM_BLEACH5 is in step <Production> and LI_622_21 < Safe Level to drain (TP002 of 622/21) and No other filter in step <DRAIN TO 622> (DRAIN_TO_622)) or (Manual mode and step <Drain to 622> selected)

Step	Description	Action	Action Description
			or (Auto mode (OP001) selected and EM_BLEACH5 is in step <Pre Drain> and no. of filters in Filtration (FILTRATION_C) is > 1 and LI_622_21 < Safe Level to drain (TP002 of 622/21) and no other filter in step <Drain to 622> (DRAIN_TO_622)) then go to step S1200
S0900	Final Filtration	A0010	Enabling Conditional Logic for DEV1
		A0020	Enabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Enabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Enabling Conditional Logic for DEV13
		A0140	Closing and Disabling Conditional Logic for DEV14
		A0150	Closing and Disabling Conditional Logic for DEV15
		A0160	Stopping and Disabling Conditional Logic for DEV16
		A0170	Closing DEV17
		A0190	Starting Drain command for EM_RF_OF_BLEACH5
		A0210	Reset bit "From Final Recirculation" (FROM_FNL_RECIRC)
T1200B	Drain to 622		If Auto Mode (OP001) and STR_BLEACH5 is in step <Fill From Filter 616Ax> then go to step S1200
S1000	Empty 604C	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2

Step	Description	Action	Action Description
		A0030	Closing DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Opening and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Opening and Disabling Conditional Logic for DEV14
		A0150	Closing and Disabling Conditional Logic for DEV15
		A0160	Enabling Conditional Logic for DEV16
		A0170	Closing DEV17
		A0180	Set Message: LSL_604C alarm is active and 5 Sec delay.
T0800B	Filtration		If LSL_604C is in alarm and 5 sec (TP001) completed then go to step S0800
S1100	Final Recirculation	A0010	Enabling Conditional Logic for DEV1
		A0020	Enabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Enabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11

Step	Description	Action	Action Description
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Enabling Conditional Logic for DEV13
		A0140	Closing and Disabling Conditional Logic for DEV14
		A0150	Closing and Disabling Conditional Logic for DEV15
		A0160	Stopping and Disabling Conditional Logic for DEV16
		A0170	Closing DEV17
		A0180	Prompt: Acknowledge prompt to proceed with Final Filtration.
		A0190	Set Final Recirculation time
		A0200	Start Final Recirculation Timer
		A0210	Final Recirculation time Message
		A0220	Set bit "From Final Recirculation" (FROM FNL RECIRC)
T0400B	Xfer 1		If (Auto mode (OP001) selected and after Final recirculation preset time (TP004) elapsed) Or (Manual mode (OP001) and <Final Filtration> step selected) then go to step S0400
S1200	Drain to 622	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Opening and Disabling Conditional Logic for DEV6
		A0070	Enabling Conditional Logic for DEV7 (TP01)
		A0080	Opening and Disabling Conditional Logic for DEV8
		A0090	Closing DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Opening and Disabling Conditional Logic for DEV13
		A0140	Closing and Disabling Conditional Logic for DEV14

Step	Description	Action	Action Description
		A0150	Closing and Disabling Conditional Logic for DEV15
		A0160	Stopping and Disabling Conditional Logic for DEV16
		A0170	Closing DEV17
		A0180	Prompt: Acknowledge prompt to proceed with Drying process.
		A0190	Acquire Shared Device
Drying	T1300A		If (Auto Mode Or Manual Mode and <Drying> step selected and LSL_616A is in alarm) then go to step S1300
S1300	Drying	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Enabling Conditional Logic for DEV7 (TP02)
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Opening DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Closing and Disabling Conditional Logic for DEV14
		A0150	Closing and Disabling Conditional Logic for DEV15
		A0160	Stopping and Disabling Conditional Logic for DEV16
		A0170	Opening DEV17
		A0180	Operator Prompt
		A0190	Set Drying time
		A0200	Start Drying Timer
		A0210	Drying time Message
T1200C	Drain to 622		If Manual Mode (OP001) and step <Drain to 622> selected and EM_BLEACH5 is in step

Step	Description	Action	Action Description
			<Production> and LI_622_21< Safe Level to drain (TP002 of 622/21) and No other filter in step <DRAIN TO 622> (DRAIN_TO_622) then go to step S1200
T1400A	Sec. Drain		If (Auto Mode (OP001) and Drying time (TP006) elapsed) Or (Manual mode (OP001) and step <Sec. Drain> selected) then go to step S1400
S1400	Sec. Drain	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Enabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Enabling Conditional Logic for DEV8
		A0090	Closing DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Opening and Disabling Conditional Logic for DEV13
		A0140	Closing and Disabling Conditional Logic for DEV14
		A0150	Closing and Disabling Conditional Logic for DEV15
		A0160	Stopping and Disabling Conditional Logic for DEV16
		A0170	Closing DEV17
		A0180	Operator Prompt
		A0190	Set Security Drain time
		A0200	Start Security Drain Timer
		A0210	Security Drain time Message
		A0220	Release Shared Device
T1200D	Drain to 622		If Manual Mode (OP001) and step <Drain to 622> selected and EM_BLEACH5 is in step <Production> and LI_622_21 < Safe Level to drain (TP002 of 622/21) and no other filter in

Step	Description	Action	Action Description
			step "DRAIN TO 622" (DRAIN_TO_622) then go to step S1200
T1300B	Drying		If Manual mode (OP001) and step <Drying> selected then go to step S1300
T1500A	Depressurisation		If (Auto Mode (OP001) selected and Sec. Drain time (TP007) elapsed and LSL_616A is in alarm) Or (Manual mode (OP001) and step <Depressurisation> selected) then go to step S1500
S1500	Depressurisation	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Opening DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Closing and Disabling Conditional Logic for DEV14
		A0150	Closing and Disabling Conditional Logic for DEV15
		A0160	Stopping and Disabling Conditional Logic for DEV16
		A0170	Closing DEV17
		A0180	Operator Prompt
		A0190	Set Depressurisation time
		A0200	Start Depressurisation Timer
		A0210	Depressurization time Message
T1300C	Drying		If Manual mode (OP001) and step <Drying> selected then go to step S1300

Step	Description	Action	Action Description
T1400B	Sec. Drain		If Manual mode (OP001) and step <Sec. Drain> selected then go to step S1400
T9900	Complete		If (Auto Mode and Depress. time (TP008) elapsed and confirmation button For Open Filter is pressed (OP003)) or (Manual mode and step <Open Filter> selected) then go to step T9900
S9900	Complete	A0010	Set Message: Filtration Command Complete
END	Command Complete		True

Table 87: Filtration Sequence

11.3.4 Command 03 – Discharge

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Discharge Command
		A0020	Reset Open Filter Button (OP003)
T0100	Open Filter		If Task Pointer = 20 then go to step S0100
T0200B	Discharge		If Task Pointer = 21 then go to step S0200
T0300B	Standby Open		If Task Pointer = 22 then go to step S0300
T0400B	Close Filter		If Task Pointer = 23 then go to step S0400
S0100	Open Filter	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Opening DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing DEV9
		A0100	Enabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13

Step	Description	Action	Action Description
		A0140	Closing and Disabling Conditional Logic for DEV14
		A0150	Closing and Disabling Conditional Logic for DEV15
		A0160	Stopping and Disabling Conditional Logic for DEV16
		A0170	Closing DEV17
		A0180	Operator Prompt
T0200A	Discharge		If Auto Mode or Manual mode (OP001) is selected and step <Discharge> selected and DEV10 open feedback is available then go to step S0200
S0200	Discharge	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Opening DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing DEV9
		A0100	Opening and Disabling Conditional Logic for DEV10
		A0110	Enabling Conditional Logic for DEV11
		A0120	Enabling Conditional Logic for DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Closing and Disabling Conditional Logic for DEV14
		A0150	Closing and Disabling Conditional Logic for DEV15
		A0160	Stopping and Disabling Conditional Logic for DEV16
		A0170	Closing DEV17
		A0180	Operator Prompt
		A0190	Set Discharge time
		A0200	Start Discharge Timer
		A0210	Discharge time Message

Step	Description	Action	Action Description
T0300A	Standby Open		If (Auto Mode and Discharge time (TP009) elapsed) or (Manual mode (OP001) and step <Standby Open> selected) then go to step S0300
S0300	Standby Open	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Opening DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing DEV9
		A0100	Opening and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Closing and Disabling Conditional Logic for DEV14
		A0150	Closing and Disabling Conditional Logic for DEV15
		A0160	Stopping and Disabling Conditional Logic for DEV16
		A0170	Closing DEV17
		A0180	Operator Prompt
		A0190	Set Standby Open time
		A0200	Start Standby Open Timer
		A0210	Standby Open time Message
T0400A	Close Filter		If (Auto Mode (OP001) and Standby Open time (TP010) elapsed) or (Manual mode (OP001) and step <Close Filter> selected) then go to step S0400
S0400	Close Filter	A0010	Stopping and Disabling Conditional Logic for DEV1

Step	Description	Action	Action Description
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Opening DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Closing and Disabling Conditional Logic for DEV14
		A0150	Closing and Disabling Conditional Logic for DEV15
		A0160	Stopping and Disabling Conditional Logic for DEV16
		A0170	Closing DEV17
		A0180	Operator Prompt
T0500	Close Devices		If Auto Mode or Manual mode (OP001) is selected and step <Standby Off> selected and DEV10 closed feedback available then go to step S0500
S0500	Close Devices	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7

Step	Description	Action	Action Description
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Closing and Disabling Conditional Logic for DEV14
		A0150	Closing and Disabling Conditional Logic for DEV15
		A0160	Stopping and Disabling Conditional Logic for DEV16
		A0170	Closing DEV17
T0600	Disable HM		If no pending actions in S0500 then go to step S0600
S0600	Disable HM	A0010	Disable HM 01 (Module Ownership Lost)
T0700	Action Complete		If no pending actions in S0600 then go to step S0700
S0700	Release Devices	A0010	Release Devices Message
		A0020	Release Private Devices
		A0030	Release Private Devices
		A0040	Release Private Devices
		A0050	Release Private Devices
		A0060	Release Private Devices
		A0070	Release Private Devices
		A0080	Release Private Devices
		A0090	Release Private Devices
		A0100	Release Private Devices
		A0110	Release Private Devices
		A0120	Release Private Devices
		A0130	Release Private Devices
		A0140	Release Private Devices
		A0150	Release Private Devices
		A0160	Release Private Devices
T9900	Complete		If no pending actions in S0700 then go to step S9900

Step	Description	Action	Action Description
S9900	Complete	A0010	Set Message: Discharge Command Complete
		A0020	Self Release the EM
END	Command Complete		True

Table 88: Discharge Sequence

11.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value/Reference
Operating Parameters				
OP001	MODE	Boolean	NA	MAN = 0 (AUTO = 1)
OP002	FILL FROM 604C	Floating Point	NA	0
OP003	Confirm Open Filter	Named Set	NA	NO
Tuning Parameters				
TP001	Delay Timer	Floating Point	Sec	5
TP002	Delay Timer	Floating Point	Sec	10
TP003	Preset Precoat Time	Floating Point	Sec	400
TP004	Final Recirculation Preset Time	Floating Point	Sec	400
TP005	Delay Timer	Floating Point	Sec	3
TP006	Drying Preset Time	Floating Point	Sec	300
TP007	Security Drain Preset Time	Floating Point	Sec	300
TP008	Depressurization Preset Time	Floating Point	Sec	200
TP009	Discharge Preset Time	Floating Point	Sec	250
TP010	Standby Open Preset Time	Floating Point	Sec	250
TP011	Maximum Bleaching Earth	Floating Point	kg	250
TP012	PI_616A1-5	Floating Point	barg	3.5
TP013	Precoating from 1-604C Preset Time	Floating Point	Sec	400

Table 89: Operating and Tuning Parameters

11.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

Parameter Name	Parameter Type	Value/Reference				
MY_TAG		RF2EQ_616A1	RF2EQ_616A2	RF2EQ_616A3	RF2EQ_616A4	RF2EQ_616A5
OPR_CMD_MASK		15				
LSH_616AX	External Reference	RF2_LS H2616A/DI1/DISC_ACT	RF2_LS H2616B/DI1/DISC_ACT	RF2_LS H2616C/DI1/DISC_ACT	RF2_LS H2616D/DI1/DISC_ACT	RF2_LS H2616E/DI1/DISC_ACT
STR_BLEACH5	String	RF2EQ_622_21				
LSLL_622	External Reference	RF2_LSLL2622/DI1/DISC_ACT				
LSL_604C	External Reference	RF2_LSL2604C/DI1/DISC_ACT				
PT_616A	External Reference	RF2_PT 2616A1/AI1/PV	RF2_PT 2616A2/AI1/PV	RF2_PT 2616A3/AI1/PV	RF2_PT 2616A4/AI1/PV	RF2_PT 2616A5/AI1/PV
LSL_616A	External Reference	RF2_LS L2616A/DI1/DISC_ACT	RF2_LS L2616B/DI1/DISC_ACT	RF2_LS L2616C/DI1/DISC_ACT	RF2_LS L2616D/DI1/DISC_ACT	RF2_LS L2616E/DI1/DISC_ACT
LI_622_21	External Reference	RF2_LT2622/AI1/PV				
EM_RF_FLT_VPRD6	External Reference	RF2EQ_604C/COMMAND_00001/ACT_STEPS				
EM_BLEACH5	External Reference	RF2EQ_622_21/COMMAND_00001/ACT_STEPS				
HOLD	External Reference	RF2EQ_STARTUP3/OP005_VALUE				
FROM_PRE_COAT	Boolean	False				
FROM_FNL_RECIRC	Boolean	False				
FROM_PRE_COAT604C	Boolean	False				
DRAIN_TO_622	External Reference	RF2_MISC/DRAIN_TO_622_C				
STEP_TIME	Floating Point	0				



Parameter Name	Parameter Type	Value/Reference
COMPLETE_D_TIME	Floating Point	0
RESET_COMP_TM	Boolean	False
FILTRATION_C	External Reference	RF2_MISC/FILTRATION_C
FILL_C	External Reference	RF2_MISC/FILL_C
DY_REF_1	Dynamic Reference	
DY_REF_2	Dynamic Reference	
DY_REF_3	Dynamic Reference	
MAX_QTY_FILTER	External Reference	RF2_MISC/MAX_BE_QTY

Table 90: Configurable Parameters

11.6 Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_RF_FILTRATION	Refinery Area: Filtration and Vacuum Production

Table 91: Named Set Details

Details of states for NS_RF_FILTRATION named set are as follows,

Named Set Value	Named Set Name
00	Hold
01	Fill
02	Filtration
03	Discharge

Table 92: Name Set State Details

12 EQUIPMENT MODULE CLASS DESIGN – EM_RF_FLT_VPRD1 (2-616S)

12.1 Functional Description

EM_RF_FLT_VPRD1 is a class based equipment module that performs filtration and vacuum production. This EM performs Filling and Filtration Operation and drains the material based on the Operator command. EM_RF_FLT_VPRD1 is derived from the Command Driven equipment module class _EM_CMD32_130. Refer FDS for the functional design requirements including faceplate and detail display. Initially Operator shall be provided with Hold and Filtration Command. All devices used in EM shall be acquired in Filtration Command and should be released on Completion of Filtration Command. Hold Request shall be enabled for all acquired devices at the end of Hold command.

12.2 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name
EM	EM_RF_FLT_VPRD1	RF2EQ_616S
DEV1	_V12_IP_130_SE	RF2_V261S1
DEV2	_V12_IP_130_SE	RF2_V262S1
DEV3	_V12_IP_130_SE	RF2_V264S1
DEV4	_V12_IP_130_SE	Ignored
DEV5	_V12_IP_130_SE	RF2_V266S1
DEV6	_V12_IP_130_SE	RF2_V269S1
DEV7	_V12_IP_130_SE	RF2_V2610S1
DEV8	_V12_IP_130_SE	RF2_V2682BBY

Table 93: Instances

12.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

12.3.1 Command 00 – Hold

Step	Description	Action	Action Description
H0010	Command Hold Start	A0010	Store Fail Message
		A0020	Reset Hold Request
		A0030	Save Previous Step Time
T1000	Actions Complete or Time Out		Go to step H1000

Step	Description	Action	Action Description
H1000	Hold Command 1	A0010	Closing DEV1
		A0020	Closing DEV2
		A0030	Closing DEV3
		A0040	Closing DEV4
		A0050	Closing DEV5
		A0060	Closing DEV6
		A0070	Closing DEV7
		A0080	Closing DEV8
T9900A	Actions Complete or Time Out & Hold=True		Go to step H9900A
T9900B	Actions Complete or Time Out & Hold=False		Go to step H9900B
H9900A	Command Hold End	A0010	Completed
END	Command Complete		True
H9900B	Release Devices	A0010	Release Private Devices
		A0020	Release Private Devices
		A0030	Release Private Devices
		A0040	Release Private Devices
		A0050	Release Private Devices
		A0060	Release Private Devices
		A0070	Release Private Devices
		A0080	Release Private Devices
		A0090	Reset Parameters
END1	Command Complete		True

Table 94: Hold Sequence

12.3.2 Command 01 – Filtration

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Filtration Command
T0010	First Pass = False		If First Pass = False then go to step S0010
T0020A	First Pass = True		If First Pass = True then go to step S0020

Step	Description	Action	Action Description
S0010	Initializations	A0010	First Pass Logic
		A0020	First Pass = True
T0020B	Actions Complete		If no pending actions in S0010 then go to step S0020
S0020	Acquire Devices	A0010	Acquire Devices Message
		A0020	Acquire Private Devices
		A0030	Acquire Private Devices
		A0040	Acquire Private Devices
		A0050	Acquire Private Devices
		A0060	Acquire Private Devices
		A0070	Acquire Private Devices
		A0080	Acquire Private Devices
		A0090	Acquire Private Devices
		A0110	Reset HOLD_REQ
T0100	Fill		If Task Pointer = 1 then go to step S0100
T0200A	Filtration		If Task Pointer = 2 then go to step S0200A
T0300A	Drain		If Task Pointer = 3 then go to step S0300A
T0400A	Drain Wait		If Task Pointer = 4 then go to step S0400A
S0100	Fill	A0010	Opening DEV1
		A0020	Opening DEV2
		A0030	Closing DEV3
		A0040	Closing DEV4
		A0050	Closing DEV5
		A0060	Closing DEV6
		A0070	Closing DEV7
		A0080	Closing DEV8
T0200	Filtration		If LSH_616S is in alarm then go to step S0200
S0200	Filtration	A0010	Opening DEV1
		A0020	Closing DEV2
		A0030	Opening DEV3
		A0040	Opening DEV4
		A0050	Closing DEV5
		A0060	Closing DEV6
		A0070	Closing DEV7
		A0080	Closing DEV8
		A0090	Checking other EM states
T0300	Drain		If PT_616S > SV (TP004) or VPRD2_OWNER is in step <Standby> and BLEACH5_OWNER is in step <Standby> then go to step S0300

Step	Description	Action	Action Description
S0300	Drain	A0010	Closing DEV1
		A0020	Closing DEV2
		A0030	Closing DEV3
		A0040	Opening DEV4
		A0050	Enabling Conditional Logic for DEV5
		A0060	Opening DEV6
		A0070	Opening DEV7
		A0080	Opening DEV8
		A0090	Set Message
		A0100	Set Drain Wait Recirculation time
		A0110	Start Drain Wait Timer
		A0120	Preset Drain time Message
T0400	Drain Wait		If LSL_616S is in alarm or Drain timer (TP001) is elapsed then go to step S0400
S0400	Drain Wait	A0010	Closing DEV1
		A0020	Closing DEV2
		A0030	Closing DEV3
		A0040	Closing DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing DEV6
		A0070	Closing DEV7
		A0080	Opening DEV8
T0100B	Fill		If BLEACH5_STEP is in step <Production> or <Pre Drain> or <Final Recirculation> or <Drain> or <Fill from Filter 616Ax> or <Empty> or <Blow> then go to step S0100
T0500	Close Devices		If VPRD2_OWNER is in step <Standby> and BLEACH5_OWNER in Step <Standby> then go to step S0500
S0500	Close Devices	A0010	Closing DEV1
		A0020	Closing DEV2
		A0030	Closing DEV3
		A0040	Closing DEV4
		A0050	Closing DEV5
		A0060	Closing DEV6
		A0070	Closing DEV7
		A0080	Closing DEV8
T0600	Disable HM		If no pending actions in S0500 then go to step S0600

Step	Description	Action	Action Description
S0600	Disable HM	A0010	Disable HM 01 (Module Ownership Lost)
T0700	Release Devices		If no pending actions in S0600 then go to step S0700
S0700	Release Devices	A0010	Release Private Devices
		A0020	Release Private Devices
		A0030	Release Private Devices
		A0040	Release Private Devices
		A0050	Release Private Devices
		A0060	Release Private Devices
		A0070	Release Private Devices
		A0080	Release Private Devices
T9900	Complete		If no pending actions in S0700 then go to step S9900
S9900	Complete	A0010	Set Message: Filtration Command Complete
		A0020	Self release the EM
END	Command Complete		True

Table 95: Filtration Sequence

12.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value/Reference
Operating Parameters				
N/A				
Tuning Parameters				
TP001	Drain time	Floating Point	Sec	400
TP002	DEV5 Open time	Floating Point	Sec	400
TP003	DEV7 Open time	Floating Point	Sec	400
TP004	PT_616S	Floating Point	barg	3.5

Table 96: Operating and Tuning Parameters

12.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

Parameter Name	Parameter Type	Value/Reference
MY_TAG		RF2EQ_616S
OPR_CMD_MASK		3
HOLD	External Reference	RF2EQ_STARTUP3/OP005_V ALUE
STEP_TIME	Floating Point	0
COMPLETED_TIME	Floating Point	0
RESET_COMP_TM	Boolean	False
VPRD2_OWNER	External Reference	RF2EQ_682B/OWNER_ID
BLEACH5_OWNER	External Reference	RF2EQ_622_21/OWNER_ID
LSH_616S	External Reference	RF2_LSH2616S/DI1/DISC_AC T
LSL_616S	External Reference	RF2_LSL2616S/DI1/DISC_ACT
PT_616S	External Reference	RF2_PT2616S/PV
BLEACH5_STEP	External Reference	RF2EQ_622_21/STEP_NAME
V1616S_FAIL_STAT	External Reference	RF2_V266S1/EDC1/FAILURE_ STATE

Table 97: Configurable Parameters

12.6 Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_RF_FLT_VPRD1	Refinery Area:Filtration & Vacuum Production EM

Table 98: Named Set Details

Details of states for NS_RF_FLT_VPRD1 named set are as follows,

Named Set Value	Named Set Name
00	Hold
01	Filtration

Table 99: Name Set State Details

13 EQUIPMENT MODULE CLASS DESIGN – EM_RF_FLT_VPRD2 (2-682B)

13.1 Functional Description

EM_RF_FLT_VPRD2 is a class based equipment module that performs filling operation and drains the material to complete vacuum production. EM_RF_FLT_VPRD2 is derived from the Command Driven equipment module class _EM_CMD32_130. Refer FDS for the functional design requirements including faceplate and detail display. Operator shall be provided with Hold and Production commands. Hold Request shall be enabled for all acquired devices at the end of Hold command.

13.2 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name
EM	EM_RF_FLT_VPRD2	RF2EQ_682B
DEV1	_MRSS11_130_SE	RF2_PU2682B
DEV2	_V12_IP_130_SE	RF2_V2682BD
DEV3	_V12_IP_130_SE	RF2_V2682BBS
DEV4	_CSD_130_SE	RF2_LT2622
DEV5	EM_RF_FLT_VPRD1	RF2EQ_616S

Table 100: Instances

13.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

13.3.1 Command 00 – Hold

Step	Description	Action	Action Description
H0010	Command Hold Start	A0010	Store Fail Message
		A0020	Reset Hold Request
		A0030	Save Previous Step Time
T1000	Actions Complete		Go to step H1000
H1000	Hold Command 1	A0010	Stopping DEV1
		A0020	Closing DEV2
		A0030	Closing DEV3
		A0040	Set DEV4 Mode to Remote Out
		A0050	Set DEV4 output to Closed

Step	Description	Action	Action Description
T9900A	Actions Complete or Time Out & Hold=True		Go to step H9900A
T9900B	Actions Complete or Time Out & Hold=False		Go to step H9900B
H9900A	Command Hold End	A0010	Completed
END	Command Complete		True
H9900B	Release Devices	A0010	Release Private Devices
		A0020	Release Private Devices
		A0030	Release Private Devices
		A0040	Release Private Devices
		A0050	Reset Parameters
END1	Command Complete		True

Table 101: Hold Sequence

13.3.2

Command 01 – Production

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Production Command
T0010	First Pass = False		If First Pass = False then go to step S0010
T0020A	First Pass = True		If First Pass = True then go to step S0020
S0010	Initializations	A0010	First Pass Logic
		A0020	First Pass = True
T0020B	Actions Complete		If no pending actions in S0010 then go to step S0020
S0020	Acquire Devices	A0010	Acquire Devices Message
		A0020	Acquire Private Devices
		A0030	Acquire Shared Devices
		A0040	Acquire Private Devices
		A0050	Acquire Private Devices
		A0060	Acquire Private Devices
		A0070	Enable HM 01 (Module Ownership Lost)



Step	Description	Action	Action Description
		A0080	Reset HOLD_REQ
T0100	Fill		If Task Pointer = 1 then go to step S0100
T0200A	Production		If Task Pointer = 2 then go to step S0200
T0300A	Drain		If Task Pointer = 3 then go to step S0300
T0400A	Drain wait		If Task Pointer = 4 then go to step S0400
S0100	Fill	A0010	Stopping DEV1
		A0020	Opening DEV2
		A0030	Closing DEV3
		A0040	Set DEV4 Mode to Remote Out
		A0050	Set DEV4 output to Closed
		A0055	LSL_682B is not in alarm then S0200
		A0060	Set Delay Time
		A0070	Start Delay Time
		A0080	Delay Time Message
T0200	Production		If LSL_682B is not in alarm and 1 min (TP003) is completed then go to step S0200
S0200	Production	A0010	Enabling Conditional Logic for DEV1
		A0020	Opening DEV2
		A0030	Closing DEV3
		A0040	Set DEV4 Mode to Remote Cascade
		A0050	Set DEV4 Setpoint to Remote Setpoint for LT_622
		A0060	Checking Private Devices 5 Owner ID
		A0070	Acquire Private Devices DEV5
		A0080	Starting DEV5
		A0090	Checking EM_RF_OF_BLEACH5 is in Fill from Filter-616Ax step
T0300	Drain		If RF_OF_BLEACH5 is in step <Fill From Filter 616Ax> then go to step S0300
S0300	Drain	A0010	Enabling Conditional Logic for DEV1
		A0020	Opening DEV2
		A0030	Closing DEV3
		A0040	Set DEV4 Mode to Remote Out
		A0050	Set DEV4 output to Full open
		A0060	Delay Time
		A0070	Delay Timer Message
		A0080	Checking LSL_682B is in alarm
T0400	Drain Wait		If LSL_682B is in alarm and 10 secs is over then go to step S0400

Step	Description	Action	Action Description
S0400	Drain Wait	A0010	Stopping and Disabling Conditional Logic for DEV1 (DEV1)
		A0020	Closing DEV2
		A0030	Opening DEV3
		A0040	Set DEV4 Mode to Remote Out
		A0050	Set DEV4 output to Full open
		A0060	Wait for Blow Time
		A0070	Set Blowing Time
		A0080	Start Preset Blowing Time
		A0090	Delay Timer Message
T0500	Close Devices		If DEV3 blowing timer elapsed and 5 secs (TP001) over then go to step S0500
S0500	Close Devices	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Closing DEV2
		A0030	Closing DEV3
		A0040	Set DEV4 Mode to Remote Out
		A0050	Set DEV4 output to Full close
T0600	Disable HM		If no pending action in S0500 then go to step S0600
S0600	Disable HM	A0010	Disable HM 01 (Module Ownership Lost)
T0700	Release Devices		If no pending action in S0600 then go to step S0700
S0700	Release Devices	A0010	Release Private Devices
		A0020	Release Private Devices
		A0030	Release Private Devices
		A0040	Release Private Devices
T9900	Complete		If no pending action in S0700 then go to step S9900
S9900	Complete	A0010	Set Message: Production Command Complete
		A0020	Self Release the EM
END1	Command Complete		True

Table 102: Production Sequence

13.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value/Reference
Operating Parameters				
NA				
Tuning Parameters				
TP001	Delay Timer 1	Floating Point	Sec	5
TP002	Delay Timer 2	Floating Point	Sec	10
TP003	Delay Timer 3	Floating Point	Sec	60
TP004	Blow Time	Floating Point	Sec	600
TP005	RSP For LT_622	Floating Point	%	10

Table 103: Operating and Tuning Parameters

13.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

Parameter Name	Parameter Type	Value/Reference
MY_TAG		RF2EQ_682B
OPR_CMD_MASK		3
RF_FILTRTN1	String	RF2EQ_616A1
RF_FILTRTN2	String	RF2EQ_616A2
RF_FILTRTN3	String	RF2EQ_616A3
RF_FILTRTN4	String	RF2EQ_616A4
RF_FILTRTN5	String	RF2EQ_616A5
RF_OF_BLEACH5	String	RF2EQ_622_21
LSL_682B	External Reference	RF2_LSL2682B/DI1/DISC_ACT
HOLD	Boolean	False
STEP_TIME	Floating Point	0
COMPLETED_TIME	Floating Point	0
RESET_COMP_TM	Boolean	False

Table 104: Configurable Parameters

13.6 Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_RF_FLT_VPRD2	Refinery Area: Filtration and Vacuum Production EM

Table 105: Named Set Details

Details of states for NS_RF_FLT_VPRD2 named set are as follows,

Named Set Value	Named Set Name
00	Hold
01	Production

Table 106: Name Set State Details

14 EQUIPMENT MODULE CLASS DESIGN – EM_RF_FLT_VPRD3 (2-682A)

14.1 Functional Description

EM_RF_FLT_VPRD3 is a class based equipment module that performs filling operation and then drains the material on operator command. EM_RF_FLT_VPRD3 is derived from the Command Driven equipment module class _EM_CMD16_130. Refer FDS for the functional design requirements including faceplate and detail display. Initially Operator shall be provided with Hold and Fill Commands. Start drain and Stop drain commands enable during the execution of Fill command. Hold Request shall be enabled for all acquired devices at the end of HOLD command.

14.2 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name
EM	EM_RF_FLT_VPRD3	RF2EQ_682A
DEV1	_V12_IP_130_SE	RF2_V2682AD
DEV2	_V12_IP_130_SE	RF2_V2682AB
DEV3	_V12_IP_130_SE	RF2_V2682AD1
DEV4	_V12_IP_130_SE	RF2_V2682AD2
DEV5	_MRSS11_130_SE	RF2_PU2682A

Table 107: Instances

14.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

14.3.1 Command 00 – Hold

Step	Description	Action	Action Description
H0010	Command Hold Start	A0010	Store Fail Message
		A0020	Reset Hold Request
		A0030	Save Previous Step Time
T1000	Actions Complete		Go to step H1000
H1000	Hold Command 1	A0010	Opening DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3

Step	Description	Action	Action Description
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Stopping and Disabling Conditional Logic for DEV5
T9900A	Actions Complete or Time Out & Hold=True		Go to step H9900A
T9900B	Actions Complete or Time Out & Hold=False		Go to step H9900B
H9900A	Command Hold End	A0010	Completed
END	Command Complete		True
H9900B	Release Devices	A0010	Release Devices Message
		A0020	Release Private Devices
		A0030	Release Private Devices
		A0040	Release Private Devices
		A0050	Release Private Devices
		A0060	Release Private Devices
		A0070	Reset Parameters
END1	Command Complete		True

Table 108: Hold Sequence

14.3.2

Command 01 – Fill

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Fill Command
T0010	First Pass = False		If First Pass = False then go to step S0010
T0020A	First Pass = True		If First Pass = True then go to step S0020
S0010	Initializations	A0010	First Pass Logic
		A0020	First Pass = True
T0020B	Actions Complete		If no pending actions in S0010 then go to step S0020
S0020	Acquire Devices	A0010	Acquire Devices Message
		A0020	Acquire Private Devices
		A0030	Acquire Private Devices



Step	Description	Action	Action Description
		A0040	Acquire Private Devices
		A0050	Acquire Private Devices
		A0060	Acquire Private Devices
		A0070	Reset HOLD_REQ
T0100	Start		If Task Pointer = 1 then go to step S0100
T0200B	Fill		If Task Pointer = 2 then go to step S0200
S0100	Start	A0010	Closing DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Stopping and Disabling Conditional Logic for DEV5
		A0060	Checking LSH_682A is not in alarm
T0200	Fill		If LSL_682A is not in alarm then go to step S0200
S0200	Fill	A0010	Closing DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Stopping and Disabling Conditional Logic for DEV5
		A0060	Start Fill Time Counter
		A0070	Message
		A0080	Set Preset Fill time
		A0090	Start Preset Fill Timer
		A0100	Preset Fill time Message
T9900	Complete		If LSH_682A is in alarm or operator select <Drain> command (OP001) or Preset fill time elapsed (TP003) then go to step S9900
S9900	Complete	A0010	Set Message: Fill Command Complete
		A0020	Setting Task pointer 3
		A0030	Disabling TP parameters
END	Command Complete		True

Table 109: Fill Sequence

14.3.3 Command 02 – Start Drain

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Drain Command
		A0020	Reset HOLD_REQ
T0100	Drain		If Task Pointer = 3 then go to step S0100
S0100	Drain	A0010	Opening DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Enabling Conditional Logic for DEV3
		A0040	Enabling Conditional Logic for DEV4
		A0050	Enabling Conditional Logic for DEV5
		A0060	Checking LSL_682A is in alarm
T9900	Complete		If LSL_682A is in alarm or Operator command <Stop Drain> (OP004) then go to step S9900
S9900	Complete	A0010	Set Message: Start Drain Command Complete
		A0020	Disabling TP parameters
END	Command Complete		True

Table 110: Start Drain Sequence

14.3.4 Command 03 – Stop Drain

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Executing Stop Drain Command
T0100	Drain Wait		If Task Pointer = 4 then go to step S0100
S0100	Drain Wait	A0010	Closing DEV1
		A0020	Enabling Conditional Logic for DEV2
		A0030	Enabling Conditional Logic for DEV3
		A0040	Enabling Conditional Logic for DEV4
		A0050	Stopping and Disabling Conditional Logic for DEV5
		A0060	Message
		A0070	Set Blowing time
		A0080	Start Blowing Timer
		A0090	Preset Blowing time Message
T0200	Delay		If DEV1 Blowing timer (TP002) elapsed then go to step S0200
S0200	Delay	A0010	Set Message: Waiting For Delay timer

Step	Description	Action	Action Description
T0300	Close Devices		If 5 sec (TP001) delay time is over then go to step S0300
S0300	Close Devices	A0010	Closing DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Stopping and Disabling Conditional Logic for DEV5
		A0060	Reset Drain Selection Button (OP001)
T0400	Disable HM		If no pending actions in S0300 then go to step S0400
S0400	Disable HM	A0010	Disable HM 01 (Module Ownership Lost)
T0500	Release Devices		If no pending actions in S0400 then go to step S0500
S0500	Release Devices	A0010	Release Devices Message
		A0020	Release Private Devices
		A0030	Release Private Devices
		A0040	Release Private Devices
		A0050	Release Private Devices
		A0060	Release Private Devices
T9900	Complete		Go to step S9900
S9900	Complete	A0010	Set Message: Stop Drain Command Complete
		A0020	Self Release the EM
END	Command Complete		True

Table 111: Stop Drain Sequence

14.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value/Reference
Operating Parameters				
OP001	Start Drain	Boolean	NA	False
OP002	Bleacher 622 Selection	Boolean	NA	False

Parameter	Description	Type	Units	Default Value/Reference
OP003	Slop oil tank selection	Boolean	NA	False
OP004	Stop Drain	Boolean	NA	False
Tuning Parameters				
TP001	Delay Timer	Floating Point	Sec	5
TP002	Blowing Preset time	Floating Point	Sec	600
TP003	Filling time	Floating Point	Sec	1000

Table 112: Operating and Tuning Parameters

14.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

Parameter Name	Parameter Type	Value/Reference
MY_TAG		RF2EQ_682A
OPR_CMD_MASK		15
BEACHER_622	Boolean	False
SLOP_OIL_TNK	Boolean	False
RF_OF_BLEACH5	External Reference	RF2EQ_622_21/STEP_NAME
LSH_682A	External Reference	RF2_LSH2682A/DI1/DISC_ACT
LSL_682A	External Reference	RF2_LSL2682A/DI1/DISC_ACT
LSH_622_21	External Reference	RF2_LSH2622/DI1/DISC_ACT
LT_622_21	External Reference	RF2_LT2622/AI1/HI_HI_ACT
DY_REF_1	Dynamic Reference	
HOLD	External Reference	RF2EQ_STARTUP3/OP005_VALUE
STEP_TIME	Floating Point	0
COMPLETED_TIME	Floating Point	0
RESET_COMP_TM	Boolean	False

Table 113: Configurable Parameters

14.6 Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_RF_FLT_VPRD3	Refinery Area: Filtration and Vacuum Production

Table 114: Named Set Details

Details of states for NS_RF_FLT_VPRD3 named set are as follows,

Named Set Value	Named Set Name
00	Hold
01	Fill
02	Start Drain
03	Stop Drain

Table 115: Name Set State Details

15 EQUIPMENT MODULE CLASS DESIGN – EM_RF_FLT_VPRD4 (2-682A2)

15.1 Functional Description

EM_RF_FLT_VPRD4 is a class based equipment module that performs filtration and vacuum production. This EM performs Filling Operation and drains the material based on the Operator command. EM_RF_FLT_VPRD4 is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. Initially Operator shall be provided with Hold, Fill and Start Drain Command. All devices used in EM shall be acquired in Fill Command and shall be released on Completion of Stop Drain Command. Hold Request shall be enabled for all acquired devices at the end of Hold command.

15.2 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name
EM	EM_RF_FLT_VPRD4	RF2EQ_682A2
DEV1	_V12_IP_130_SE	RF2_V2682A2D
DEV2	_V12_IP_130_SE	RF2_V2682A2B

Table 116: Instances

15.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

15.3.1 Command 00 – Hold

Step	Description	Action	Action Description
H0010	Command Hold Start	A0010	Store Fail Message
		A0020	Reset Hold Request
		A0030	Save Previous Step Time
T1000	Actions Complete		Go to step H1000
H1000	Hold Command 1	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Closing DEV2
T9900A	Actions Complete or Time Out & Hold=True		Go to step H9900A
T9900B	Actions Complete or		Go to step H9900B

Step	Description	Action	Action Description
	Time Out & Hold=False		
H9900A	Command Hold End	A0010	Completed
END	Command Complete		True
H9900B	Release Devices	A0010	Release Devices Message
		A0020	Release Private Devices
		A0030	Release Private Devices
		A0040	Reset Parametrs
END1	Command Complete		True

Table 117: Hold Sequence

15.3.2 Command 01 –Fill

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Start Fill Command
T0010	First Pass = False		If First Pass = False then go to step S0010
T0020A	First Pass = True		If First Pass = True then go to step S0020
S0010	Initializations	A0010	First Pass Logic
		A0020	First Pass = True
T0020B	Actions Complete		If no pending actions in S0010 then go to step S0020
S0020	Acquire Devices	A0010	Acquire Devices Message
		A0020	Acquire Private Devices
		A0030	Acquire Private Devices
		A0040	Reset HOLD_REQ
		A0050	Checking LSL_682A2 is not in alarm
T0100	Fill		If LSL_682A2 is not in alarm then go to step S0100
S0100	Fill	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Closing DEV2
		A0030	Set Preset Fill time
		A0040	Start Preset Fill Timer
		A0050	Preset Fill time Message
		A0060	Start Fill Time Counter



Step	Description	Action	Action Description
		A0070	Message
T9900	Complete		If LSM_682A2 is in alarm or operator select <Drain> command (OP001) or Preset Fill Time (TP001) elapsed then go to step S0100
S9900	Complete	A0010	Set Message: Start Fill Command Complete
END	Command Complete		True

Table 118: Fill Sequence

15.3.3 Command 02 – Start Drain

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Start Drain Command
		A0020	Reset HOLD_REQ
T0100	Drain		Go to step S0100
S0100	Drain	A0010	Enabling Conditional Logic for DEV1
		A0020	Closing DEV2
		A0030	Checking LSL_682A2 is in alarm
T9900	Complete		If LSL_682A2 is in alarm or Operator select <Stop Drain> Command then go to step S9900
S9900	Complete	A0010	Set Message: Start Drain Command Complete
END	Command Complete		True

Table 119: Start Drain Sequence

15.3.4 Command 03 – Stop Drain

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Executing Stop Drain Command
		A0020	Reset HOLD_REQ
T0100	Blow		Go to step S0100
S0100	Blow	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Opening DEV2
		A0030	Set Preset Blowing time
		A0040	Start Preset Blowing Timer
		A0050	Preset Blowing time Message

Step	Description	Action	Action Description
T0200	Close Devices		If DEV2 blowing time elapsed (TP002) then go to step S0200
S0200	Close Devices	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Closing DEV2
		A0030	Reset Start/Stop Drain
T0300	Disable HM		If no pending actions in S0200 then go to step S0300
S0300	Disable HM	A0010	Disable HM 01 (Module Ownership Lost)
T0400	Release Devices		If no pending actions in S0300 then go to step S0400
S0400	Release Devices	A0010	Release Devices Message
		A0020	Release Private Devices
		A0030	Release Private Devices
T9900	Complete		If no pending actions in S0400 then go to step S9900
S9900	Complete	A0010	Set Message: Stop Drain Command Complete
		A0020	Self Release the EM
END	Command Complete		True

Table 120: Stop Drain Sequence

15.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value/Reference
Operating Parameters				
OP001	Start Drain	Boolean	NA	False
OP002	Stop Drain	Boolean	NA	False
Tuning Parameters				
TP001	Preset Fill time	15	Sec	Yes
TP002	Blowing time	600	Sec	Yes

Table 121: Operating and Tuning Parameters

15.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

Parameter Name	Parameter Type	Value/Reference
MY_TAG		RF2EQ_682A2
OPR_CMD_MASK		3
RF_OF_BLEACH5	External Reference	RF2EQ_622_21/STEP_NAME
LSL_682A2	External Reference	RF2_LSL2682C/DI1/DISC_ACT
LSM_682A2	External Reference	RF2_LSM2682A/DI1/DISC_ACT
LSH_622_21	External Reference	RF2_LSH2622/DI1/DISC_ACT
LT_622_21	External Reference	RF2_LT2622/AI1/HI_HI_LIM
VACCUM_ALM	Boolean	False
DY_REF_1	Dynamic Reference	
STEP_TIME	Floating Point	0
COMPLETED_TIME	Floating Point	0
RESET_COMP_TM	Boolean	False
HOLD	External Reference	RF2EQ_STARTUP3/OP005_VALUE

Table 122: Configurable Parameters

15.6 Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_RF_FLT_VPRD4	Refinery Area: Filtration & Vacuum Production EM

Table 123: Named Set Details

Details of states for NS_RF_FLT_VPRD4 named set are as follows,

Named Set Value	Named Set Name
00	Hold
01	Fill
02	Start Drain
03	Stop Drain

Table 124: Name Set State Details

16 EQUIPMENT MODULE CLASS DESIGN – EM_RF_FLT_VPRD5 (2-606C1/C2)

16.1 Functional Description

EM_RF_FLT_VPRD5 is a class based equipment module that performs filling and discharge for a pre-defined time. EM_RF_FLT_VPRD5 module class drives two instances RF2EQ-2/606C1 and RF2EQ-2/606C2. This EM Class is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. Operator shall be provided only with Hold and Fill Commands. Discharge operation shall be executed as part of Fill State. Hold Request shall be enabled for all acquired devices at the end of Hold command.

16.2 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name	
EM	EM_RF_FLT_VPRD5	RF2EQ_606C1	RF2EQ_606C2
DEV1	_V12_IP_130_SE	RF2_V2606C1F	RF2_V2606C2F
DEV2	_V12_IP_130_SE	RF2_V2606C1D	RF2_V2606C2D
DEV3	_V12_IP_130_SE	RF2_V2630C	

Table 125: Instances

16.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

16.3.1 Command 00 – Hold

Step	Description	Action	Action Description
H0010	Command Hold Start	A0010	Store Fail Message
		A0020	Reset Hold Request
		A0030	Save Previous Step Time
T1000	Actions Complete or Time Out		Go to step H1000
H1000	Hold Command 1	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2

Step	Description	Action	Action Description
		A0030	Closing and Disabling Conditional Logic for DEV3
T9900A	Actions Complete or Time Out & Hold=True		Go to step H9900A
T9900B	Actions Complete or Time Out & Hold=False		Go to step H9900B
H9900A	Command Hold End	A0010	Completed
END	Command Complete		True
H9900B	Release Devices	A0010	Disable HM 01 (Module Ownership Lost)
		A0020	Release Devices Message
		A0030	Release Private Devices
		A0040	Release Private Devices
		A0050	Release Private Devices
		A0060	Reset Parameters
END1	Command Complete		True

Table 126: Hold Sequence

16.3.2 Command 01 – Fill

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Fill Command
T0010	First Pass = False		If First Pass = False then go to step S0010
T0020A	First Pass = True		If First Pass = True then go to step S0020
S0010	Initiaizations	A0010	First Pass Logicc
		A0020	First Pass = True
T0020B	Actions Complete		If no pending actions in S0010 then go to step S0020
S0020	Acquire Devices	A0010	Acquire Devices Message
		A0020	Acquire Private Devices
		A0030	Acquire Private Devices
		A0040	Acquire Shared Devices
		A0050	Reset HOLD_REQ

Step	Description	Action	Action Description
		A0060	Set Message: Waiting for 604C to be in Filter Aid Dosing
		A0070	Set Message: Checking Dosing Filter Aid is Enable
T0100	Standby		If RF_FLT_VPRD6 is in <Filter Aid Dosing> steps and Demand for the Dosing Filter Aid > 0 (DEMAND_DOSING) and Dosing Filter Aid enabled for dosing (FILTER_ENABLE) and Task Pointer = 1 then go to step S0100
T0200A	Fill		If Task Pointer = 2 then go to step S0200
T0300A	Wait		If Task Pointer = 3 then go to step S0300
T0400A	Discharge		If Task Pointer = 4 then go to step S0400
S0100	Evacuate	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Enabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Set Preset Evacuate Time
		A0050	Start Preset Evacuate Time
		A0060	Preset Evacuate Time Message
T0200	Fill		If Preset Evacuate time (TP004) is over then go to step S0200
S0200	Fill	A0010	Closing and Disabling Conditional Logic for DEV2
		A0020	Enabling Conditional Logic for DEV1
		A0030	Enabling Conditional Logic for DEV3
		A0040	Set Preset Fill Time
		A0050	Start Preset Fill Time
		A0060	Preset Fill Time Message
T0300	Wait		If Preset Fill time (TP003) is over then go to step S0300
S0300	Wait	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Set Preset Wait Time
		A0050	Start Preset Wait Time
		A0060	Preset Wait Time Message
		A0070	Integrate Quantity Dosed
T0400	Discharge		If Preset Wait time (TP005 or TP006) is over then go to step S0400

Step	Description	Action	Action Description
S0400	Discharge	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Enabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Set Preset Discharge Time
		A0050	Start Preset Discharge Time
		A0060	Preset Discharge Time Message
		A0070	Waiting for Discharge Time
T0500	Close Devices		If Preset Discharge time (TP007) is over then go to step S0500
S0500	Close Devices	A0010	Closing DEV2
T0600	Disable HM		If no pending actions in S0500 then go to step S0600
S0600	Disable HM	A0010	Disable HM 01 (Module Ownership Lost)
T0700	Release Devices		If no pending actions in S0600 then go to step S0700
S0700	Release Devices	A0010	Disable HM 01 (Module Ownership Lost)
		A0020	Release Devices Message
		A0030	Release Private Devices
		A0040	Release Private Devices
		A0050	Release Private Devices
T9900	Complete		If no pending actions in S0700 then go to step S9900
S9900	Complete	A0010	Set Message: Fill Command Complete
		A0020	Self Release the EM
END	Command Complete		True

Table 127: Fill Sequence

16.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value/Reference
Operating Parameters				
OP001	Dosing Arm	String	N/A	606C

Parameter	Description	Type	Units	Default Value/Reference
OP002	Reset dosed quantity	Boolean	N/A	Yes
OP003	Actual Step	String	N/A	Fill
Tuning Parameters				
TP001	Delay Timer	Floating Point	Sec	5
TP002	Dosing Amount Set Point	Floating Point	Kg	50
TP003	Preset Fill time	Floating Point	Sec	10
TP004	Preset Evacuate time	Floating Point	Sec	5
TP005	Preset Waiting time Slow	Floating Point	Sec	5
TP006	Preset Waiting time Fast	Floating Point	Sec	2
TP007	Discharge time	Floating Point	Sec	10
TP008	Volume 606C	Floating Point	Kg	4.0
TP009	Call demand	Floating Point	Kg	0.2
TP010	Remain Time	Floating Point	Sec	10
TP011	Dosed Quantity	Floating Point	Kg	100
TP012	Initial Volume	Floating Point	Kg	0
TP013	Final Volume	Floating Point	Kg	0
TP014	Demand Limit	Floating Point	Kg	0

Table 128: Operating and Tuning Parameters

16.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

Parameter Name	Parameter Type	Value/Reference	
MY_TAG		RF2EQ_606C1	RF2EQ_606C2
OPR_CMD_MASK		3	
RF_FLT_VPRD6	String	RF2EQ_604C	
FILTER_ENABLE	Boolean	False	
WT_630C	External Reference	RF2_WE2630C/PV	
DEMAND_DOSING	Floating Point	0	
COMPLETED_TIME	Floating Point	0	
RESET_COMP_TM	Boolean	False	

Parameter Name	Parameter Type	Value/Reference
STEP_TIME	Floating Point	0
HOLD	External Reference	RF2EQ_STARTUP3/OP005_VALUEr
DY_REF_1	Dynamic Reference	

Table 129: Configurable Parameters

16.6 Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_RF_FLT_VPRD5	Refinery Area: Filtration and Vacuum Production

Table 130: Named Set Details

Details of states NS_RF_FLT_VPRD5 named set are as follows,

Named Set Value	Named Set Name
00	Hold
01	Fill

Table 131: Name Set State Details

17 EQUIPMENT MODULE CLASS DESIGN – EM_RF_FLT_VPRD6 (2-604C)

17.1 Functional Description

EM_RF_FLT_VPRD6 is a class based equipment module that performs filling operation followed by filter aid dosing and then drains the material. EM_RF_FLT_VPRD6 is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. All devices used in EM shall be acquired in Fill Command and shall be released on completion of Drain Operation. Hold Request shall be enabled for all acquired devices at the end of Hold command.

17.2 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name
EM	EM_RF_FLT_VPRD6	RF2EQ_604C
DEV1	_V12_IP_130_SE	RF2_V2692BD
DEV2	_V12_IP_130_SE	RF2_V2604CV
DEV3	_V12_IP_130_SE	Ignored
DEV4	_V12_IP_130_SE	RF2_V2604CS
DEV5	_MRSS11_130_SE	RF2_PU2604C
DEV6	_MRSS11_130_SE	RF2_AG2604C
DEV7	EM_RF_FLT_VPRD5	RF2EQ_606C1
DEV8	EM_RF_FLT_VPRD5	RF2EQ_606C2

Table 132: Instances

17.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

17.3.1 Command 00 – Hold

Step	Description	Action	Action Description
H0010	Command Hold Start	A0010	Store Fail Message
		A0020	Reset Hold Request
T1000	Actions Complete & Hold=True		Go to step H1000
H1000	Hold Command 1	A0010	Closing and Disabling Conditional Logic for DEV1

Step	Description	Action	Action Description
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Stopping and Disabling Conditional logic for DEV5
		A0060	Stopping and Disabling Conditional logic for DEV6
T9900A	Actions Complete or Time Out & Hold=True		Go to step H9900A
T9900B	Actions Complete or Time Out & Hold=False		Go to step H9900B
H9900A	Command Hold End	A0010	Completed
END	Command Complete		True
H9900B	Release Devices	A0010	Release Devices Message
		A0020	Release Private Devices
		A0030	Release Private Devices
		A0040	Release Private Devices
		A0050	Release Private Devices
		A0060	Release Private Devices
		A0070	Release Private Devices
		A0080	Reset Parameters
END1	Command Complete		True

Table 133: Hold Sequence

17.3.2 Command 01 –Fill

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Fill Command
T0010	First Pass = False		If First Pass = False then go to step S0010
T0020A	First Pass = True		If First Pass = True then go to step S0020
S0010	Initializations	A0010	First Pass Logic

Step	Description	Action	Action Description
		A0020	First Pass = True
T0020B	Actions Complete		If no pending actions in S0010 then go to step S0020
S0020	Acquire Devices	A0010	Acquire Devices Message
		A0020	Acquire Shared Devices
		A0030	Acquire Private Devices
		A0040	Acquire Private Devices
		A0050	Acquire Private Devices
		A0060	Acquire Shared Devices
		A0070	Acquire Private Devices
		A0080	Acquire Private Devices
		A0090	Enable HM 01 (Module Ownership Lost)
		A0100	Reset HOLD_REQ
T0100	Fill		If Filling Button (OP001) is pressed and Task Pointer = 1 then go to step S0100
T0200A	Filter Aid Dosing		If Task Pointer = 2 then go to step S0200
T0300A	Ready		If Task Pointer = 3 then go to step S0300
S0100	Fill	A0010	Enabling Conditional Logic for DEV1
		A0020	Enabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Enabling Conditional Logic for DEV4
		A0050	Stopping and Disabling Conditional logic for DEV5
		A0060	Enabling Conditional Logic for DEV6
		A0070	Msg: Waiting for LSM_604C or LSH_604C to be in alarm
T0200	Filter Aid Dosing		If LSM_604C or LSH_604C is in alarm then go to step S0200
S0200	Filter Aid Dosing	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Enabling Conditional Logic for DEV4
		A0050	Stopping and Disabling Conditional logic for DEV5
		A0060	Enabling Conditional Logic for DEV6
		A0070	Provide Fill Command to DEV7 and DEV8
		A0080	Waiting for Filter Aid Demand < 0 Kg

Step	Description	Action	Action Description
T0300	Ready		If Filter Aid Demand (FILTER_AID_DEM) < 0 kg and Temperature (TT_604C) > = Set point (TP001) then go to step S0300
S0300	Ready	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Enabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Enabling Conditional Logic for DEV4
		A0050	Stopping and Disabling Conditional logic for DEV5
		A0060	Enabling Conditional Logic for DEV6
T9900	Complete		If no pending actions in S0300 then go to step S9900
S9900	Complete	A0010	Set Message: Start Fill Command Complete
END	Command Complete		True

Table 134: Fill Sequence

17.3.3

Command 02 – Drain

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Stop Fill Command
T0100	Drain		Go to step S0100
S0100	Drain	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Opening DEV3
		A0040	Enabling Conditional Logic for DEV4
		A0050	Enabling Conditional Logic for DEV5
		A0060	Enabling Conditional Logic for DEV6
		A0070	Waiting for LSL_604C to be in alarm
T0200	Close Devices		If LSL_604C is in alarm then go to step S0200
S0200	Close Devices	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3

Step	Description	Action	Action Description
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Stopping and Disabling Conditional logic for DEV5
		A0060	Stopping and Disabling Conditional logic for DEV6
		A0070	Reset Start & Stop Fill
T0300	Disable HM		If no pending actions in S0200 then go to step S0300
S0300	Disable HM	A0010	Disable HM 01 (Module Ownership Lost)
T0400	Release Devices		If no pending actions in S0300 then go to step S0400
S0400	Release Devices	A0010	Release Devices Message
		A0020	Release Private Devices
		A0030	Release Private Devices
		A0040	Release Private Devices
		A0050	Release Private Devices
		A0060	Release Private Devices
		A0070	Release Private Devices
T9900	Complete		If no pending actions in S0400 then go to step S9900
S9900	Complete	A0010	Set Message: Stop Fill Command Complete
		A0020	Self Release the EM
END	Command Complete		True

Table 135: Drain Sequence

17.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value/Reference
Operating Parameters				
OP001	Start Fill	Boolean	NA	False
OP002	Stop Fill	Boolean	NA	False
Tuning Parameters				
TP001	Temperature Setpoint	Floating Point	°C	70
TP002	Hysteresis	Floating Point	°C	5

Table 136: Operating and Tuning Parameters**17.5 Configurable Parameters**

The following table provides the information of configurable parameters for the equipment module instances:

Parameter Name	Parameter Type	Value/Reference
MY_TAG		RF2EQ_604C
OPR_CMD_MASK		7
DY_REF_1	Dynamic Reference	
LSL_604C	External Reference	RF2_LSL2604C/DI1/DISC_ACT
LSM_604C	External Reference	RF2_LSM2604C/DI1/DISC_ACT
LSH_604C	External Reference	RF2_LSH2604C/DI1/DISC_ACT
RF_FILTRTN	External Reference	#IGNORE
TT_604C	External Reference	RF2_TT2604C/PV
FILTER_AID_DEM	External Reference	#IGNORE
VACUUM	External Reference	#IGNORE
XV_616A1_1A	External Reference	RF2_V2616A1A/PV_D
P682B_RUN	External Reference	RF2_PU2682B/PV_D
COMPLETED_TIME	Floating Point	0
RESET_COMP_TM	Boolean	False
STEP_TIME	Floating Point	0
HOLD	External Reference	RF2EQ_STARTUP3/OP005_VALUE

Table 137: Configurable Parameters**17.6 Named Set**

Following Named set shall be used for configuration of EM.

Name	Description
NS_RF_FLT_VPRD6	Refinery Area: Filtration and Vacuum Production

Table 138: Named Set Details

Details of states for NS_RF_FLT_VPRD6 named set are as follows,

Named Set Value	Named Set Name
00	Hold
01	Fill
02	Drain

Table 139: Name Set State Details

18 EQUIPMENT MODULE CLASS DESIGN – EM_RF_OF_BLEACH6 (2-635)

18.1 Functional Description

EM_RF_OF_BLEACH6 is a class based equipment module. EM_RF_OF_BLEACH6 is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. All devices used in EM shall be acquired in Production Command and shall be released on completion of Production Operation. Hold Request shall be enabled for all acquired devices at the end of Hold command.

18.2 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name
EM	EM_RF_OF_BLEACH6	RF2EQ_635
DEV1	_V12_IP_130_SE	Ignored
DEV2	_V12_IP_130_SE	Ignored
DEV3	_V12_IP_130_SE	Ignored
DEV4	_MRSS11_130_SE	Ignored
DEV5	_CSD_130_SE	Ignored

Table 140: Instances

18.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

18.3.1 Command 00 – Hold

Step	Description	Action	Action Description
H0010	Command Hold Start	A0010	Store Fail Message
		A0020	Reset Hold Request
		A0030	Save Previous Step Time
T1000	Actions Complete or Time Out		Go to step H1000
H1000	Hold Command 1	A0010	Closing and Disabling Conditional Logic for DEV1and Set Task Pointer
		A0020	Closing DEV2
		A0030	Opening DEV3

Step	Description	Action	Action Description
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Set DEV5 Mode to Remote Out
		A0060	Set DEV5 output to Closed
T9900A	Actions Complete or Time Out & Hold=True		Go to step H9900A
T9900B	Actions Complete or Time Out & Hold=False		Go to step H9900B
H9900A	Command Hold End	A0010	Completed
END	Command Complete		True
H9900B	Release Devices	A0010	Release Devices Message
		A0020	Release Private Device
		A0030	Release Private Device
		A0040	Release Private Device
		A0050	Release Private Device
		A0060	Release Private Device
		A0170	Reset Parameters
END1	Command Complete		True

Table 141: Hold Sequence

18.3.2 Command 01 –Production

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Production Command
T0010	First Pass = False		If First Pass = False then go to step S0010
T0020A	First Pass = True		If First Pass = True then go to step S0020
S0010	Initializations	A0010	First Pass Logic
		A0020	First Pass = True
		A0030	Set Task Pointer
T0020B	Actions Complete		If no pending actions in S0010 then go to step S0020
S0020	Acquire Devices	A0010	Acquire Devices Message
		A0020	Acquire Private Device



Step	Description	Action	Action Description
		A0030	Acquire Private Device
		A0040	Acquire Private Device
		A0050	Acquire Private Device
		A0060	Acquire Private Device
		A0120	Set Hold Request
T0100	Standby		If Task Pointer = 1 then go to step S0100
T0200B	Fill		If Task Pointer = 2 then go to step S0200
T0300B	Production		If Task Pointer = 3 then go to step S0300
T0400B	Drain		If Task Pointer = 4 then go to step S0400
T0500B	Drain Wait		If Task Pointer = 5 then go to step S0500
S0100	Standby	A0010	Closing DEV1
		A0020	Closing DEV2
		A0030	Closing DEV3
		A0040	Closing DEV4
		A0050	Set DEV5 Mode to Remote Out
		A0060	Set DEV5 output to Closed
		A0070	Set Message
T0200A	Fill		If EM_BLEACH3 is in step <Production> then go to step S0200
S0200	Fill	A0010	Enabling Conditional Logic for DEV1 and Set Task Pointer
		A0020	Opening DEV2
		A0030	Closing DEV3
		A0040	Closing DEV4
		A0050	Set DEV5 Mode to Remote Out
		A0060	Set DEV5 output to Closed
		A0070	Set Message
T0300A	Production		If DEV5 >= LT_635 reach the setpoint (TP001) then go to step S0300
S0300	Production	A0010	Enabling Conditional Logic for DEV1 and Set Task Pointer
		A0020	Opening DEV2
		A0030	Closing DEV3
		A0040	Enabling Conditional Logic for DEV4
		A0050	Set DEV5 Mode to Remote Cascade
		A0060	Setting DEV5 setpoint to LT_635 RSP
		A0070	Set Message
T0400A	Drain		If EM_BLEACH3 is in step <Drain Wait> then go to step S0400

Step	Description	Action	Action Description
S0400	Drain	A0010	Enabling Conditional Logic for DEV1and Set Task Pointer
		A0020	Opening DEV2
		A0030	Closing DEV3
		A0040	Enabling Conditional Logic for DEV4
		A0050	Set DEV5 Mode to Remote Out
		A0060	Set DEV5 output to Open
		A0070	Set Message: Checking LSL_635 is in alarm
		A0080	Set Scratch Time
T0500A	Drain Wait		If LSL_635 is in alarm and 30 secs (DELAY_TIMER) time is over then go to step S0500
S0500	Drain Wait	A0010	Closing and Disabling Conditional Logic for DEV1and Set Task Pointer
		A0020	Closing DEV2
		A0030	Enabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Set DEV5 Mode to Remote Out
		A0060	Set DEV5 output to Open
		A0070	Set Blow Time
		A0080	Start Blow Time
		A0090	Blow Time Message
T0600	Close Devices		If Blowing time XV 2-635-B (TP004) is elapsed then go to step S0600
S0600	Close Devices	A0010	Closing and Disabling Conditional Logic for DEV1and Set Task Pointer
		A0020	Closing DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Set DEV5 Mode to Remote Out
		A0060	Set DEV5 output to Closed
T0700	Disable HM		If no pending actions in S0600 then go to step S0700
S0700	Disable HM	A0010	Disable HM 01 (Module Ownership Lost)
T0800	Release Devices		If no pending actions in S0700 then go to step S0800
S0800	Release Devices	A0010	Release Devices Message
		A0020	Release Private Device

Step	Description	Action	Action Description
		A0030	Release Private Device
		A0040	Release Private Device
		A0050	Release Private Device
		A0060	Release Private Device
T9900	Complete		If no pending actions in S0800 then go to step S9900
S9900	Complete	A0010	Set Message: Production Command Complete
		A0020	Self Release the EM
END	Command Complete		True

Table 142: Production Sequence

18.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value/Reference
Operating Parameters				
NA				
Tuning Parameters				
TP001	LT_635 RSP	Floating Point	%	5
TP002	TT_635 RSP	Floating Point	°C	70
TP003	Hysteresis	Floating Point	NA	5
TP004	Blow Time	Floating Point	sec	0

Table 143: Operating and Tuning Parameters

18.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

Parameter Name	Parameter Type	Value/Reference
MY_TAG		RF2EQ_635
OPR_CMD_MASK		3
EM_BLEACH3	String	RF2EQ_T503
DY_REF_1		
HOLD	External Reference	RF2EQ_STARTUP3/OP005_VALUE
DELAY_TIMER	Floating Point	30

Parameter Name	Parameter Type	Value/Reference
LSL_635	External Reference	RF2_LSL2635/DI1/DISC_ACT
STEP_TIME	Floating Point	0
COMPLETED_TIME	Floating Point	0
RESET_COMP_TM	Boolean	False

Table 144: Configurable Parameters

18.6 Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_RF_OF_BLEACH6	Refinery Area: Filtration and Vacuum Production

Table 145: Named Set Details

Details of states for NS_RF_OF_BLEACH6 named set are as follows,

Named Set Value	Named Set Name
00	Hold
01	Production

Table 146: Name Set State Details

19 MODULE INTERLOCKS

19.1 Functional Description

Module	Name	Description	Expression	Force SP State	Delay On (s)
RF2_PU2750	Auto Logic	Condition 1	'/TP01.CV' = TRUE AND <CR>('/RF2_PU2714AG/PV_D.CV' = ' EDC MTR2_PV:RUNNING')	START	NO
RF2_PU2750	Auto Logic	Condition 2	'/TP01.CV' = TRUE AND <CR>('/RF2_PU2714AG/PV_D.CV' != ' EDC MTR2_PV:RUNNING')	STOP	NO
RF2_PU2722C2	Auto Logic	Condition 1	('/TP02.CV' = TRUE) AND <CR>('/RF2_LSL2722C/DI1/DISC_ACT' != TRUE AND <CR>('/RF2_LT2722C/PID1/LO_LO_ACT.CV' != TRUE AND <CR>('/RF2_LSH2780B/DI1/DISC_ACT' != TRUE AND <CR>('/RF2EQ_822QE/OP001_VALUE.CV' = 'NS_DEOD1_PUMP:RF2_PU2722C2' AND <CR>('/RF2_V2722CD1/PV_D.CV' = ' EDC VLV_PV:OPEN')	START	NO
RF2_PU2722C2	Auto Logic	Condition 2	('/TP02.CV' = TRUE) AND <CR>('/RF2_LSL2722C/DI1/DISC_ACT' = TRUE OR <CR>('/RF2_LT2722C/PID1/LO_LO_ACT.CV' = TRUE OR <CR>('/RF2_LSH2780B/DI1/DISC_ACT' = TRUE OR <CR>('/RF2EQ_822QE/OP001_VALUE.CV' != 'NS_DEOD1_PUMP:RF2_PU2722C2' OR <CR>('/RF2_V2722CD1/PV_D.CV' != ' EDC VLV_PV:OPEN')	STOP	NO
RF2_PU2722C1	Auto Logic	Condition 1	('/TP02.CV' = TRUE) AND <CR>('/RF2_LSL2722C/DI1/DISC_ACT' != TRUE AND <CR>('/RF2_LT2722C/PID1/LO_LO_ACT.CV' != TRUE AND <CR>('/RF2_LSH2780B/DI1/DISC_ACT' != TRUE AND <CR>('/RF2EQ_822QE/OP001_VALUE.CV' = 'NS_DEOD1_PUMP:RF2_PU2722C1' AND <CR>('/RF2_V2722CD1/PV_D.CV' = ' EDC VLV_PV:OPEN')	START	NO
RF2_PU2722C1	Auto Logic	Condition 2	('/TP02.CV' = TRUE) AND <CR>('/RF2_LSL2722C/DI1/DISC_ACT' = TRUE OR <CR>('/RF2_LT2722C/PID1/LO_LO_ACT.CV' = TRUE	STOP	NO

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Module	Name	Description	Expression	Force SP State	Delay On (s)
			OR<CR>'//RF2_LSH2780B/DI1/DISC_ACT' = TRUE OR<CR>'//RF2EQ_822QE/OP001_VALUE.CV' != 'NS_DEOD1_PUMP:RF2_PU2722C1' OR<CR>'//RF2_V2722CD1/PV_D.CV' != ' EDC_VLV_PV:OPEN')		
RF2_PU27222	Auto Logic	Condition 1	('TP01.CV' = TRUE) AND <CR>'//RF2_LSL2780B/DI1/DISC_ACT' != TRUE AND <CR>'//RF2_LT2780B/PID1/LO_LO_ACT.CV' != TRUE AND<CR>'//RF2EQ_822QE/OP002_VALUE.CV' = 'NS_DEOD2_PUMP:RF2_PU27222' AND<CR>'//RF2_V2780BD/PV_D.CV' = ' EDC_VLV_PV:OPEN')	START	NO
RF2_PU27222	Auto Logic	Condition 2	('TP01.CV' = TRUE) AND <CR>'//RF2_LSL2780B/DI1/DISC_ACT' = TRUE OR <CR>'//RF2_LT2780B/PID1/LO_LO_ACT.CV' = TRUE OR<CR>'//RF2EQ_822QE/OP002_VALUE.CV' != 'NS_DEOD2_PUMP:RF2_PU27221' OR<CR>'//RF2_V2780BD/PV_D.CV' != ' EDC_VLV_PV:OPEN')	STOP	NO
RF2_PU27221	Auto Logic	Condition 1	('TP01.CV' = TRUE) AND <CR>'//RF2_LSL2780B/DI1/DISC_ACT' != TRUE AND <CR>'//RF2_LT2780B/PID1/LO_LO_ACT.CV' != TRUE AND<CR>'//RF2EQ_822QE/OP002_VALUE.CV' = 'NS_DEOD2_PUMP:RF2_PU27221' AND<CR>'//RF2_V2780BD/PV_D.CV' = ' EDC_VLV_PV:OPEN')	START	NO
RF2_PU27221	Auto Logic	Condition 2	('TP01.CV' = TRUE) AND <CR>'//RF2_LSL2780B/DI1/DISC_ACT' = TRUE OR <CR>'//RF2_LT2780B/PID1/LO_LO_ACT.CV' = TRUE OR<CR>'//RF2EQ_822QE/OP002_VALUE.CV' != 'NS_DEOD2_PUMP:RF2_PU27221' OR<CR>'//RF2_V2780BD/PV_D.CV' != ' EDC_VLV_PV:OPEN')	STOP	NO
RF2_PU2702	Auto Logic	Condition 1	('//RF2EQ_802/COMMAND_00001/S0300/ACTIVE.CV' = TRUE) AND <CR>'//RF2_LSL2702/DI1/DISC_ACT' != TRUE) AND <CR>'//TP01.CV' = TRUE) AND <CR>'//RF2_V2712D/PV_D.CV' = ' EDC_VLV_PV:OPEN')AND <CR>'//RF2_LSH2722C/DI1/DISC_ACT.CV'!= TRUE) AND	START	NO

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Module	Name	Description	Expression	Force SP State	Delay On (s)
			<CR>('//RF2_LT2722C/PID1/HI_HI_ACT.CV'!= TRUE) AND<CR>('//RF2_LT2702/AI1/LO_LO_ACT.CV'!= TRUE)		
RF2_PU2702	Auto Logic	Condition 2	('//RF2EQ_802/COMMAND_00001/S0300/ACTIVE.CV'!= TRUE OR <CR>'//RF2_LSL2702/DI1/DISC_ACT' = TRUE OR <CR>'//RF2_V2712D/PV_D.CV'!= '_EDC_VLV_PV:OPEN' OR <CR>'//RF2_LSH2722C/DI1/DISC_ACT.CV' = TRUE OR <CR>'//RF2_LT2722C/PID1/HI_HI_ACT.CV' = TRUE OR <CR>'//RF2_LT2702/AI1/LO_LO_ACT.CV' = TRUE) AND <CR>('/TP01.CV' = TRUE)	STOP	NO
RF2_PU2702	Auto Logic	Condition 3	('//RF2EQ_802/COMMAND_00001/S0300/ACTIVE.CV' = TRUE AND <CR>'//RF2_LSH2722C/DI1/DISC_ACT.CV'!= TRUE AND <CR>'//RF2_LSL2702/DI1/DISC_ACT.CV'!= TRUE AND<CR>'//RF2_V2712D/PV_D.CV' = '_EDC_VLV_PV:OPEN') AND<CR>('/TP02.CV' = TRUE)AND <CR>('//RF2_LT2722C/PID1/HI_HI_ACT.CV'!= TRUE) AND <CR>('//RF2_LT2702/AI1/LO_LO_ACT.CV'!= TRUE)	START	NO
RF2_PU2702	Auto Logic	Condition 4	('//RF2EQ_802/COMMAND_00001/S0300/ACTIVE.CV'!= TRUE OR <CR>'//RF2_LSH2722C/DI1/DISC_ACT.CV' = TRUE OR <CR>'//RF2_LSL2702/DI1/DISC_ACT.CV' = TRUE OR<CR>'//RF2_V2712D/PV_D.CV'!= '_EDC_VLV_PV:OPEN' OR <CR>'//RF2_LT2722C/PID1/HI_HI_ACT.CV' = TRUE OR <CR>'//RF2_LT2702/AI1/LO_LO_ACT.CV' = TRUE) AND <CR>('/TP02.CV' = TRUE)	STOP	NO
RF2_PU2702	Auto Logic	Condition 5	('//RF2EQ_802/COMMAND_00001/S0300/ACTIVE.CV' = TRUE AND<CR>'//RF2_LSH2722C/DI1/DISC_ACT.CV'!= TRUE AND<CR>'//RF2_V2712D/PV_D.CV' = '_EDC_VLV_PV:OPEN') AND <CR>('/TP03.CV' = TRUE)	START	NO
RF2_PU2702	Auto Logic	Condition 6	('//RF2EQ_802/COMMAND_00001/S0300/ACTIVE.CV' = TRUE OR<CR>'//RF2_LSH2722C/DI1/DISC_ACT.CV' = TRUE	STOP	NO

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			OR<CR>'//RF2_V2712D/PV_D.CV' != '_EDC_VLV_PV:OPEN') AND <CR>('/TP03.CV' = TRUE)		
RF2_PU2741X	Auto Logic	Condition 1	'/TP01.CV' = TRUE	START	NO
RF2_PU2741X	Auto Logic	Condition 2	'/TP01.CV' = FALSE	STOP	NO
RF2_PU2734CA	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND ('//RF2_LT2734CA/AI1/LO_LO_ACT.CV' != TRUE);	START	NO
RF2_PU2734CA	Auto Logic	Condition 2	('/TP01.CV' = TRUE) AND ('//RF1_LT1734CA/AI1/LO_LO_ACT.CV' = TRUE);	STOP	NO
RF2_AG2734CA	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND ('//RF2_LT2714/AI1/HI_HI_ACT.CV' != TRUE)	START	NO
RF2_AG2734CA	Auto Logic	Condition 2	('/TP01.CV' = TRUE) AND ('//RF2_LT2714/AI1/HI_HI_ACT.CV' = TRUE)	STOP	NO
RF2_AG2003A	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND <CR>('/RF2_MX2004AC/EDC1/PV_D.CV' = '_EDC_MTR2_PV:RUNNING')	START	NO
RF2_AG2003A	Auto Logic	Condition 2	('/TP01.CV' = TRUE) AND <CR>('/RF2_MX2004AC/EDC1/PV_D.CV' = '_EDC_MTR2_PV:STOPPED')	STOP	NO
RF2_AG2003A	Auto Logic	Condition 3	('/TP02.CV' = TRUE) AND <CR>('/RF2_LSL2003/DI1/DISC_ACT.CV' != TRUE)	START	NO
RF2_AG2003A	Auto Logic	Condition 4	('/TP02.CV' = TRUE) AND <CR>('/RF2_LSL2003/DI1/DISC_ACT.CV' = TRUE)	STOP	NO
RF2_PU2034PA	Auto Logic	Condition 1	('/TP01.CV' = TRUE AND<CR>'//RF2_LT2006/PID1/PV' > 30 AND<CR>'//RF2_PU2006/PV_D.CV' = '_EDC_MTR2_PV:RUNNING' AND <CR>'//RF2_LT2034PA/AI1/LO_LO_ACT.CV' != TRUE) AND <CR>('/RF2EQ_T506/OP001_VALUE.CV' = 'NS_RF_ACID_SELECTN:Phosphoric Acid')	START	NO
RF2_PU2034PA	Auto Logic	Condition 2	('/TP01.CV' = TRUE) AND <CR>('/RF2_LT2006/PID1/PV' <= 30 OR<CR>'//RF2_PU2006/PV_D.CV' = '_EDC_MTR2_PV:STOPPED'OR	STOP	NO

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Module	Name	Description	Expression	Force SP State	Delay On (s)
			<CR>'//RF2_LT2034PA/AI1/LO_LO_ACT.CV' = TRUE) OR <CR>'//RF2EQ_T506/OP001_VALUE.CV' != 'NS_RF_ACID_SELECTN:Phosphoric Acid')		
RF2_PU2006	Auto Logic	Condition 1	('//TP01.CV' = TRUE) AND <CR>'//RF2_LSL2006/DI1/DISC_ACT' != TRUE AND <CR>'//RF2_LSH2622/DI1/DISC_ACT' != TRUE AND <CR>'//RF2_V2006D/PV_D.CV' != ' EDC_VLV_PV:CLOSED')	START	NO
RF2_PU2006	Auto Logic	Condition 2	('//TP01.CV' = TRUE) AND <CR>'//RF2_LSL2006/DI1/DISC_ACT' = TRUE OR <CR>'//RF2_LSH2622/DI1/DISC_ACT' = TRUE OR <CR>'//RF2_V2006D/PV_D.CV' = ' EDC_VLV_PV:CLOSED')	STOP	NO
RF2_PU2006	Auto Logic	Condition 3	('//TP02.CV' = TRUE) AND <CR>'//RF2_LSL2006/DI1/DISC_ACT' != TRUE)	START	NO
RF2_PU2006	Auto Logic	Condition 4	('//TP02.CV' = TRUE) AND <CR>'//RF2_LSL2006/DI1/DISC_ACT' = TRUE)	STOP	NO
RF2_MX2004AC	Auto Logic	Condition 1	('//TP01.CV' = TRUE AND <CR>'//RF2_LT2006/PID1/PV' > 30 AND <CR>'//RF2_PU2006/PV_D.CV' = ' EDC_MTR2_PV:RUNNING')	START	NO
RF2_MX2004AC	Auto Logic	Condition 2	('//TP01.CV' = TRUE) AND <CR>'//RF2_LT2006/PID1/PV' <= 30 OR <CR>'//RF2_PU2006/PV_D.CV' = ' EDC_MTR2_PV:STOPPED')	STOP	NO
RF2_PU2682B	Auto Logic	Condition 1	('//TP01.CV' = TRUE) AND <CR>'//RF2_LSL2682B/DI1/DISC_ACT.CV' != TRUE) AND '//RF2_V2682BD/PV_D' = ' EDC_VLV_PV:OPEN'))	START	NO
RF2_PU2682B	Auto Logic	Condition 2	('//TP01.CV' = TRUE) AND <CR>'//RF2_LSL2682B/DI1/DISC_ACT.CV' = TRUE) OR '//RF2_V2682BD/PV_D' = ' EDC_VLV_PV:CLOSED'))	STOP	NO
RF2_PU2682A	Auto Logic	Condition 1	('//TP01.CV' = TRUE) AND <CR>'//RF2_LSH2622/DI1/DISC_ACT.CV' != TRUE AND <CR>'//RF2_LT2622/PID1/HI_HI_ACT.CV' != TRUE AND <CR>'//RF2_PT2622/AI1/LO_LO_ACT.CV' != TRUE)	START	NO
RF2_PU2682A	Auto Logic	Condition 2	('//TP01.CV' = TRUE) AND <CR>'//RF2_LSH2622/DI1/DISC_ACT.CV' = TRUE OR <CR>'//RF2_LT2622/PID1/HI_HI_ACT.CV' = TRUE OR <CR>'//RF2_PT2622/AI1/LO_LO_ACT.CV' = TRUE)	STOP	NO

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Module	Name	Description	Expression	Force SP State	Delay On (s)
RF2_PU26225	Auto Logic	Condition 1	('TP01.CV' = TRUE) AND <CR>'//RF2_LT2622/AI1/LO_LO_ACT.CV' != TRUE AND <CR>'//RF2_LSLL2622/DI1/DISC_ACT' != TRUE AND<CR>'//RF2_V2622C/EDC1/PV_D' = ' EDC_VLV_PV:OPEN')	START	NO
RF2_PU26225	Auto Logic	Condition 2	('TP01.CV' = TRUE) AND <CR>'//RF2_LT2622/AI1/LO_LO_ACT.CV' = TRUE OR <CR>'//RF2_LSLL2622/DI1/DISC_ACT' = TRUE OR <CR>'//RF2_V2622C/EDC1/PV_D' = ' EDC_VLV_PV:CLOSED')	STOP	NO
RF2_PU26225	Auto Logic	Condition 3	('TP02.CV' = TRUE) AND <CR>'//RF2_LT2622/AI1/LO_LO_ACT.CV' != TRUE AND <CR>'//RF2_LSLL2622/DI1/DISC_ACT' != TRUE)	START	NO
RF2_PU26225	Auto Logic	Condition 4	('TP02.CV' = TRUE) AND <CR>'//RF2EQ_622_21/COMMAND_00002/S0400/ACTIVE.CV' = TRUE)	STOP	NO
RF2_PU26224	Auto Logic	Condition 1	('TP01.CV' = TRUE) AND <CR>'//RF2_LT2622/AI1/LO_LO_ACT.CV' != TRUE AND <CR>'//RF2_LSLL2622/DI1/DISC_ACT' != TRUE AND<CR>'//RF2_V2622C/EDC1/PV_D' = ' EDC_VLV_PV:OPEN')	START	NO
RF2_PU26224	Auto Logic	Condition 2	('TP01.CV' = TRUE) AND <CR>'//RF2_LT2622/AI1/LO_LO_ACT.CV' = TRUE OR <CR>'//RF2_LSLL2622/DI1/DISC_ACT' = TRUE OR <CR>'//RF2_V2622C/EDC1/PV_D' = ' EDC_VLV_PV:CLOSED')	STOP	NO
RF2_PU26223	Auto Logic	Condition 1	('TP01.CV' = TRUE) AND <CR>'//RF2_LT2622/AI1/LO_LO_ACT.CV' != TRUE AND <CR>'//RF2_LSLL2622/DI1/DISC_ACT' != TRUE AND<CR>'//RF2_V2622C/EDC1/PV_D' = ' EDC_VLV_PV:OPEN')	START	NO
RF2_PU26223	Auto Logic	Condition 2	('TP01.CV' = TRUE) AND <CR>'//RF2_LT2622/AI1/LO_LO_ACT.CV' = TRUE OR <CR>'//RF2_LSLL2622/DI1/DISC_ACT' = TRUE OR <CR>'//RF2_V2622C/EDC1/PV_D' = ' EDC_VLV_PV:CLOSED')	STOP	NO
RF2_PU26222	Auto Logic	Condition 1	('TP01.CV' = TRUE) AND <CR>'//RF2_LT2622/AI1/LO_LO_ACT.CV' != TRUE AND <CR>'//RF2_LSLL2622/DI1/DISC_ACT' != TRUE AND<CR>'//RF2_V2622C/EDC1/PV_D' = ' EDC_VLV_PV:OPEN')	START	NO
RF2_PU26222	Auto Logic	Condition 2	('TP01.CV' = TRUE) AND <CR>'//RF2_LT2622/AI1/LO_LO_ACT.CV' = TRUE OR <CR>'//RF2_LSLL2622/DI1/DISC_ACT' = TRUE OR <CR>'//RF2_V2622C/EDC1/PV_D' = ' EDC_VLV_PV:CLOSED')	STOP	NO

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Module	Name	Description	Expression	Force SP State	Delay On (s)
RF2_PU26221	Auto Logic	Condition 1	('TP01.CV' = TRUE) AND <CR>('RF2_LT2622/AI1/LO_LO_ACT.CV' != TRUE AND <CR>'RF2_LSL2622/DI1/DISC_ACT' != TRUE AND<CR>'RF2_V2622C/EDC1/PV_D' = ' _EDC_VLV_PV:OPEN')	START	NO
RF2_PU26221	Auto Logic	Condition 2	('TP01.CV' = TRUE) AND <CR>('RF2_LT2622/AI1/LO_LO_ACT.CV' = TRUE OR <CR>'RF2_LSL2622/DI1/DISC_ACT' = TRUE OR <CR>'RF2_V2622C/EDC1/PV_D' = ' _EDC_VLV_PV:CLOSED')	STOP	NO
RF2_PU2604C	Auto Logic	Condition 1	('TP01.CV' = TRUE) AND <CR>('RF2_LSL2604C/DI1/DISC_ACT.CV' != TRUE) AND <CR>('RF2_V2616A1B/PV_D' = ' _EDC_VLV_PV:OPEN' OR <CR>'RF2_V2616A2B/PV_D' = ' _EDC_VLV_PV:OPEN' OR<CR>'RF2_V2616A3B/PV_D' = ' _EDC_VLV_PV:OPEN' OR<CR>'RF2_V2616A4B/PV_D' = ' _EDC_VLV_PV:OPEN' OR<CR>'RF2_V2616A5B/PV_D' = ' _EDC_VLV_PV:OPEN')	START	NO
RF2_PU2604C	Auto Logic	Condition 2	('TP01.CV' = TRUE) AND <CR>('RF2_LSL2604C/DI1/DISC_ACT.CV' = TRUE OR <CR>'RF2_V2616A1B/PV_D' != ' _EDC_VLV_PV:OPEN' AND <CR>'RF2_V2616A2B/PV_D' != ' _EDC_VLV_PV:OPEN' AND <CR>'RF2_V2616A3B/PV_D' != ' _EDC_VLV_PV:OPEN' AND <CR>'RF2_V2616A4B/PV_D' != ' _EDC_VLV_PV:OPEN' AND <CR>'RF2_V2616A5B/PV_D' != ' _EDC_VLV_PV:OPEN'))	STOP	NO
RF2_AG2604C	Auto Logic	Condition 1	('TP01.CV' = TRUE) AND <CR>('RF2_LSL2604C/DI1/DISC_ACT.CV' != TRUE)	START	NO
RF2_AG2604C	Auto Logic	Condition 2	('TP01.CV' = TRUE) AND <CR>('RF2_LSL2604C/DI1/DISC_ACT.CV' = TRUE)	STOP	NO
RF2_V2714AGB	Auto Logic	Condition 1	'TP01.CV' = TRUE AND 'PV_D.CV' = ' _EDC_VLV_PV:CLOSED'	OPEN	5
RF2_V2714AGB	Auto Logic	Condition 2	'TP01.CV' = TRUE AND 'PV_D.CV' = ' _EDC_VLV_PV:OPEN'	CLOSE	30
RF2_V2781AB	Auto Logic	Condition 1	'TP01.CV' = TRUE AND 'PV_D.CV' = ' _EDC_VLV_PV:CLOSED'	OPEN	5

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RF2_V2781AB	Auto Logic	Condition 2	'/TP01.CV' = TRUE AND '/PV_D.CV' = '_EDC_VLV_PV:OPEN'	CLOSE	30
RF2_V2780BB	Auto Logic	Condition 1	'/TP01.CV' = TRUE AND '/PV_D.CV' = '_EDC_VLV_PV:CLOSED'	OPEN	5
RF2_V2780BB	Auto Logic	Condition 2	'/TP01.CV' = TRUE AND '/PV_D.CV' = '_EDC_VLV_PV:OPEN'	CLOSE	30
RF2_V2780A2B	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND <CR>('//RF2_LT2780B/PID1/HI_HI_ACT.CV' != TRUE OR <CR>'//RF2_LSH2780B/DI1/DISC_ACT.CV' != TRUE)	OPEN	NO
RF2_V2780A2B	Auto Logic	Condition 2	('/TP01.CV' = TRUE) AND <CR>('//RF2_LT2780B/PID1/HI_HI_ACT.CV' = TRUE AND <CR>'//RF2_LSH2780B/DI1/DISC_ACT.CV' = TRUE)	CLOSE	NO
RF2_V2780A2A	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND <CR>('//RF2_LT2780B/PID1/HI_HI_ACT.CV' != TRUE OR <CR>'//RF2_LSH2780B/DI1/DISC_ACT.CV' != TRUE)	OPEN	NO
RF2_V2780A2A	Auto Logic	Condition 2	('/TP01.CV' = TRUE) AND <CR>('//RF2_LT2780B/PID1/HI_HI_ACT.CV' = TRUE AND <CR>'//RF2_LSH2780B/DI1/DISC_ACT.CV' = TRUE)	CLOSE	NO
RF2_V2780A1B	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND <CR>('//RF2_LT2780B/PID1/HI_HI_ACT.CV' != TRUE OR <CR>'//RF2_LSH2780B/DI1/DISC_ACT.CV' != TRUE)	OPEN	NO
RF2_V2780A1B	Auto Logic	Condition 2	('/TP01.CV' = TRUE) AND <CR>('//RF2_LT2780B/PID1/HI_HI_ACT.CV' = TRUE AND <CR>'//RF2_LSH2780B/DI1/DISC_ACT.CV' = TRUE)	CLOSE	NO
RF2_V2780A1A	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND <CR>('//RF2_LT2780B/PID1/HI_HI_ACT.CV' != TRUE OR <CR>'//RF2_LSH2780B/DI1/DISC_ACT.CV' != TRUE)	OPEN	NO
RF2_V2780A1A	Auto Logic	Condition 2	('/TP01.CV' = TRUE) AND <CR>('//RF2_LT2780B/PID1/HI_HI_ACT.CV' = TRUE AND <CR>'//RF2_LSH2780B/DI1/DISC_ACT.CV' = TRUE)	CLOSE	NO

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RF2_V2722QEC	Auto Logic	Condition 1	'/TP01.CV' = TRUE AND '/RF2_V2722QED/PV_D.CV' = ' _EDC_VLV_PV:CLOSED'	OPEN	NO
RF2_V2722QEC	Auto Logic	Condition 2	'/TP01.CV' = TRUE AND '/RF2_V2722QED/PV_D.CV' = ' _EDC_VLV_PV:OPEN'	CLOSE	NO
RF2_V2722QEA	Auto Logic	Condition 1	'/TP01.CV' = TRUE AND '/PV_D.CV' = ' _EDC_VLV_PV:CLOSED'	OPEN	5
RF2_V2722QEA	Auto Logic	Condition 2	'/TP01.CV' = TRUE AND '/PV_D.CV' = ' _EDC_VLV_PV:OPEN'	CLOSE	30
RF2_V2722QE	Auto Logic	Condition 1	'/RF1EQ_822QE/TP017_VALUE.CV' = 1 AND '/TP01.CV' = TRUE	OPEN	NO
RF2_V2722QE	Auto Logic	Condition 2	'/RF1EQ_822QE/TP017_VALUE.CV' != 1 AND '/TP01.CV' = TRUE	CLOSE	NO
RF2_V2722CB	Auto Logic	Condition 1	'/TP01.CV' = TRUE AND '/PV_D.CV' = ' _EDC_VLV_PV:CLOSED'	OPEN	5
RF2_V2722CB	Auto Logic	Condition 2	'/TP01.CV' = TRUE AND '/PV_D.CV' = ' _EDC_VLV_PV:OPEN'	CLOSE	30
RF2_V2722A5D	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND <CR>(' /RF2_LT2780B/PID1/HI_HI_ACT.CV' != TRUE OR <CR>' /RF2_LSH2780B/DI1/DISC_ACT.CV' != TRUE)	OPEN	NO
RF2_V2722A5D	Auto Logic	Condition 2	('/TP01.CV' = TRUE) AND <CR>(' /RF2_LT2780B/PID1/HI_HI_ACT.CV' = TRUE AND <CR>' /RF2_LSH2780B/DI1/DISC_ACT.CV' = TRUE)	CLOSE	NO
RF2_V2722A4D	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND <CR>(' /RF2_LT2780B/PID1/HI_HI_ACT.CV' != TRUE OR <CR>' /RF2_LSH2780B/DI1/DISC_ACT.CV' != TRUE)	OPEN	NO
RF2_V2722A4D	Auto Logic	Condition 2	('/TP01.CV' = TRUE) AND <CR>(' /RF2_LT2780B/PID1/HI_HI_ACT.CV' = TRUE AND <CR>' /RF2_LSH2780B/DI1/DISC_ACT.CV' = TRUE)	CLOSE	NO
RF2_V2722A3D	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND <CR>(' /RF2_LT2780B/PID1/HI_HI_ACT.CV' != TRUE OR <CR>' /RF2_LSH2780B/DI1/DISC_ACT.CV' != TRUE)	OPEN	NO

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RF2_V2722A3D	Auto Logic	Condition 2	('TP01.CV' = TRUE) AND <CR>('RF2_LT2780B/PID1/HI_HI_ACT.CV' = TRUE AND <CR>'RF2_LSH2780B/DI1/DISC_ACT.CV' = TRUE)	CLOSE	NO
RF2_V2722A2D	Auto Logic	Condition 1	('TP01.CV' = TRUE) AND <CR>('RF2_LT2780B/PID1/HI_HI_ACT.CV' != TRUE OR <CR>'RF2_LSH2780B/DI1/DISC_ACT.CV' != TRUE)	OPEN	NO
RF2_V2722A2D	Auto Logic	Condition 2	('TP01.CV' = TRUE) AND <CR>('RF2_LT2780B/PID1/HI_HI_ACT.CV' = TRUE AND <CR>'RF2_LSH2780B/DI1/DISC_ACT.CV' = TRUE)	CLOSE	NO
RF2_V2722A1D	Auto Logic	Condition 1	('TP01.CV' = TRUE) AND <CR>('RF2_LT2780B/PID1/HI_HI_ACT.CV' != TRUE OR <CR>'RF2_LSH2780B/DI1/DISC_ACT.CV' != TRUE)	OPEN	NO
RF2_V2722A1D	Auto Logic	Condition 2	('TP01.CV' = TRUE) AND <CR>('RF2_LT2780B/PID1/HI_HI_ACT.CV' = TRUE AND <CR>'RF2_LSH2780B/DI1/DISC_ACT.CV' = TRUE)	CLOSE	NO
RF2_V2721A2D	Auto Logic	Condition 1	('TP01.CV' = TRUE) AND <CR>('RF2_LSH2722C/DI1/DISC_ACT' != TRUE OR <CR>'RF2_LT2722C/AI1/HI_HI_ACT.CV' != TRUE)	OPEN	NO
RF2_V2721A2D	Auto Logic	Condition 2	('TP01.CV' = TRUE) AND <CR>('RF2_LSH2722C/DI1/DISC_ACT' = TRUE AND <CR>'RF2_LT2722C/AI1/HI_HI_ACT.CV' = TRUE)	CLOSE	NO
RF2_V2721A1D	Auto Logic	Condition 1	('TP02.CV' = TRUE) AND <CR>('RF2_LSH2722C/DI1/DISC_ACT' != TRUE AND <CR>'RF1_LT1722C/PID1/HI_HI_ACT' != TRUE)	OPEN	NO
RF2_V2721A1D	Auto Logic	Condition 2	('TP02.CV' = TRUE) AND <CR>('RF2_LSH2722C/DI1/DISC_ACT' = TRUE OR <CR>'RF1_LT1722C/PID1/HI_HI_ACT' = TRUE)	CLOSE	NO
RF2_V2702B	Auto Logic	Condition 1	'TP01.CV' = TRUE AND 'PV_D.CV' = '_EDC_VLV_PV:CLOSED'	OPEN	5
RF2_V2702B	Auto Logic	Condition 2	'TP01.CV' = TRUE AND 'PV_D.CV' = '_EDC_VLV_PV:OPEN'	CLOSE	30
RF2_V2721XS	Auto Logic	Condition 1	'TP01.CV' = TRUE AND <CR>'RF2EQ_802/OP001_VALUE.CV' = TRUE	OPEN	NO

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Module	Name	Description	Expression	Force SP State	Delay On (s)
RF2_V2721XS	Auto Logic	Condition 2	'/TP01.CV' = TRUE AND <CR>'//RF2EQ_802/OP001_VALUE.CV' = FALSE	CLOSE	NO
RF2_V2702F	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND <CR>'//RF2_LSH2702/DI1/DISC_ACT' != TRUE AND '/RF2_LT2702/AI1/HI_HI_ACT.CV' != TRUE)	OPEN	NO
RF2_V2702F	Auto Logic	Condition 2	('/TP01.CV' = TRUE) AND <CR>'//RF2_LSH2702/DI1/DISC_ACT' = TRUE OR '/RF2_LT2702/AI1/HI_HI_ACT.CV' = TRUE)	CLOSE	NO
RF2_V2702F	Auto Logic	Condition 3	('/TP02.CV' = TRUE) AND <CR>'//RF2_LSH2702/DI1/DISC_ACT' != TRUE AND <CR>'//RF2EQ_822QE/COMMAND_00002/S0500/ACTIVE.CV' != TRUE	OPEN	NO
RF2_V2702F	Auto Logic	Condition 4	('/TP02.CV' = TRUE) AND <CR>'//RF2_LSH2702/DI1/DISC_ACT' = TRUE OR <CR>'//RF2EQ_822QE/COMMAND_00002/S0500/ACTIVE.CV' = TRUE)	CLOSE	NO
RF2_V266S1	Auto Logic	Condition 1	'/TP01.CV' = TRUE AND '/RF2_V266S1/PV_D' = 'EDC_VLV_PV:CLOSED'	OPEN	5
RF2_V266S1	Auto Logic	Condition 2	'/TP01.CV' = TRUE AND '/RF2_V266S1/PV_D' = 'EDC_VLV_PV:OPEN'	CLOSE	5
RF2_V2682A2D	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND<CR>'//RF2_LSH2622/DI1/DISC_ACT.CV' != TRUE AND<CR>'//RF2_LT2622/PID1/HI_HI_ACT.CV' != TRUE AND <CR>'//RF2_PT2622/AI1/HI_HI_ACT.CV' != TRUE)	OPEN	NO
RF2_V2682A2D	Auto Logic	Condition 2	('/TP01.CV' = TRUE) AND<CR>'//RF2_LSH2622/DI1/DISC_ACT.CV' = TRUE OR <CR>'//RF2_LT2622/PID1/HI_HI_ACT.CV' = TRUE OR <CR>'//RF2_PT2622/AI1/HI_HI_ACT.CV' = TRUE)	CLOSE	NO
RF2_V2682AD2	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND '/RF2EQ_682A/OP003_VALUE' = 1	OPEN	NO
RF2_V2682AD2	Auto Logic	Condition 2	('/TP01.CV' = TRUE) AND '/RF2EQ_682A/OP003_VALUE' = 0	CLOSE	NO
RF2_V2682AD1	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND '/RF2EQ_682A/OP002_VALUE' = 1	OPEN	NO

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Module	Name	Description	Expression	Force SP State	Delay On (s)
RF2_V2682AD1	Auto Logic	Condition 2	('TP01.CV' = TRUE) AND '//RF2EQ_682A/OP002_VALUE' = 0	CLOSE	NO
RF2_V2682AB	Auto Logic	Condition 1	('TP01.CV' = TRUE) AND<CR><CR>('RF2_LSH2622/DI1/DISC_ACT.CV' != TRUE AND<CR><CR>'//RF2_LT2622/PID1/HI_HI_ACT.CV' != TRUE AND <CR><CR>'//RF2_PT2622/AI1/LO_LO_ACT.CV' != TRUE)	OPEN	NO
RF2_V2682AB	Auto Logic	Condition 2	('TP01.CV' = TRUE) AND<CR>('RF2_LSH2622/DI1/DISC_ACT.CV' = TRUE OR <CR>'//RF2_LT2622/PID1/HI_HI_ACT.CV' = TRUE OR <CR>'//RF2_PT2622/AI1/LO_LO_ACT.CV' = TRUE)	CLOSE	NO
RF2_V2616A5N	Auto Logic	Condition 1	('TP01.CV' = TRUE) AND ('//RF2_PU26225/PV_D' = 'EDC_MTR2_PV:STOPPED')	OPEN	NO
RF2_V2616A5N	Auto Logic	Condition 2	('TP01.CV' = TRUE) AND ('//RF2_PU26225/PV_D' != 'EDC_MTR2_PV:STOPPED')	CLOSE	NO
RF2_V2616A5M	Auto Logic	Condition 1	('TP01.CV' = TRUE) AND<CR>('RF2_V2616A5J/PV_D.CV' = 'EDC_VLV_PV:OPEN') AND <CR>('PV_D.CV' = 'EDC_VLV_PV:CLOSED')	OPEN	10
RF2_V2616A5M	Auto Logic	Condition 2	('TP01.CV' = TRUE) AND <CR>('RF2_V2616A5J/PV_D.CV' = 'EDC_VLV_PV:OPEN') AND <CR>('PV_D.CV' = 'EDC_VLV_PV:OPEN')	CLOSE	10
RF2_V2616A5M	Auto Logic	Condition 3	('TP01.CV' = TRUE) AND <CR>('RF2_V2616A5J/PV_D.CV' = 'EDC_VLV_PV:CLOSED')	CLOSE	NO
RF2_V2616A5L	Auto Logic	Condition 1	('TP01.CV' = TRUE) AND<CR>('RF2_V2616A5J/PV_D.CV' = 'EDC_VLV_PV:OPEN') AND <CR>('PV_D.CV' = 'EDC_VLV_PV:CLOSED')	OPEN	10
RF2_V2616A5L	Auto Logic	Condition 2	('TP01.CV' = TRUE) AND <CR>('RF2_V2616A5J/PV_D.CV' = 'EDC_VLV_PV:OPEN') AND <CR>('PV_D.CV' = 'EDC_VLV_PV:OPEN')	CLOSE	10
RF2_V2616A5L	Auto Logic	Condition 3	('TP01.CV' = TRUE) AND <CR>('RF2_V2616A5J/PV_D.CV' = 'EDC_VLV_PV:CLOSED')	CLOSE	NO

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RF2_V2616A5K	Auto Logic	Condition 1	'/TP01.CV' = TRUE AND '/RF2EQ_616A5/OP003_VALUE.CV' = 'NS_YES_NO:YES'	OPEN	NO
RF2_V2616A5K	Auto Logic	Condition 2	'/TP01.CV' = TRUE AND '/RF2EQ_616A1/OP003_VALUE.CV' = 'NS_YES_NO:NO'	CLOSE	NO
RF2_V2616A5J	Auto Logic	Condition 1	'/TP01.CV' = TRUE AND '/RF2EQ_616A5/OP003_VALUE.CV' = 'NS_YES_NO:YES'	OPEN	NO
RF2_V2616A5J	Auto Logic	Condition 2	'/TP01.CV' = TRUE AND '/RF2EQ_616A5/OP003_VALUE.CV' = 'NS_YES_NO:NO'	CLOSE	NO
RF2_V2616A5I	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND ('/RF2_LSH2622/DI1/DISC_ACT' != TRUE)	OPEN	NO
RF2_V2616A5I	Auto Logic	Condition 2	('/TP01.CV' = TRUE) AND ('/RF2_LSH2622/DI1/DISC_ACT' = TRUE)	CLOSE	NO
RF2_V2616A5H	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND<CR>('/RF2_V2616A5G/PV_D.CV' = '_EDC_VLV_PV:OPEN' AND<CR>'/RF2_V2616A5I/PV_D.CV' = '_EDC_VLV_PV:OPEN')	OPEN	3
RF2_V2616A5H	Auto Logic	Condition 2	('/TP01.CV' = TRUE) AND<CR>('/RF2_V2616A5G/PV_D.CV' != '_EDC_VLV_PV:OPEN' OR <CR>'/RF2_V2616A5I/PV_D.CV' != '_EDC_VLV_PV:OPEN')	CLOSE	NO
RF2_V2616A5H	Auto Logic	Condition 3	('/TP02.CV' = TRUE) AND<CR>('/RF2_V2616A1J/PV_D.CV' = '_EDC_VLV_PV:OPEN')	OPEN	3
RF2_V2616A5H	Auto Logic	Condition 4	('/TP02.CV' = TRUE) AND<CR>('/RF2_V2616A1J/PV_D.CV' != '_EDC_VLV_PV:OPEN')	CLOSE	NO
RF2_V2616A5G	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND<CR>('/RF2_V2616A5A/PV_D.CV' = '_EDC_VLV_PV:OPEN' AND<CR>'/RF2_V2616A5C/PV_D.CV' = '_EDC_VLV_PV:OPEN' AND<CR>'/RF2_V2616A5D/PV_D.CV' = '_EDC_VLV_PV:OPEN' AND<CR>'/RF2_V2616A5F/PV_D.CV' = '_EDC_VLV_PV:OPEN' AND<CR>'/RF2_V2616A5H/PV_D.CV' = '_EDC_VLV_PV:OPEN')	OPEN	NO
RF2_V2616A5G	Auto Logic	Condition 2	('/TP01.CV' = TRUE) AND<CR>('/RF2_V2616A5A/PV_D.CV' != '_EDC_VLV_PV:OPEN' OR<CR>'/RF2_V2616A5C/PV_D.CV' != '_EDC_VLV_PV:OPEN')	CLOSE	NO

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Module	Name	Description	Expression	Force SP State	Delay On (s)
			'_EDC_VLV_PV:OPEN' OR<CR>'//RF2_V2616A5D/PV_D.CV' != '_EDC_VLV_PV:OPEN' OR<CR>'//RF2_V2616A5F/PV_D.CV' != '_EDC_VLV_PV:OPEN' OR<CR>'//RF2_V2616A5H/PV_D.CV' != '_EDC_VLV_PV:OPEN')		
RF2_V2616A5G	Auto Logic	Condition 3	('TP02.CV' = TRUE) AND<CR>('//RF2_V2616A5A/PV_D.CV' = '_EDC_VLV_PV:OPEN' AND<CR>'//RF2_V2616A5C/PV_D.CV' = '_EDC_VLV_PV:OPEN' AND<CR>'//RF2_V2616A5D/PV_D.CV' = '_EDC_VLV_PV:OPEN' AND<CR>'//RF2_V2616A5F/PV_D.CV' = '_EDC_VLV_PV:OPEN' AND<CR>'//RF2_V2616A5H/PV_D.CV' = '_EDC_VLV_PV:OPEN' AND<CR>'//RF2_V2616A5I/PV_D.CV' = '_EDC_VLV_PV:OPEN')	OPEN	NO
RF2_V2616A5G	Auto Logic	Condition 4	('TP02.CV' = TRUE) AND<CR>('//RF2_V2616A5A/PV_D.CV' != '_EDC_VLV_PV:OPEN' OR<CR>'//RF2_V2616A5C/PV_D.CV' != '_EDC_VLV_PV:OPEN' OR<CR>'//RF2_V2616A5D/PV_D.CV' != '_EDC_VLV_PV:OPEN' OR<CR>'//RF2_V2616A5F/PV_D.CV' != '_EDC_VLV_PV:OPEN' OR<CR>'//RF2_V2616A5H/PV_D.CV' != '_EDC_VLV_PV:OPEN' OR<CR>'//RF2_V2616A5I/PV_D.CV' != '_EDC_VLV_PV:OPEN')	CLOSE	NO
RF2_V2616A5F	Auto Logic	Condition 1	('TP01.CV' = TRUE) AND ('//RF2_V2616A5N/PV_D.CV' = '_EDC_VLV_PV:CLOSED')	OPEN	NO
RF2_V2616A5F	Auto Logic	Condition 2	('TP01.CV' = TRUE) AND ('//RF2_V2616A5N/PV_D.CV' != '_EDC_VLV_PV:CLOSED')	CLOSE	NO
RF2_V2616A5E	Auto Logic	Condition 1	('TP01.CV' = TRUE) AND ('//RF2_V2616A5N/PV_D.CV' = '_EDC_VLV_PV:CLOSED')	OPEN	NO
RF2_V2616A5E	Auto Logic	Condition 2	('TP01.CV' = TRUE) AND ('//RF2_V2616A5N/PV_D.CV' != '_EDC_VLV_PV:CLOSED')	CLOSE	NO
RF2_V2616A5D	Auto Logic	Condition 1	('TP01.CV' = TRUE) AND ('//RF2_V2616A5N/PV_D.CV' = '_EDC_VLV_PV:CLOSED')	OPEN	NO
RF2_V2616A5D	Auto Logic	Condition 2	('TP01.CV' = TRUE) AND ('//RF2_V2616A5N/PV_D.CV' != '_EDC_VLV_PV:CLOSED')	CLOSE	NO

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RF2_V2616A5B	Auto Logic	Condition 1	('TP01.CV' = TRUE) AND ('/RF2_V2616A5N/PV_D.CV' = 'EDC_VLV_PV:CLOSED')	OPEN	NO
RF2_V2616A5B	Auto Logic	Condition 2	('TP01.CV' = TRUE) AND ('/RF2_V2616A5N/PV_D.CV' != 'EDC_VLV_PV:CLOSED')	CLOSE	NO
RF2_V2616A5A	Auto Logic	Condition 1	('TP01.CV' = TRUE) AND ('/RF2_V2616A5N/PV_D.CV' = 'EDC_VLV_PV:CLOSED')	OPEN	NO
RF2_V2616A5A	Auto Logic	Condition 2	('TP01.CV' = TRUE) AND ('/RF2_V2616A5N/PV_D.CV' != 'EDC_VLV_PV:CLOSED')	CLOSE	NO
RF2_V2616A4M	Auto Logic	Condition 1	('TP01.CV' = TRUE) AND ('/RF2_PU26224/PV_D' = 'EDC_MTR2_PV:STOPPED')	OPEN	NO
RF2_V2616A4M	Auto Logic	Condition 2	('TP01.CV' = TRUE) AND ('/RF2_PU26224/PV_D' != 'EDC_MTR2_PV:STOPPED')	CLOSE	NO
RF2_V2616A4L	Auto Logic	Condition 1	('TP01.CV' = TRUE) AND <CR>('/RF2_V2616A4J/PV_D.CV' = 'EDC_VLV_PV:OPEN') AND <CR>('/PV_D.CV' = 'EDC_VLV_PV:CLOSED')	OPEN	NO
RF2_V2616A4L	Auto Logic	Condition 2	('TP01.CV' = TRUE) AND <CR>('/RF2_V2616A4J/PV_D.CV' = 'EDC_VLV_PV:OPEN') AND <CR>('/PV_D.CV' = 'EDC_VLV_PV:OPEN')	CLOSE	NO
RF2_V2616A4L	Auto Logic	Condition 3	('TP01.CV' = TRUE) AND <CR>('/RF2_V2616A4J/PV_D.CV' = 'EDC_VLV_PV:CLOSED')	CLOSE	NO
RF2_V2616A4K	Auto Logic	Condition 1	('TP01.CV' = TRUE) AND <CR>('/RF2_V2616A4J/PV_D.CV' = 'EDC_VLV_PV:OPEN') AND <CR>('/PV_D.CV' = 'EDC_VLV_PV:CLOSED')	OPEN	10
RF2_V2616A4K	Auto Logic	Condition 2	('TP01.CV' = TRUE) AND <CR>('/RF2_V2616A4J/PV_D.CV' = 'EDC_VLV_PV:OPEN') AND <CR>('/PV_D.CV' = 'EDC_VLV_PV:OPEN')	CLOSE	10
RF2_V2616A4K	Auto Logic	Condition 3	('TP01.CV' = TRUE) AND <CR>('/RF2_V2616A4J/PV_D.CV' = 'EDC_VLV_PV:CLOSED')	CLOSE	NO
RF2_V2616A4J	Auto Logic	Condition 1	'TP01.CV' = TRUE AND '/RF2EQ_616A4/OP003_VALUE.CV' = 'NS_YES_NO:YES'	OPEN	NO

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RF2_V2616A4J	Auto Logic	Condition 2	'/TP01.CV' = TRUE AND '/RF2EQ_616A4/OP003_VALUE.CV' != 'NS_YES_NO:YES'	CLOSE	NO
RF2_V2616A4H	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND ('/RF2_LSH2622/DI1/DISC_ACT' != TRUE) AND <CR>('/RF2_V2616A3H/PV_D' = ' EDC_VLV_PV:CLOSED')	OPEN	NO
RF2_V2616A4H	Auto Logic	Condition 2	('/TP01.CV' = TRUE) AND <CR>('/RF2_LSH2622/DI1/DISC_ACT' = TRUE OR <CR>'/RF2_V2616A3H/PV_D' != ' EDC_VLV_PV:CLOSED')	CLOSE	NO
RF2_V2616A4G	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND<CR>('/RF2_V2616A4H/PV_D.CV' = ' EDC_VLV_PV:OPEN') AND <CR>('/RF2_V2616A3N/PV_D' = ' EDC_VLV_PV:OPEN')	OPEN	3
RF2_V2616A4G	Auto Logic	Condition 2	('/TP01.CV' = TRUE) AND<CR>('/RF2_V2616A4H/PV_D.CV' != ' EDC_VLV_PV:OPEN' OR <CR>'/RF2_V2616A3N/PV_D' != ' EDC_VLV_PV:OPEN')	CLOSE	NO
RF2_V2616A4G	Auto Logic	Condition 3	('/TP02.CV' = TRUE) AND<CR>('/RF2_V2616A4I/PV_D.CV' = ' EDC_VLV_PV:OPEN')	OPEN	3
RF2_V2616A4G	Auto Logic	Condition 4	('/TP02.CV' = TRUE) AND<CR>('/RF2_V2616A4I/PV_D.CV' = ' EDC_VLV_PV:OPEN')	CLOSE	NO
RF2_V2616A4F	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND ('/RF2_V2616A4M/PV_D.CV' = ' EDC_VLV_PV:CLOSED')	OPEN	NO
RF2_V2616A4F	Auto Logic	Condition 2	('/TP01.CV' = TRUE) AND ('/RF2_V2616A4M/PV_D.CV' != ' EDC_VLV_PV:CLOSED')	CLOSE	NO
RF2_V2616A4E	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND ('/RF2_V2616A4M/PV_D.CV' = ' EDC_VLV_PV:CLOSED')	OPEN	NO
RF2_V2616A4E	Auto Logic	Condition 2	('/TP01.CV' = TRUE) AND ('/RF2_V2616A4M/PV_D.CV' != ' EDC_VLV_PV:CLOSED')	CLOSE	NO
RF2_V2616A4D	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND ('/RF2_V2616A4M/PV_D.CV' = ' EDC_VLV_PV:CLOSED')	OPEN	NO
RF2_V2616A4D	Auto Logic	Condition 2	('/TP01.CV' = TRUE) AND ('/RF2_V2616A4M/PV_D.CV' != ' EDC_VLV_PV:CLOSED')	CLOSE	NO

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RF2_V2616A4B	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND ('/RF2_V2616A4M/PV_D.CV' = 'EDC_VLV_PV:CLOSED')	OPEN	NO
RF2_V2616A4B	Auto Logic	Condition 2	('/TP01.CV' = TRUE) AND ('/RF2_V2616A4M/PV_D.CV' != 'EDC_VLV_PV:CLOSED')	CLOSE	NO
RF2_V2616A4A	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND ('/RF2_V2616A4M/PV_D.CV' = 'EDC_VLV_PV:CLOSED')	OPEN	NO
RF2_V2616A4A	Auto Logic	Condition 2	('/TP01.CV' = TRUE) AND ('/RF2_V2616A4M/PV_D.CV' != 'EDC_VLV_PV:CLOSED')	CLOSE	NO
RF2_V2616A3M	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND <CR>('/RF2_V2616A3J/PV_D.CV' = 'EDC_VLV_PV:OPEN') AND <CR>('/PV_D.CV' = 'EDC_VLV_PV:CLOSED')	OPEN	NO
RF2_V2616A3M	Auto Logic	Condition 2	('/TP01.CV' = TRUE) AND <CR>('/RF2_V2616A3J/PV_D.CV' != 'EDC_VLV_PV:CLOSED') AND <CR>('/PV_D.CV' != 'EDC_VLV_PV:CLOSED')	CLOSE	NO
RF2_V2616A3M	Auto Logic	Condition 3	('/TP01.CV' = TRUE) AND <CR>('/RF2_V2616A3J/PV_D.CV' = 'EDC_VLV_PV:CLOSED')	CLOSE	NO
RF2_V2616A3L	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND <CR>('/RF2_V2616A3J/PV_D.CV' = 'EDC_VLV_PV:OPEN') AND <CR>('/PV_D.CV' = 'EDC_VLV_PV:CLOSED')	OPEN	NO
RF2_V2616A3L	Auto Logic	Condition 2	('/TP01.CV' = TRUE) AND <CR>('/RF2_V2616A3J/PV_D.CV' = 'EDC_VLV_PV:OPEN') AND <CR>('/PV_D.CV' = 'EDC_VLV_PV:OPEN')	CLOSE	NO
RF2_V2616A3L	Auto Logic	Condition 3	('/TP01.CV' = TRUE) AND <CR>('/RF2_V2616A3J/PV_D.CV' = 'EDC_VLV_PV:CLOSED')	CLOSE	NO
RF2_V2616A3K	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND ('/RF2_PU26223/PV_D' = 'EDC_MTR2_PV:STOPPED')	OPEN	NO
RF2_V2616A3K	Auto Logic	Condition 2	('/TP01.CV' = TRUE) AND ('/RF2_PU26223/PV_D' != 'EDC_MTR2_PV:STOPPED')	CLOSE	NO
RF2_V2616A3J	Auto Logic	Condition 1	'/TP01.CV' = TRUE AND '/RF2EQ_616A3/OP003_VALUE.CV' = 'NS_YES_NO:YES'	OPEN	NO

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RF2_V2616A3J	Auto Logic	Condition 2	'/TP01.CV' = TRUE AND '/RF2EQ_616A3/OP003_VALUE.CV' = 'NS_YES_NO:NO'	CLOSE	NO
RF2_V2616A3H	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND ('/RF2_LSH2622/DI1/DISC_ACT' != TRUE) AND <CR>('/RF2_V2616A4H/PV_D' = ' EDC_VLV_PV:CLOSED')	OPEN	NO
RF2_V2616A3H	Auto Logic	Condition 2	('/TP01.CV' = TRUE) AND <CR>('/RF2_LSH2622/DI1/DISC_ACT' = TRUE OR<CR>'/RF2_V2616A4H/PV_D' != ' EDC_VLV_PV:CLOSED')	CLOSE	NO
RF2_V2616A3G	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND<CR>('/RF2_V2616A3H/PV_D.CV' = ' EDC_VLV_PV:OPEN') AND <CR>('/RF2_V2616A3N/PV_D.CV' = ' EDC_VLV_PV:OPEN')	OPEN	3
RF2_V2616A3G	Auto Logic	Condition 2	('/TP01.CV' = TRUE) AND<CR>('/RF2_V2616A3H/PV_D.CV' != ' EDC_VLV_PV:OPEN' OR <CR>'/RF2_V2616A3N/PV_D.CV' != ' EDC_VLV_PV:OPEN')	CLOSE	NO
RF2_V2616A3G	Auto Logic	Condition 3	('/TP02.CV' = TRUE) AND<CR>('/RF2_V2616A3I/PV_D.CV' = ' EDC_VLV_PV:OPEN')	OPEN	3
RF2_V2616A3G	Auto Logic	Condition 4	('/TP02.CV' = TRUE) AND<CR>('/RF2_V2616A3I/PV_D.CV' != ' EDC_VLV_PV:OPEN')	CLOSE	NO
RF2_V2616A3F	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND ('/RF2_V2616A3K/PV_D.CV' = ' EDC_VLV_PV:CLOSED')	OPEN	NO
RF2_V2616A3F	Auto Logic	Condition 2	('/TP01.CV' = TRUE) AND ('/RF2_V2616A3K/PV_D.CV' != ' EDC_VLV_PV:CLOSED')	CLOSE	NO
RF2_V2616A3E	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND ('/RF2_V2616A3K/PV_D.CV' = ' EDC_VLV_PV:CLOSED')	OPEN	NO
RF2_V2616A3E	Auto Logic	Condition 2	('/TP01.CV' = TRUE) AND ('/RF2_V2616A3K/PV_D.CV' != ' EDC_VLV_PV:CLOSED')	CLOSE	NO
RF2_V2616A3D	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND ('/RF2_V2616A3K/PV_D.CV' = ' EDC_VLV_PV:CLOSED')	OPEN	NO
RF2_V2616A3D	Auto Logic	Condition 2	('/TP01.CV' = TRUE) AND ('/RF2_V2616A3K/PV_D.CV' != ' EDC_VLV_PV:CLOSED')	CLOSE	NO

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Module	Name	Description	Expression	Force SP State	Delay On (s)
RF2_V2616A3B	Auto Logic	Condition 1	('TP01.CV' = TRUE) AND ('/RF2_V2616A3K/PV_D.CV' = 'EDC_VLV_PV:CLOSED')	OPEN	NO
RF2_V2616A3B	Auto Logic	Condition 2	('TP01.CV' = TRUE) AND ('/RF2_V2616A3K/PV_D.CV' != 'EDC_VLV_PV:CLOSED')	CLOSE	NO
RF2_V2616A3A	Auto Logic	Condition 1	('TP01.CV' = TRUE) AND ('/RF2_V2616A3K/PV_D.CV' = 'EDC_VLV_PV:CLOSED')	OPEN	NO
RF2_V2616A3A	Auto Logic	Condition 2	('TP01.CV' = TRUE) AND ('/RF2_V2616A3K/PV_D.CV' != 'EDC_VLV_PV:CLOSED')	CLOSE	NO
RF2_V2616A2M	Auto Logic	Condition 1	('TP01.CV' = TRUE) AND <CR>('/RF2_V2616A2J/PV_D.CV' = 'EDC_VLV_PV:OPEN') AND <CR>('/PV_D.CV' = 'EDC_VLV_PV:CLOSED')	OPEN	10
RF2_V2616A2M	Auto Logic	Condition 2	('TP01.CV' = TRUE) AND <CR>('/RF2_V2616A2J/PV_D.CV' = 'EDC_VLV_PV:OPEN') AND <CR>('/PV_D.CV' = 'EDC_VLV_PV:OPEN')	CLOSE	10
RF2_V2616A2M	Auto Logic	Condition 3	('TP01.CV' = TRUE) AND <CR>('/RF2_V2616A2J/PV_D.CV' = 'EDC_VLV_PV:CLOSED')	CLOSE	NO
RF2_V2616A2L	Auto Logic	Condition 1	('TP01.CV' = TRUE) AND <CR>('/RF2_V2616A2J/PV_D.CV' = 'EDC_VLV_PV:OPEN') AND <CR>('/PV_D.CV' = 'EDC_VLV_PV:CLOSED')	OPEN	10
RF2_V2616A2L	Auto Logic	Condition 2	('TP01.CV' = TRUE) AND <CR>('/RF2_V2616A2J/PV_D.CV' = 'EDC_VLV_PV:OPEN') AND <CR>('/PV_D.CV' = 'EDC_VLV_PV:OPEN')	CLOSE	10
RF2_V2616A2L	Auto Logic	Condition 3	('TP01.CV' = TRUE) AND <CR>('/RF2_V2616A2J/PV_D.CV' = 'EDC_VLV_PV:CLOSED')	CLOSE	NO
RF2_V2616A2K	Auto Logic	Condition 1	('TP01.CV' = TRUE) AND ('/RF2_PU26222/PV_D' = 'EDC_MTR2_PV:STOPPED')	OPEN	NO
RF2_V2616A2K	Auto Logic	Condition 2	('TP01.CV' = TRUE) AND ('/RF2_PU26222/PV_D' != 'EDC_MTR2_PV:STOPPED')	CLOSE	NO
RF2_V2616A2J	Auto Logic	Condition 1	'TP01.CV' = TRUE AND '/RF2EQ_616A2/OP003_VALUE.CV' = 'NS_YES_NO:YES'	OPEN	NO

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Module	Name	Description	Expression	Force SP State	Delay On (s)
RF2_V2616A2J	Auto Logic	Condition 2	'/TP01.CV' = TRUE AND '/RF2EQ_616A2/OP003_VALUE.CV' = 'NS_YES_NO:NO'	CLOSE	NO
RF2_V2616A2H	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND ('/RF2_LSH2622/DI1/DISC_ACT' != TRUE)AND <CR>('/RF2_V2616A1H/PV_D' = ' EDC_VLV_PV:CLOSED')	OPEN	NO
RF2_V2616A2H	Auto Logic	Condition 2	('/TP01.CV' = TRUE) AND <CR>('/RF2_LSH2622/DI1/DISC_ACT' = TRUE OR <CR>'/RF2_V2616A1H/PV_D' != ' EDC_VLV_PV:CLOSED')	CLOSE	NO
RF2_V2616A2G	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND<CR>('/RF2_V2616A2H/PV_D.CV' = ' EDC_VLV_PV:OPEN' AND<CR>'/RF2_V2616A1N/PV_D.CV' = ' EDC_VLV_PV:OPEN')	OPEN	NO
RF2_V2616A2G	Auto Logic	Condition 2	('/TP01.CV' = TRUE) AND<CR>('/RF2_V2616A2H/PV_D.CV' != ' EDC_VLV_PV:OPEN' OR <CR>'/RF2_V2616A1N/PV_D.CV' != ' EDC_VLV_PV:OPEN')	CLOSE	NO
RF2_V2616A2G	Auto Logic	Condition 3	('/TP02.CV' = TRUE) AND<CR>('/RF2_V2616A2I/PV_D.CV' = ' EDC_VLV_PV:OPEN')	OPEN	NO
RF2_V2616A2G	Auto Logic	Condition 4	('/TP02.CV' = TRUE) AND<CR>('/RF2_V2616A2I/PV_D.CV' != ' EDC_VLV_PV:OPEN')	CLOSE	NO
RF2_V2616A2F	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND ('/RF2_V2616A2K/PV_D.CV' = ' EDC_VLV_PV:CLOSED')	OPEN	NO
RF2_V2616A2F	Auto Logic	Condition 2	('/TP01.CV' = TRUE) AND ('/RF2_V2616A2K/PV_D.CV' != ' EDC_VLV_PV:CLOSED')	CLOSE	NO
RF2_V2616A2E	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND ('/RF2_V2616A2K/PV_D.CV' = ' EDC_VLV_PV:CLOSED')	OPEN	NO
RF2_V2616A2E	Auto Logic	Condition 2	('/TP01.CV' = TRUE) AND ('/RF2_V2616A2K/PV_D.CV' != ' EDC_VLV_PV:CLOSED')	CLOSE	NO
RF2_V2616A2D	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND ('/RF2_V2616A2K/PV_D.CV' = ' EDC_VLV_PV:CLOSED')	OPEN	NO
RF2_V2616A2D	Auto Logic	Condition 2	('/TP01.CV' = TRUE) AND ('/RF2_V2616A2K/PV_D.CV' != ' EDC_VLV_PV:CLOSED')	CLOSE	NO

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RF2_V2616A2B	Auto Logic	Condition 1	('TP01.CV' = TRUE) AND ('/RF2_V2616A2K/PV_D.CV' = 'EDC_VLV_PV:CLOSED')	OPEN	NO
RF2_V2616A2B	Auto Logic	Condition 2	('TP01.CV' = TRUE) AND ('/RF2_V2616A2K/PV_D.CV' != 'EDC_VLV_PV:CLOSED')	CLOSE	NO
RF2_V2616A2A	Auto Logic	Condition 1	('TP01.CV' = TRUE) AND ('/RF2_V2616A2K/PV_D.CV' = 'EDC_VLV_PV:CLOSED')	OPEN	NO
RF2_V2616A2A	Auto Logic	Condition 2	('TP01.CV' = TRUE) AND ('/RF2_V2616A2K/PV_D.CV' != 'EDC_VLV_PV:CLOSED')	CLOSE	NO
RF2_V2616A1M	Auto Logic	Condition 1	('TP01.CV' = TRUE) AND ('/RF2_PU26221/PV_D' = 'EDC_MTR2_PV:STOPPED')	OPEN	NO
RF2_V2616A1M	Auto Logic	Condition 2	('TP01.CV' = TRUE) AND ('/RF2_PU26221/PV_D' != 'EDC_MTR2_PV:STOPPED')	CLOSE	NO
RF2_V2616A1L	Auto Logic	Condition 1	('TP01.CV' = TRUE) AND <CR>('/RF2_V2616A1J/PV_D.CV' = 'EDC_VLV_PV:OPEN') AND <CR>('/PV_D.CV' = 'EDC_VLV_PV:CLOSED')	OPEN	NO
RF2_V2616A1L	Auto Logic	Condition 2	('TP01.CV' = TRUE) AND <CR>('/RF2_V2616A1J/PV_D.CV' = 'EDC_VLV_PV:OPEN') AND <CR>('/PV_D.CV' = 'EDC_VLV_PV:OPEN')	CLOSE	NO
RF2_V2616A1L	Auto Logic	Condition 3	('TP01.CV' = TRUE) AND <CR>('/RF2_V2616A1J/PV_D.CV' = 'EDC_VLV_PV:CLOSED')	CLOSE	NO
RF2_V2616A1K	Auto Logic	Condition 1	('TP01.CV' = TRUE) AND <CR>('/RF2_V2616A1J/PV_D.CV' = 'EDC_VLV_PV:OPEN') AND <CR>('/PV_D.CV' = 'EDC_VLV_PV:CLOSED')	OPEN	10
RF2_V2616A1K	Auto Logic	Condition 2	('TP01.CV' = TRUE) AND <CR>('/RF2_V2616A1J/PV_D.CV' = 'EDC_VLV_PV:OPEN') AND <CR>('/PV_D.CV' = 'EDC_VLV_PV:OPEN')	CLOSE	10
RF2_V2616A1K	Auto Logic	Condition 3	('TP01.CV' = TRUE) AND <CR>('/RF2_V2616A1J/PV_D.CV' = 'EDC_VLV_PV:CLOSED')	CLOSE	NO
RF2_V2616A1J	Auto Logic	Condition 1	'TP01.CV' = TRUE AND '/RF2EQ_616A1/OP003_VALUE.CV' = 'NS_YES_NO:YES'	OPEN	NO

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RF2_V2616A1J	Auto Logic	Condition 2	'/TP01.CV' = TRUE AND '/RF2EQ_616A1/OP003_VALUE.CV' != 'NS_YES_NO:YES'	CLOSE	NO
RF2_V2616A1H	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND ('/RF2_LSH2622/DI1/DISC_ACT' != TRUE) AND <CR>('/RF2_V2616A2H/PV_D' = ' EDC_VLV_PV:CLOSED')	OPEN	NO
RF2_V2616A1H	Auto Logic	Condition 2	('/TP01.CV' = TRUE) AND<CR>('/RF2_LSH2622/DI1/DISC_ACT' = TRUE OR <CR>'/RF2_V2616A2H/PV_D' != ' EDC_VLV_PV:CLOSED')	CLOSE	NO
RF2_V2616A1G	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND<CR>('/RF2_V2616A1N/PV_D.CV' = ' EDC_VLV_PV:OPEN' AND<CR>'/RF2_V2616A1H/PV_D.CV' = ' EDC_VLV_PV:OPEN')	OPEN	3
RF2_V2616A1G	Auto Logic	Condition 2	('/TP01.CV' = TRUE) AND<CR>('/RF2_V2616A1N/PV_D.CV' != ' EDC_VLV_PV:OPEN' AND<CR>'/RF2_V2616A1H/PV_D.CV' != ' EDC_VLV_PV:OPEN')	CLOSE	NO
RF2_V2616A1G	Auto Logic	Condition 3	('/TP02.CV' = TRUE) AND<CR>('/RF2_V2616A1I/PV_D.CV' = ' EDC_VLV_PV:OPEN')	OPEN	3
RF2_V2616A1G	Auto Logic	Condition 4	('/TP02.CV' = TRUE) AND<CR>('/RF2_V2616A1I/PV_D.CV' != ' EDC_VLV_PV:OPEN')	CLOSE	NO
RF2_V2616A1F	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND ('/RF2_V2616A1M/PV_D.CV' = ' EDC_VLV_PV:CLOSED')	OPEN	NO
RF2_V2616A1F	Auto Logic	Condition 2	('/TP01.CV' = TRUE) AND ('/RF2_V2616A1M/PV_D.CV' != ' EDC_VLV_PV:CLOSED')	CLOSE	NO
RF2_V2616A1E	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND ('/RF2_V2616A1M/PV_D.CV' = ' EDC_VLV_PV:CLOSED')	OPEN	NO
RF2_V2616A1E	Auto Logic	Condition 2	('/TP01.CV' = TRUE) AND ('/RF2_V2616A1M/PV_D.CV' != ' EDC_VLV_PV:CLOSED')	CLOSE	NO
RF2_V2616A1D	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND ('/RF2_V2616A1M/PV_D.CV' = ' EDC_VLV_PV:CLOSED')	OPEN	NO
RF2_V2616A1D	Auto Logic	Condition 2	('/TP01.CV' = TRUE) AND ('/RF2_V2616A1M/PV_D.CV' != ' EDC_VLV_PV:CLOSED')	CLOSE	NO

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RF2_V2616A1B	Auto Logic	Condition 1	('TP01.CV' = TRUE) AND ('/RF2_V2616A1M/PV_D.CV' = ' EDC_VLV_PV:CLOSED')	OPEN	NO
RF2_V2616A1B	Auto Logic	Condition 2	('TP01.CV' = TRUE) AND ('/RF2_V2616A1M/PV_D.CV' != ' EDC_VLV_PV:CLOSED')	CLOSE	NO
RF2_V2616A1A	Auto Logic	Condition 1	('TP01.CV' = TRUE) AND ('/RF2_V2616A1M/PV_D.CV' = ' EDC_VLV_PV:CLOSED')	OPEN	NO
RF2_V2616A1A	Auto Logic	Condition 2	('TP01.CV' = TRUE) AND ('/RF2_V2616A1M/PV_D.CV' != ' EDC_VLV_PV:CLOSED')	CLOSE	NO
RF2_V2622B	Auto Logic	Condition 1	('TP01.CV' = TRUE) AND <CR>('/RF2_LSH2622/DI1/DISC_ACT' != TRUE)	OPEN	NO
RF2_V2622B	Auto Logic	Condition 2	('TP01.CV' = TRUE) AND <CR>('/RF2_LSH2622/DI1/DISC_ACT' = TRUE)	CLOSE	NO
RF2_V2606A6F	Auto Logic	Condition 1	'TP01.CV' = TRUE AND <CR>('/RF2_V2606A6D/PV_D.CV' = ' EDC_VLV_PV:CLOSED')	OPEN	NO
RF2_V2606A6F	Auto Logic	Condition 2	'TP01.CV' = TRUE AND <CR>('/RF2_V2606A6D/PV_D.CV' = ' EDC_VLV_PV:OPEN')	CLOSE	NO
RF2_V2606A6D	Auto Logic	Condition 1	'TP01.CV' = TRUE AND <CR>('/RF2_V2606A6F/PV_D.CV' = ' EDC_VLV_PV:CLOSED')	OPEN	NO
RF2_V2606A6D	Auto Logic	Condition 2	'TP01.CV' = TRUE AND <CR>('/RF2_V2606A6F/PV_D.CV' = ' EDC_VLV_PV:OPEN')	CLOSE	NO
RF2_V2606A5F	Auto Logic	Condition 1	'TP01.CV' = TRUE AND <CR>('/RF2_V2606A5D/PV_D.CV' = ' EDC_VLV_PV:CLOSED')	OPEN	NO
RF2_V2606A5F	Auto Logic	Condition 2	'TP01.CV' = TRUE AND <CR>('/RF2_V2606A5D/PV_D.CV' = ' EDC_VLV_PV:OPEN')	CLOSE	NO
RF2_V2606A5D	Auto Logic	Condition 1	'TP01.CV' = TRUE AND <CR>('/RF2_V2606A5F/PV_D.CV' = ' EDC_VLV_PV:CLOSED')	OPEN	NO
RF2_V2606A5D	Auto Logic	Condition 2	'TP01.CV' = TRUE AND <CR>('/RF2_V2606A5F/PV_D.CV' = ' EDC_VLV_PV:OPEN')	CLOSE	NO
RF2_V2606A4F	Auto Logic	Condition 1	'TP01.CV' = TRUE AND <CR>('/RF2_V2606A4D/PV_D.CV' = ' EDC_VLV_PV:CLOSED')	OPEN	NO

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RF2_V2606A4F	Auto Logic	Condition 2	'/TP01.CV' = TRUE AND <CR>'//RF2_V2606A4D/PV_D.CV' = ' EDC_VLV_PV:OPEN'	CLOSE	NO
RF2_V2606A4D	Auto Logic	Condition 1	'/TP01.CV' = TRUE AND <CR>'//RF2_V2606A4F/PV_D.CV' = ' EDC_VLV_PV:CLOSED'	OPEN	NO
RF2_V2606A4D	Auto Logic	Condition 2	'/TP01.CV' = TRUE AND <CR>'//RF2_V2606A4F/PV_D.CV' = ' EDC_VLV_PV:OPEN'	CLOSE	NO
RF2_V2606A3F	Auto Logic	Condition 1	'/TP01.CV' = TRUE AND <CR>'//RF2_V2606A3D/PV_D.CV' = ' EDC_VLV_PV:CLOSED'	OPEN	NO
RF2_V2606A3F	Auto Logic	Condition 2	'/TP01.CV' = TRUE AND <CR>'//RF2_V2606A3D/PV_D.CV' = ' EDC_VLV_PV:OPEN'	CLOSE	NO
RF2_V2606A3D	Auto Logic	Condition 1	'/TP01.CV' = TRUE AND <CR>'//RF2_V2606A3F/PV_D.CV' = ' EDC_VLV_PV:CLOSED'	OPEN	NO
RF2_V2606A3D	Auto Logic	Condition 2	'/TP01.CV' = TRUE AND <CR>'//RF2_V2606A3F/PV_D.CV' = ' EDC_VLV_PV:OPEN'	CLOSE	NO
RF2_V2606A2F	Auto Logic	Condition 1	'/TP01.CV' = TRUE AND <CR>'//RF2_V2606A2D/PV_D.CV' = ' EDC_VLV_PV:CLOSED'	OPEN	NO
RF2_V2606A2F	Auto Logic	Condition 2	'/TP01.CV' = TRUE AND <CR>'//RF2_V2606A2D/PV_D.CV' = ' EDC_VLV_PV:OPEN'	CLOSE	NO
RF2_V2606A2D	Auto Logic	Condition 1	'/TP01.CV' = TRUE AND <CR>'//RF2_V2606A2F/PV_D.CV' = ' EDC_VLV_PV:CLOSED'	OPEN	NO
RF2_V2606A2D	Auto Logic	Condition 2	'/TP01.CV' = TRUE AND <CR>'//RF2_V2606A2F/PV_D.CV' = ' EDC_VLV_PV:OPEN'	CLOSE	NO
RF2_V2606A1F	Auto Logic	Condition 1	'/TP01.CV' = TRUE AND <CR>'//RF2_V2606A1D/PV_D.CV' = ' EDC_VLV_PV:CLOSED'	OPEN	NO
RF2_V2606A1F	Auto Logic	Condition 2	'/TP01.CV' = TRUE AND <CR>'//RF2_V2606A1D/PV_D.CV' = ' EDC_VLV_PV:OPEN'	CLOSE	NO
RF2_V2606A1D	Auto Logic	Condition 1	'/TP01.CV' = TRUE AND <CR>'//RF2_V2606A1F/PV_D.CV' = ' EDC_VLV_PV:CLOSED'	OPEN	NO
RF2_V2606A1D	Auto Logic	Condition 2	'/TP01.CV' = TRUE AND <CR>'//RF2_V2606A1F/PV_D.CV' = ' EDC_VLV_PV:OPEN'	CLOSE	NO

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RF2_V2606C2F	Auto Logic	Condition 1	'^/TP01'=1 AND ('//RF2_V2606C2D/PV_D' = '_EDC_VLV_PV:CLOSED')	OPEN	NO
RF2_V2606C2F	Auto Logic	Condition 2	'^/TP01'=1 AND ('//RF2_V2606C2D/PV_D' = '_EDC_VLV_PV:OPEN')	CLOSE	NO
RF2_V2606C2D	Auto Logic	Condition 1	'^/TP01'=1 AND '//RF2_V2606C2F/PV_D' = '_EDC_VLV_PV:CLOSED'	OPEN	NO
RF2_V2606C2D	Auto Logic	Condition 2	'^/TP01'=1 AND '//RF2_V2606C2F/PV_D' = '_EDC_VLV_PV:OPEN'	CLOSE	NO
RF2_V2606C1F	Auto Logic	Condition 1	'^/TP01'=1 AND ('//RF2_V2606C1D/PV_D' = '_EDC_VLV_PV:CLOSED')	OPEN	NO
RF2_V2606C1F	Auto Logic	Condition 2	'^/TP01'=1 AND ('//RF2_V2606C1D/PV_D' = '_EDC_VLV_PV:OPEN')	CLOSE	NO
RF2_V2606C1D	Auto Logic	Condition 1	'^/TP01'=1 AND '//RF2_V2606C1F/PV_D' = '_EDC_VLV_PV:CLOSED'	OPEN	NO
RF2_V2606C1D	Auto Logic	Condition 2	'^/TP01'=1 AND '//RF2_V2606C1F/PV_D' = '_EDC_VLV_PV:OPEN'	CLOSE	NO
RF2_V2604CV	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND ('//RF2_PT2622/AI1/HI_HI_ACT.CV' != TRUE)	OPEN	NO
RF2_V2604CV	Auto Logic	Condition 2	('/TP01.CV' = TRUE) AND ('//RF2_PT2622/AI1/HI_HI_ACT.CV' = TRUE)	CLOSE	NO
RF2_V2604CS	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND <CR>('//RF2_TT2604C/PV' < ('//RF2EQ_604C/TP001_VALUE.CV' + '//RF2EQ_604C/TP002_VALUE.CV'))	OPEN	NO
RF2_V2604CS	Auto Logic	Condition 2	('/TP01.CV' = TRUE) AND <CR>('//RF2_TT2604C/PV' >= ('//RF2EQ_604C/TP001_VALUE.CV' + '//RF2EQ_604C/TP002_VALUE.CV'))	CLOSE	NO
RF2_V2682BD	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND <CR>('//RF2_LSM2604C/DI1/DISC_ACT' != TRUE OR <CR>'//RF2_LSH2604C/DI1/DISC_ACT.CV' != TRUE) AND <CR>'//RF2_PU2682B/EDC1/PV_D.CV' = '_EDC_MTR2_PV:RUNNING')	OPEN	NO

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RF2_V2682BD	Auto Logic	Condition 2	('TP01.CV' = TRUE) AND <CR>('RF2_LSM2604C/DI1/DISC_ACT.CV' = TRUE AND <CR>'RF2_LSH2604C/DI1/DISC_ACT.CV' = TRUE) OR <CR>'RF2_PU2682B/EDC1/PV_D.CV' = ' EDC_MTR2_PV:STOPPED')	CLOSE	NO
RF2_V2630C	Auto Logic	Condition 1	'^TP01'=1 AND ('RF2_V2606C1D/PV_D' = ' EDC_VLV_PV:CLOSED' OR 'RF2_V2606C2D/PV_D' = ' EDC_VLV_PV:CLOSED')	OPEN	NO
RF2_V2630C	Auto Logic	Condition 2	'^TP01'=1 AND ('RF2_V2606C1D/PV_D' = ' EDC_VLV_PV:OPEN' OR 'RF2_V2606C2D/PV_D' = ' EDC_VLV_PV:OPEN')	CLOSE	NO
RF2_V2630A34	Auto Logic	Condition 1	'TP01.CV' = TRUE AND <CR>'RF2_V2606A1D/PV_D.CV' = ' EDC_VLV_PV:CLOSED' AND <CR>'RF2_V2606A2D/PV_D.CV' = ' EDC_VLV_PV:CLOSED' AND <CR>'RF2_V2606A3D/PV_D.CV' = ' EDC_VLV_PV:CLOSED' AND <CR>'RF2_V2606A4D/PV_D.CV' = ' EDC_VLV_PV:CLOSED' AND <CR>'RF2_V2606A5D/PV_D.CV' = ' EDC_VLV_PV:CLOSED' AND <CR>'RF2_V2606A6D/PV_D.CV' = ' EDC_VLV_PV:CLOSED'	OPEN	NO
RF2_V2630A34	Auto Logic	Condition 2	'TP01.CV' = TRUE AND <CR>'RF2_V2606A1D/PV_D.CV' = ' EDC_VLV_PV:OPEN' OR <CR>'RF2_V2606A2D/PV_D.CV' = ' EDC_VLV_PV:OPEN' OR <CR>'RF2_V2606A3D/PV_D.CV' = ' EDC_VLV_PV:OPEN' OR <CR>'RF2_V2606A4D/PV_D.CV' = ' EDC_VLV_PV:OPEN' OR <CR>'RF2_V2606A5D/PV_D.CV' = ' EDC_VLV_PV:OPEN' OR <CR>'RF2_V2606A6D/PV_D.CV' = ' EDC_VLV_PV:OPEN')	CLOSE	NO
RF2_V2630A12	Auto Logic	Condition 1	'TP01.CV' = TRUE AND <CR>'RF2_V2606A1D/PV_D.CV' = ' EDC_VLV_PV:CLOSED' AND <CR>'RF2_V2606A2D/PV_D.CV' = ' EDC_VLV_PV:CLOSED' AND <CR>'RF2_V2606A3D/PV_D.CV' = ' EDC_VLV_PV:CLOSED' AND <CR>'RF2_V2606A4D/PV_D.CV' = ' EDC_VLV_PV:CLOSED' AND <CR>'RF2_V2606A5D/PV_D.CV' = ' EDC_VLV_PV:CLOSED' AND <CR>'RF2_V2606A6D/PV_D.CV' = ' EDC_VLV_PV:CLOSED'	OPEN	NO

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			'_EDC_VLV_PV:CLOSED' AND <CR>'//RF2_V2606A6D/PV_D.CV' = '_EDC_VLV_PV:CLOSED'		
RF2_V2630A12	Auto Logic	Condition 2	'/TP01.CV' = TRUE AND <CR>('//RF2_V2606A1D/PV_D.CV' = '_EDC_VLV_PV:OPEN' OR <CR>'//RF2_V2606A2D/PV_D.CV' = '_EDC_VLV_PV:OPEN' OR <CR>'//RF2_V2606A3D/PV_D.CV' = '_EDC_VLV_PV:OPEN' OR <CR>'//RF2_V2606A4D/PV_D.CV' = '_EDC_VLV_PV:OPEN' OR <CR>'//RF2_V2606A5D/PV_D.CV' = '_EDC_VLV_PV:OPEN' OR <CR>'//RF2_V2606A6D/PV_D.CV' = '_EDC_VLV_PV:OPEN')	CLOSE	NO
RF2_V2024F	Auto Logic	Condition 1	('/TP01.CV' = TRUE AND '//RF2_LSH2006/DI1/DISC_ACT' != TRUE);	OPEN	NO
RF2_V2024F	Auto Logic	Condition 2	('/TP01.CV' = TRUE AND '//RF2_LSH2006/DI1/DISC_ACT' = TRUE);	CLOSE	NO
RF2_V2024F	Auto Logic	Condition 3	('/TP02.CV' = TRUE) AND<CR>('//RF1EQ_501/OP001_VALUE.CV' = 'NS_MNA_SELECTION:MNA2' AND ('//RF1_LSH1001/DI1/DISC_ACT' != TRUE OR '//RF1_LT1001/AI1/HI_HI_ACT.CV' != TRUE))	OPEN	NO
RF2_V2024F	Auto Logic	Condition 4	('/TP02.CV' = TRUE) AND<CR>('//RF1EQ_501/OP001_VALUE.CV' = 'NS_MNA_SELECTION:MNA1' OR ('//RF1_LSH1001/DI1/DISC_ACT' = TRUE AND '//RF1_LT1001/AI1/HI_HI_ACT.CV' = TRUE))	CLOSE	NO
RF2_TT2781AA	Auto Logic	Condition 1	'/TP01.CV' = TRUE AND<CR>('//RF2_PU2714AG/PV_D.CV' != '_EDC_MTR2_PV:RUNNING') AND<CR>'//MODELOCK_OVR.CV' = FALSE	0	NO
RF2_LT2714	Auto Logic	Condition 1	'/TP01.CV' = TRUE AND<CR>'//RF2_LSL2714/DI1/DISC_ACT' = TRUE AND<CR>'//MODELOCK_OVR.CV' = FALSE	0	NO
RF2_TT2781B	Auto Logic	Condition 1	('//RF2_PU27221/PV_D.CV' != '_EDC_MTR2_PV:RUNNING' AND <CR>'//RF2_PU27222/PV_D.CV' != '_EDC_MTR2_PV:RUNNING') AND<CR>'//TP01.CV' = TRUE	0	NO
RF2_TT2722C	Auto Logic	Condition 1	'/TP01.CV' = TRUE AND <CR>'//PV.CV' >= '//RF2EQ_822QE/TP012_VALUE.CV' AND <CR>'//RF2EQ_822QE/OP009_VALUE.CV' = TRUE AND	0	NO

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Module	Name	Description	Expression	Force SP State	Delay On (s)
			<CR>'//RF2EQ_822QE/OP010_VALUE.CV' = TRUE AND <CR>'//MODELOCK_OVR.CV' = FALSE);		
RF2_TT2722C	Auto Logic	Condition 2	'/TP01.CV' = TRUE AND <CR>'//RF2EQ_822QE/OP009_VALUE.CV' != TRUE AND <CR>'//MODELOCK_OVR.CV' = FALSE);	1	NO
RF2_LT2780B	Auto Logic	Condition 1	('//RF2_PU27221/PV_D.CV' != ' _EDC_MTR2_PV:RUNNING' AND <CR>'//RF2_PU27222/PV_D.CV' != ' _EDC_MTR2_PV:RUNNING') AND<CR>'//TP01.CV' = TRUE	0	NO
RF2_LT2722C	Auto Logic	Condition 1	('//RF2_PU2722C1/PV_D.CV' != ' _EDC_MTR2_PV:RUNNING' AND <CR>'//RF2_PU2722C2/PV_D.CV' != ' _EDC_MTR2_PV:RUNNING') AND<CR>'//TP01.CV' = TRUE	0	NO
RF2_FT2702	Auto Logic	Condition 1	'//RF2_PU2702/PV_D.CV' != ' _EDC_MTR2_PV:RUNNING' AND<CR>'//TP01.CV' = TRUE	0	NO
RF2_FT2734CA	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND ('//RF2_LT2734CA/AI1/LO_LO_ACT.CV' = TRUE)AND <CR>'//MODELOCK_OVR.CV' = FALSE);	0	NO
RF2_TT2021B	Auto Logic	Condition 1	'/TP01.CV' = TRUE AND '//RF2_FT2001/PID1/PV.CV' <= 1000 AND <CR>'//MODELOCK_OVR.CV' = FALSE;	0	NO
RF2_LT2006	Auto Logic	Condition 1	('/TP01.CV' = TRUE) AND<CR>'//RF2_PU2006/PV_D.CV' = ' _EDC_MTR2_PV:STOPPED' AND <CR>'//MODELOCK_OVR.CV' = FALSE	0	NO
RF2_FT2001	Auto Logic	Condition 1	('/TP01.CV' = TRUE AND <CR>'//RF2_LSH2006/DI1/DISC_ACT.CV' = TRUE AND <CR>'//MODELOCK_OVR.CV' = FALSE);	0	NO

Table 147: Module Force Setpoints